

NPVE

NanoPatterning and Visualization Engine

User Guide

Version 4.0 April 2013

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Ottawa, Canada

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NPVE
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Introduction and Overview

The NanoPatterning and Visualization Engine (NPVE) provides a comprehensive and user friendly means to define, execute and observe NanoPatterning operations on a FIB or SEM system. The NPVE is a powerful tool to enable both straightforward and complex NanoPatterning operations, and at the same time provide real-time visualization of the signals produced by the patterning beam. This combination of comprehensive patterning and real-time visualization enables a wide range of solutions and greatly increases the likelihood of a successful result.

This manual is divided into several parts. Part I discusses aspects of beam control pertaining to Image Acquisition and NanoPatterning – how to define, through the NPVE interface, various image resolutions and shapes to be patterned, organizing these shapes and changing their patterning styles and properties, and executing the NanoPatterning operations.

Part II discusses aspects of Visualization – how to apply various image processing functions during imaging and how to best visualize all NanoPatterning operations so as to maximize the available information. This section includes numerous image processing and enhancement functions, digital image zoom, false colour and other palette manipulation, and image overlay, to name but a few of the Visualization features.

Part III discusses the benefits of several optional ease-of-use and efficiency enhancement modules including the NPVE Project Analysis and Recording engine. The Project Analysis and Recording engine can automatically record pertinent data regarding all aspects of the process so as to provide a complete log of all experimental parameters. This is organized chronologically and referenced to automatically acquired images, and can be readily reviewed online or offline at a later date. Part IV also provides more detail concerning the Project Analysis tool which includes detailed visualization of patterning operations, recording of real-time thumbnail image series from these operations, production and recording of endpoint signal graphs, digitally reconstructed cross-sections, etc. This is achieved through the analysis of information generated by the microscope and selected detector(s) related to the beam dwelling on the sample, coupled with signal and image processing and analysis.

This version of the NPVE is designed to work with ion and electron beam microscopes from Carl Zeiss Microscopy. The NPVE consists of hardware and software developed by Fibics Incorporated along with a PC running the Microsoft Windows™ operating system.

It is assumed that the reader has a good working knowledge of the FIB/SEM system; this document is not intended to in any way replace the SmartSEM® operations manual.

While the **Standard** NPVE system is quite powerful, a number of additional capabilities are available in the **Advanced** version of the software which provide specific solutions and enhancements to address a wide range of applications. Throughout this manual, the features that are only available in the **Advanced** version are described where most relevant, and are indicated by the symbol **ADV** so as to indicate to the user that these features are available only in the **Advanced** version. Note that the “Mosaic Tool” capability described in Part III is virtually identical to the “ATLAS” system available for Zeiss Scanning Electron Microscopes, and is only available in the **Advanced** version of the NPVE.

At the instrument level, during any beam control procedure involving imaging or patterning, with or without gases, the microscope hardware deflects the beam in a predetermined pattern across the surface of a sample, in a process variously referred to as milling, rastering, scanning or imaging. Throughout this manual, we will

refer to the processes of Imaging and NanoPatterning. Imaging is the action of guiding the beam, typically but not necessarily in a contiguous rectangular fashion, with the goal of obtaining an image by correlating the resultant signals emitted by the interaction of the beam and the sample at the known location of the beam for each dwell point. NanoPatterning will refer to guiding the beam in a potentially much more complex and defined manner to create a desired pattern or patterns on the sample. Note that the NPVE is also able to display images of the signal resulting from these more complex patterns as they proceed.

The NPVE system contains a powerful and versatile NanoPatterning engine suitable for both FIB and SEM patterning operations. Often during SEM patterning, and without the use of gases, a typical objective is to pattern and transform a blanket material such as photoresist by selected exposure to the electron beam to enable further processing (typically etching). The NPVE makes experimentation of this sort straightforward and its simple yet powerful shape editing capability makes rapid prototyping of a range of patterns easy to undertake.

However, for applications where the sample is more substantially modified, particularly over a period of time, as is typically the case for all FIB operations and for SEM operations using gases, the combined NanoPatterning and Visualization capabilities of the NPVE make it a truly powerful system. This is both for patterning and for gauging the correct endpoint conditions in order to precisely stop or modify patterning based on visual and graphical feedback in real time.

Note:

For information on installing and configuring the NPVE hardware and software system, please refer to the NPVE Installation Manual.

Part I - Image Acquisition and NanoPatterning Control

Part I describes the control interface for acquiring images and defining and executing NanoPatterns. These functions are located in the Imaging Control Panel, the NanoPatterning Parameters Panel and the Visualization Panel, as highlighted in Figure 1.

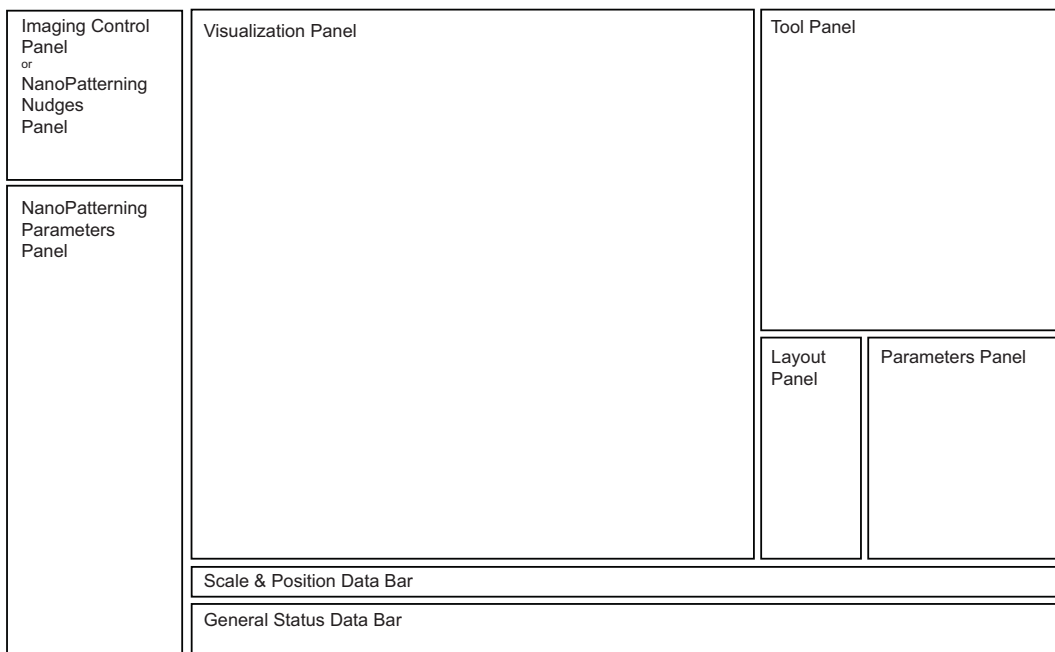
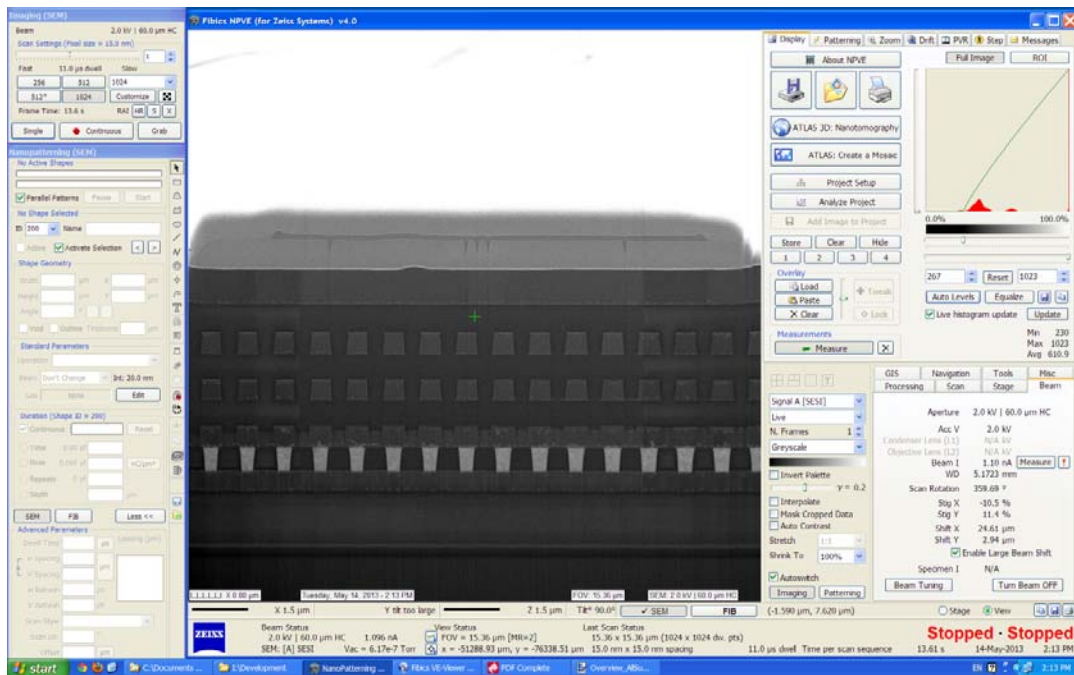


Figure 1: Overall layout of the NPVE interface, with schematic representation of key regions related to general Image Acquisition and Visualization.

1. Imaging Control Panel

The parameters for acquiring images with the NPVE are controlled from the Imaging Control Panel located in the top left corner at software start up, as shown in Figure 1.

From this panel, the following can be adjusted:

- Beam Conditions (effecting the beam spot size and in certain cases the aperture)
- Pixel Dwell Time
- Image Dimensions (pixels)
- Line Averaging
- Pixel Size
- Reduced Area Imaging Control
- Imaging Status (Continuous, Freeze)

Note:

The beam list is different whether the NPVE controls the ion beam (FIB) or the electron beam (SEM). In the case of the FIB, the list is populated using the apertures defined in the FIB Probe Table in SmartSEM®. When the NPVE controls the SEM, the beam list is not selectable and only reads the current state of the SEM aperture (Physical aperture and EHT).

1.1. Imaging with the NPVE

In the NPVE, images are displayed in the Visualization Panel, which is 1024 x 1024 pixels in size. Images that are acquired at lower or higher resolution are still scaled to fit the Visualization Panel with or without interpolation. When an image with a dimension larger than 2048 pixels is acquired, the user is prompted to select a filename where the full high resolution image will be saved. The resultant image will be displayed in the Visualization Panel, appropriately binned and/or scaled to fit.

1.2. Imaging Control Panel

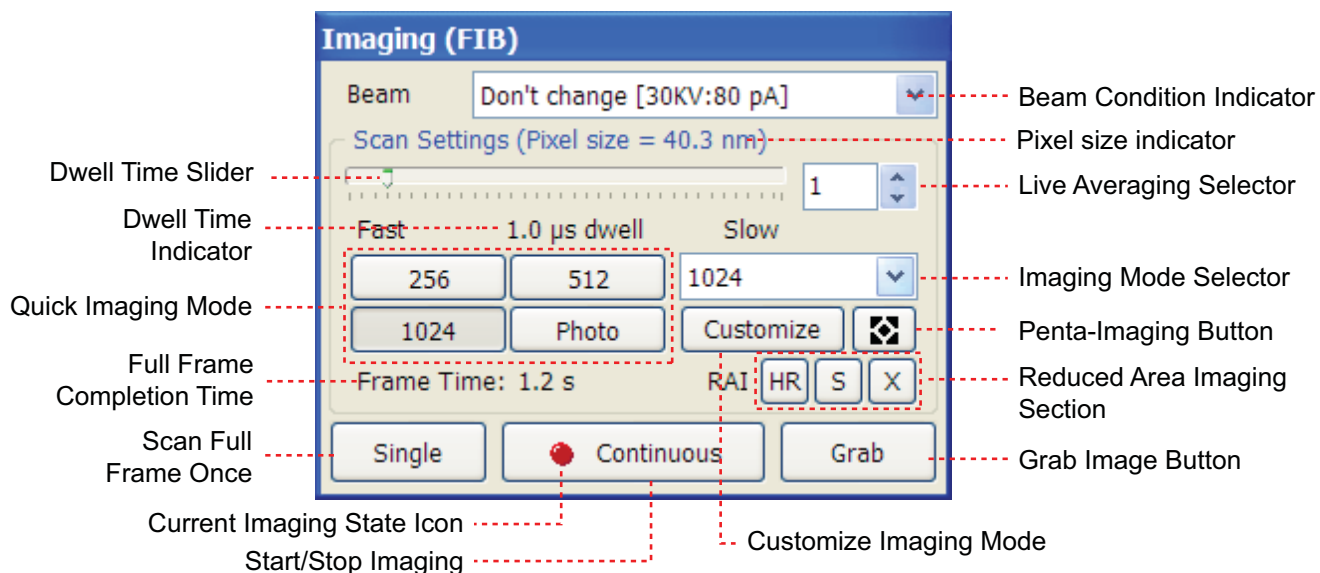


Figure 2: The Imaging Control Panel interface.

Beam Condition Selector

The probe used to image can be changed in the NPVE. When imaging with the FIB, the user chooses from a list of discrete probes. For SEM imaging, the EHT, the aperture size and the condenser setting (high or normal) can be set by the user.

Relevant changes made under the FIB Control Panel or the SEM Control Panel in SmartSEM® will be reflected in the Beam Condition Selector of the NPVE.

Changing the FIB Beam

Use the NPVE Beam Condition Indicator to select the beam conditions to acquire the image. When the beam conditions are changing, a window will appear informing the user that an aperture change is in progress.

Note:

When changing beam conditions a physical change of aperture may not always occur. Conditions pertaining to the beam such as lens settings and acceleration voltage may be changed without necessarily changing the physical aperture position.

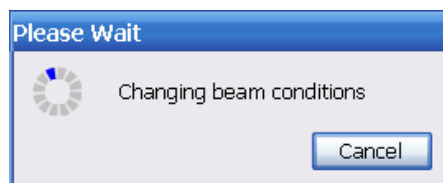


Figure 3: The Changing Beam Conditions notification window.

Once the beam conditions have changed, the NPVE will continue with the action that was underway before the change occurred. For example, if the NPVE was performing imaging, it will continue imaging. When the Beam Conditions selector is set to “Leave Unchanged”, the imaging will be performed using the currently selected beam conditions as defined in either the SmartSEM® software or in the NPVE interface. It is always possible to check what the desired beam conditions are by looking in the General Status Data Bar of the NPVE Main Panel.

Changing the SEM Beam Conditions

When using the NPVE to image with the SEM, the SEM Beam Conditions can be modified. These conditions consist of three parameters:

- The accelerating voltage (EHT)
- The aperture size
- The condenser setting

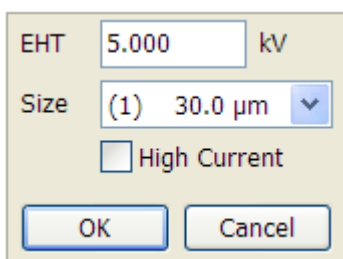
A dialog box titled "SEM Beam Conditions Form" with a light beige background. It contains three input fields: "EHT" with a text box containing "5.000" and a unit label "kV"; "Size" with a dropdown menu showing "(1) 30.0 µm"; and a checkbox labeled "High Current" which is currently unchecked. At the bottom are two buttons: "OK" and "Cancel".

Figure 4: SEM Beam Conditions Form.

Hint:

The SEM Beam Conditions can also be changed by clicking on the SEM Beam Conditions Form displayed in the General Status Data Bar.

Relevant changes made under the SEM Control Panel in SmartSEM® will be reflected in the Beam Condition Selector of the NPVE Imaging Control Panel.

Dwell Time Slider and Dwell Time Indicator

In order to improve the signal to noise ratio of the image, it is possible (and often advantageous) to vary the dwell time. The dwell time is defined as the time the beam stays on a dwell point before moving to the next dwell point in the image. Under the simplest conditions, each dwell point corresponds directly to a pixel in the image. The signal from the selected detector is supersampled and processed throughout the dwell time. In general, longer dwell times will improve the signal to noise ratio and consequently the image quality, although the properties of a particular specimen may be better suited to other approaches, such as combining shorter dwell times with frame or line averaging.

Using the **Dwell Time Slider**, a dwell time between 0.1 µs (100 ns) and 500 µs can be selected. The increment varies as a function of the cursor, as shown in Figure 5.

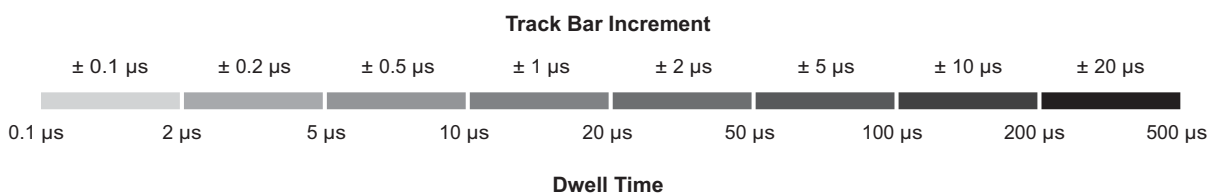


Figure 5: Dwell Time Slider increments as a function of the dwell time.

The dwell time can be dynamically changed during the acquisition of any frame, which will “speed up” or “slow down” the image scan within the frame itself.

Hint:

The scan speed can be changed dynamically while imaging by positioning the mouse anywhere within the Imaging Control Panel and using the mouse wheel to change the scan speed. This, for example, allows a portion of the image to be scanned very slowly, while the rest is scanned quickly.

The dwell time can also be input directly by clicking on the dwell time indicator of the Imaging Panel and entering the desired value.

Line Averaging

Line Averaging is the number of times the scan of a line will be repeated and the resultant signal averaged, before moving to the next line. The Line Averaging property can be changed using the line averaging selector.

Full Frame Completion Time

The full frame completion time indicator displays the time required to complete a whole frame under the conditions selected. This gives insight to the user to determine the impact of different parameters on the acquisition time.

Imaging Mode Selector

Imaging Modes define the width, height, default dwell time and line averaging conditions used for imaging. All of the Imaging Modes are accessible through the **Imaging Mode Selector**. Once a given mode is selected, only the dwell time can be modified. A new mode should be created to define different imaging conditions. Switching between modes while imaging will cause the acquisition to restart the current frame.

Quick Imaging Mode

The **Quick Imaging Mode** buttons are programmable and any predefined Imaging Mode can be stored in them for quick access. Advanced users may wish to modify the standard settings by clicking the **Customize** button. As described below, the first four Imaging Modes in the Imaging Mode List (after the standard [Grab] mode) become the modes which are displayed in the **Quick Imaging Mode** buttons.

The Grab Image Button

Unlike the **Single** and **Continuous Imaging** buttons, the **Grab Image** button always has the same user selectable image parameters assigned to it. The imaging parameters of the **Grab Image** button are defined in the Imaging Mode Form. Clicking the **Grab Image** button will ignore the current imaging conditions and will acquire a single image with the imaging parameters assigned to the Grab mode in the Imaging Mode list. This button can be useful to grab an overview image before acquiring a high-resolution image.

Penta-Imaging Mode

The Penta-Imaging Mode allows the user to simultaneously view all four corners and the center of the Field of View (FOV) as shown in Figure 6. Penta-Imaging Mode is useful when optimizing beam focus and stigmation across the entire image, especially for high resolution large area images. During Penta-Imaging, when the selected Imaging Mode is more than 1024 pixels in width, each pixel on the screen corresponds to one imaging dwell point on the sample, allowing the user to see a one-to-one correlation between the pixels displayed in Penta-Imaging Mode and the pixels as they will be acquired in the as-acquired high resolution image.

When the selected Imaging Mode has 1024 pixels in width or less, the Pixel Size Divider will be applied. For example, if the selected imaging mode has 1024 pixels across and a pixel size of 16 nm, having a Pixel Size Divider of 8, (FOV of 16.384 μm), the Penta-Imaging mode will set the imaging pixel size to 2 nm, effectively imaging at eight (8) times the current magnification.

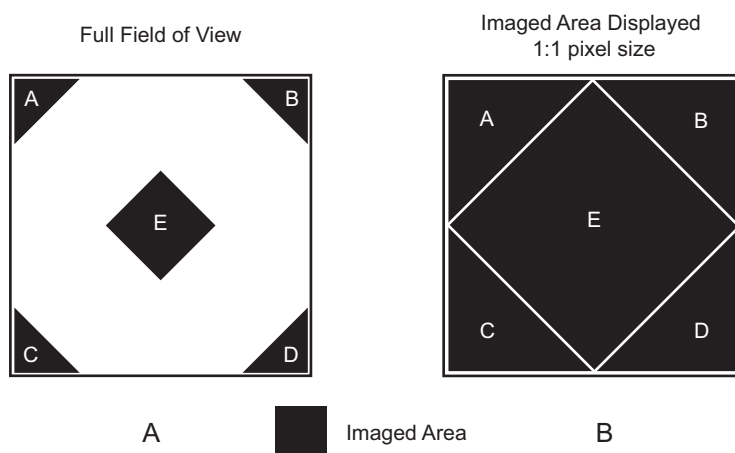


Figure 6: Penta-Imaging. A) The imaged areas account for a subset of the full field of view. B) When Penta-Imaging is initiated, the imaged areas are displayed so that they fill the Visualisation Panel.

Reduced Area Imaging (RAI)

It is often advantageous to image a smaller portion of the full image display such as when adjusting beam focus and stigmation. NPVE allows one to image a sub-region of the Field of View using the Reduced Area Imaging (RAI) function.

The RAI is only available for imaging mode with a width of 1024 pixels or less. For imaging modes with a greater number of pixels, the High Resolution Reduced Area Imaging is automatically selected.

High Resolution Reduced Area Imaging (HR RAI)

HR RAI allows the user to acquire a reduced area image that will be displayed at a 1024 x 1024 pixel resolution. Unlike the normal RAI, when HR RAI is selected the cursor changes to a predefined green box which outlines the area to be imaged. Pixel spacing of this reduced area is a fraction of the pixel spacing of the original image and can be modified using the Pixel Size Divider found in the Imaging Modes Form by pressing the **Customize** button. If the mouse wheel is scrolled up, the number of pixels used for imaging is reduced, effectively allowing for a smaller area of the image to be scanned at the same reduced pixel spacing.

Reduced Area Imaging Setup

Reduced Area Imaging functionality is dependent on the desired image resolution. RAI setup instructions for each mode are as follows:

Normal Resolution Imaging:

1. Press the button marked with an **S** (**S**etup Reduced Area Imaging) in the Imaging Panel.
2. Left click and drag the mouse cursor within the Visualization Panel to select the reduced area region of interest of the image. When the mouse button is released, the RAI should start automatically in Continuous imaging mode and the button will display "Freeze".
3. To stop the RAI, press the **Freeze Imaging** button.

High Resolution Imaging:

1. Press the button marked with an **HR** in the Imaging Panel.
2. Move the mouse over the desired imaging area. A green square representing by default a 1024 x 1024 pixel image will outline the location and size of the RAI. When the left mouse button is clicked, the RAI should start automatically in Continuous imaging mode and fill the Visualization Panel. The RAI size can be altered prior to imaging by scrolling up and down with the mouse wheel.
3. To stop the RAI, press the **Freeze Imaging** button.

Once the RAI has been set up, it can be imaged again by pressing the button marked with an **X** (**X**ecute Reduced Area Imaging).

Single Scan and Continuous Scan

Use the **Single Scan** button to acquire a single complete frame. The beam is automatically blanked and the image is frozen at the end of the frame. The **Continuous Scan** button will acquire multiple images (Continuous imaging) until stopped by the user (see Start/Stop Imaging, below). If the **Single Scan** button is pressed during Continuous imaging, the current imaging frame will be completed and then the beam will be blanked.

Note:

All of the images acquired using the **Single Scan** button are stored in the Project directory automatically when acquisition has been completed, if **Project Recording** is enabled.

Continuous Imaging

Starting and stopping image acquisition is performed using a single button. Once the **Continuous** button is clicked, the image acquisition begins and the **Continuous** button toggles to read “Freeze”. Image acquisition will immediately stop when the user presses the **Freeze** button regardless of where in the field of view the beam is scanning at that instant. If the user wishes to stop when a complete image has been acquired, pressing the **Single Scan** button will cause NPVE to finish the current frame and then stop.

The **Continuous** button has a coloured bullet associated with it. It is called the Current Imaging State icon. The colour of that icon informs the user of the present state of the scan:

	Red	The scan is stopped
	Green	The scan is continuous
	Yellow	The scan will stop at the end of the current image frame

It is important to keep in mind that scanning with the FIB beam induces changes in the sample (charging, contamination, the ballistic nature of the ion beam, etc.). It is often useful to minimize the dose delivered to the sample to prevent or reduce gallium implantation or topographic changes caused by ion sputtering. The user should determine whether to perform Single or Continuous scanning on a sample by sample basis.

1.3. Imaging Modes Form

The Imaging Modes Form is used to manage the different Imaging Modes. It is accessible by clicking the **Customize** button in the Imaging Panel.

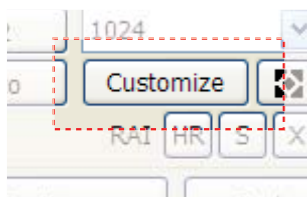


Figure 7: The Customize button provides access to the Imaging Modes Panel.

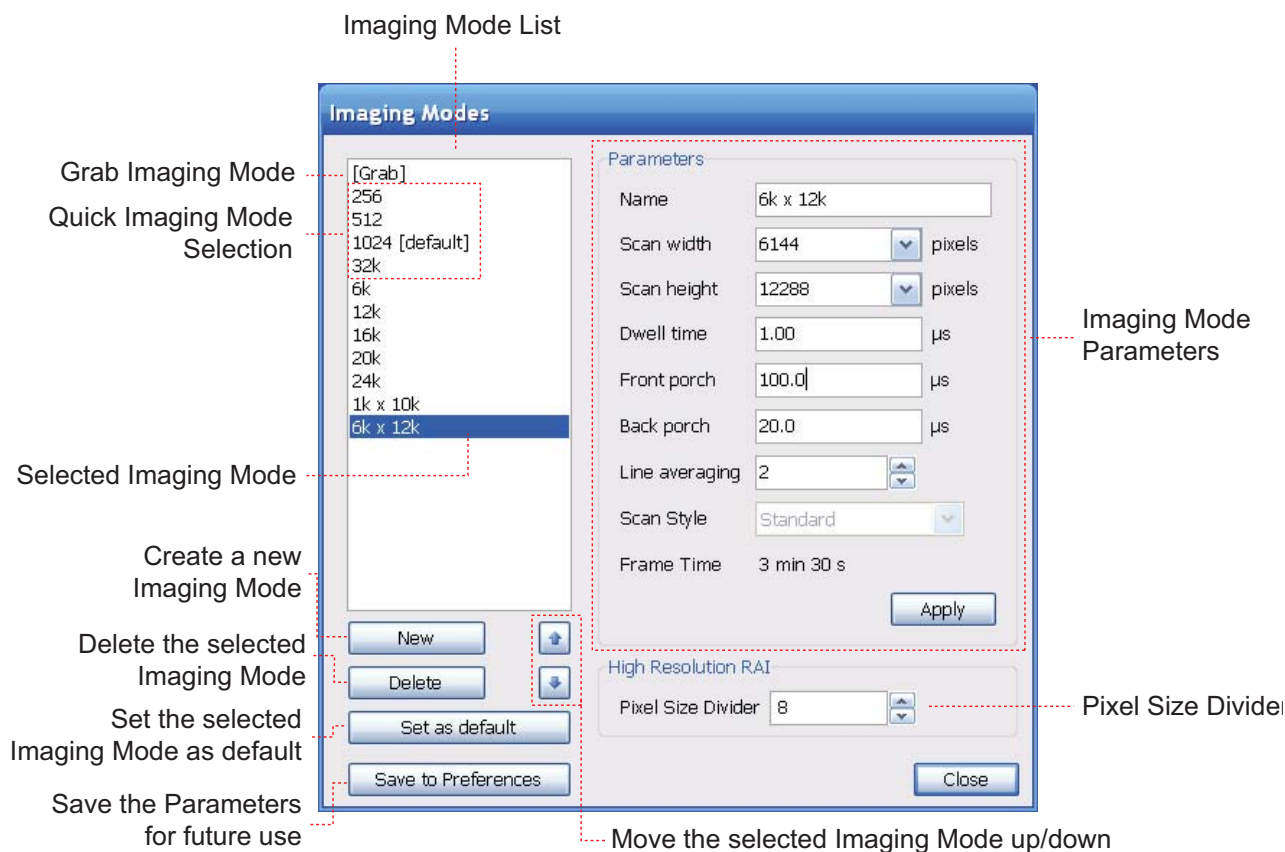


Figure 8: Imaging Modes Panel.

Imaging Mode Parameters

Name

A customizable name describing the imaging mode.

Scan Width and Scan Height

The Scan Width and Scan Height represent the dimensions of the image in pixels. The width and height dimensions of the image may be varied up to 32,768 pixels in size.

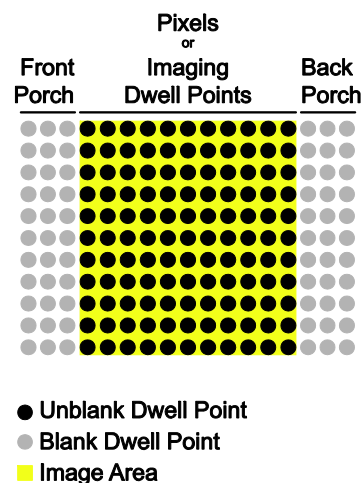
Note that images that are acquired at resolutions lower or higher than 1024 x 1024 pixels are scaled to fit the Visualization Panel, with or without interpolation, as the user prefers. Images with a width greater than 1024 pixels are saved at full resolution on the hard drive of the PC and will be downsampled with appropriate data binning and scaling to 1024 pixels for display purposes.

Dwell Time

A dwell point is defined as a point in space (X, Y) where the beam is kept still for a fixed amount of time. This length of time is called the Dwell Time. Under the simplest conditions, each dwell point corresponds directly to a pixel in the image.

Front Porch and Back Porch

The Front Porch and Back Porch scan regions are allocated for beam stabilization. On systems equipped with fast beam blanking electronics, they correspond to dwell points where the beam has not yet been unblanked. The Front Porch is located at the beginning of the scan lines (assuming “standard” left to right, top to bottom, scanning) and the Back Porch is at the end of the lines, as shown in the schematic to the right. The Front Porch and Back Porch are not part of the resulting image.



Line Averaging

In general, increasing the Dwell Time will increase the signal to noise ratio of an image. There are, however, cases where increasing the Dwell Time is not desirable, such as with charging samples. Line Averaging offers the possibility of an improved signal to noise ratio without changing the Dwell Time by repeatedly scanning a line and accumulating the signal from multiple scans. The Line Averaging value corresponds to the number of successive times a line will be repeated before moving to the next line. The accumulated signal from multiple line scans is then averaged for display purposes.

Pixel Size Divider

The Pixel Size divider determines the fraction of the image pixel size that is used during High Resolution Reduced Area Imaging. For example, if the divider is set to 16 for an image with 160 nm pixel spacing, the HR RAI image would be acquired with a pixel spacing of 10nm. The Pixel Size Divider is also used in the Penta-Imaging Mode.

2. NanoPatterning Tweaks Panel

When a pattern is started, the Imaging Panel is replaced by the NanoPatterning Tweaks Panel shown in Figure 9.

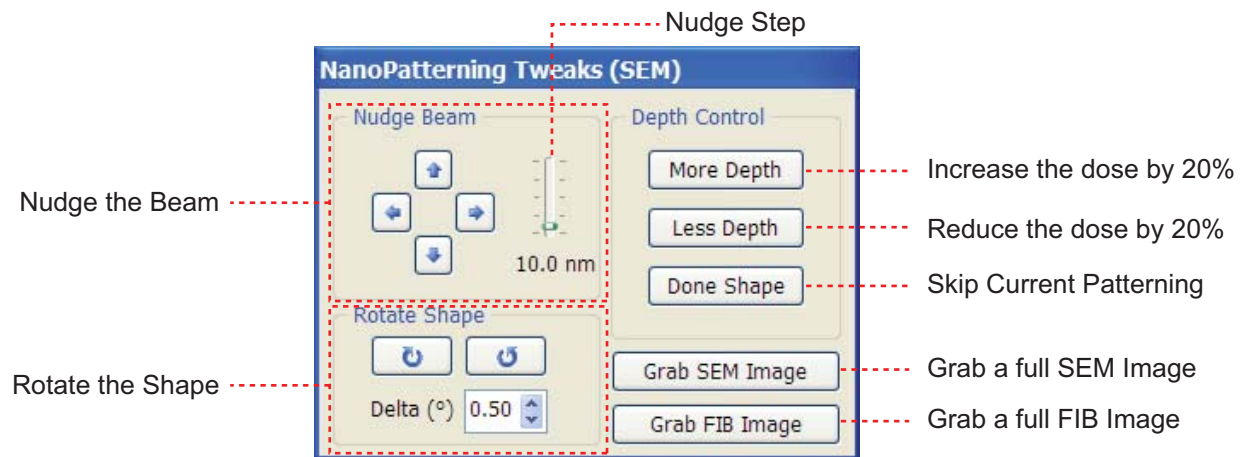


Figure 9: The NanoPatterning Tweaks Panel. This panel is used to control a few aspects of the patterning while it is taking place.

The NanoPatterning Tweaks Panel allows the user adjust the patterned shapes as patterning takes place. A summary of its functionality follows:

- Increase the target dose by 20%
- Decrease the target dose by 20%
- Skip the current pattern (stop the current patterning and move to the next shape)
- Adjust the patterning position
- Rotate the shape
- Grab a full SEM/FIB image (this will temporarily pause the patterning)

The behaviour of those functionalities varies depending on whether the NPVE is in Parallel Patterning mode or Serial Patterning mode.

Parallel Patterning

When the NPVE is in Parallel Patterning mode, any action is applied to the whole set of shapes being patterned.

Serial Patterning

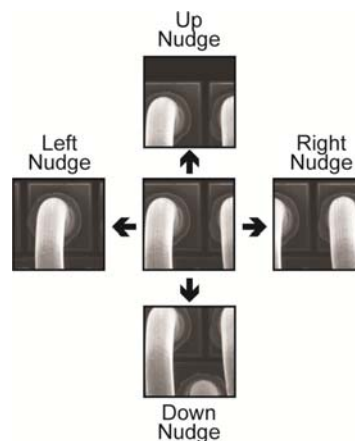
In the case of Serial Patterning mode, the dose variation is strictly applied to the specific shape being patterned by the NPVE at the moment the button is pressed. The rotation adjustments also only occur on the current shape being patterned.

The Beam Tweaks, however, acts as a global beam shift: It is applied to all the shapes yet to be patterned during the patterning session.

2.1. Beam Nudge

The position of the pattern relative to the sample can be modified as the patterning takes place by shifting the beam.

The effect of the nudge directions are shown in the figure on the left. When nudging the beam downwards, for example, the content displayed in the patterned box is actually shifted upwards. Similarly, nudging the beam to the left will cause the content of the box to appear to be shifted to the right. This can be visualized if the user sees the beam nudge from the perspective of the shape as opposed to from the perspective of the visualization of the features within the shape. Please keep in mind it is a beam nudge and not a sample nudge.



Note:

The Beam Nudge will affect all of the shapes in the queue list and not only the shape being patterned when the **Nudge** buttons are pressed.

2.2. Shape Rotation

The Shape Rotation acts directly on the angle of the shapes being patterned, as opposed to the Beam Nudge where the geometric properties of the shape are not modified. When a rotation adjustment is made, the NPVE will pause the pattern, apply the change in rotation and then resume the patterning. See section on Patterning for more details about Pause and Resume of patterning.

The rotation will only affect the shapes being patterned at the time the rotation occurs. The remaining shapes in the patterning queue will not be rotated.

2.3. Skip Current Patterning

Only available under serial patterning, the **Skip Current Patterning** button allows the user to stop the current pattern and move to the next shape in the queue. Skipping the current pattern will assign the delivered dose to the target dose of the shape.

3. Visualization Panel

The Visualization Panel is located in the middle of the application. It primarily serves to display acquired images, draw and display shapes, and display live patterning.

3.1. Frozen Image Dynamic Updates

Once an image has been acquired with the NPVE, it remains in the Visualization Panel as long as a new image is not acquired. Even though this image is static in time, the Visualization Panel will update its position, angle and size to reflect any changes that occur in the microscope after the image has been acquired – a process referred to as Frozen Image Dynamic Updating. Using dynamic updating, it is possible to set up the microscope without requiring multiple image scans, thereby reducing charging or other beam induced artefacts.

The Visualization Panel keeps track of several parameters that trigger dynamic updates to the frozen image:

- Magnification / Field of View (FOV) Changes
- Beam Shifts
- Scan Rotations
- Stage Movements in X and Y

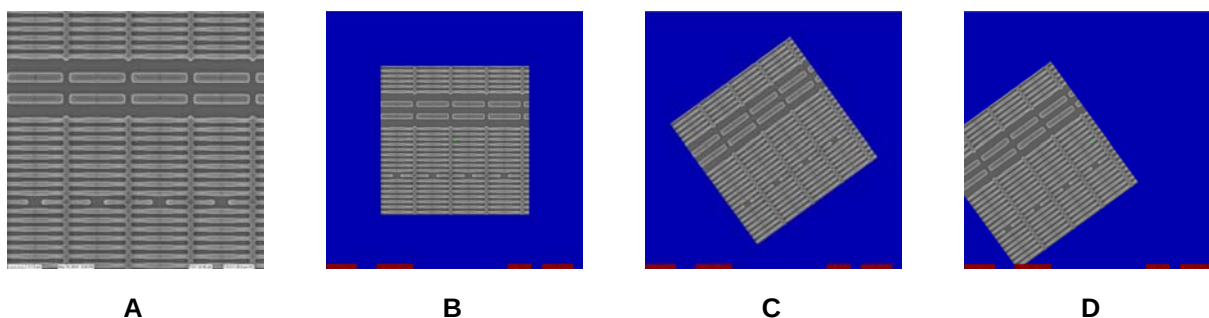


Figure 10: The impact of Frozen Image Dynamic Updating on the Visualization Panel under different conditions.

(A) The original image. (B) The image after the Magnification / FOV has changed. (C) FOV changed and Scan Rotation of roughly 40°. (D) FOV changed, Scan Rotation and a Beam Shift were applied after the Image was acquired.

Note that the image scale and information markers along the bottom of the image turn red when the image displayed is not at the same scale or position as when it was acquired.

The information at the bottom of the Visualization Panel will be displayed on a red background if the current state of the Microscope is different than the state in which it was when the image was originally acquired. A red indicator will appear in the top left hand corner of the Visualization Panel if the image parameters have changed since the last image was acquired.

Frozen Image Dynamic Updating based on FOV and Stage Tracking can be disabled by clicking the appropriate checkbox in the MISC Tab of the Parameters Panel, as described in a later section. Note that the absolute accuracy of the position of the image displayed during Frozen Image Dynamic Updating is influenced by the stage accuracy of your particular microscope.

4. Scale and Position Data Bar

The Scale and Position Data Bar is located below the Visualization Panel.

Scale bars

The Scale and Position data bar displays X, Y and Z scale bars relating to the image displayed in the Visualization Panel as well as the stage tilt relative to horizontal. On the right hand side of the bar, mouse coordinates are displayed relative to either the stage position at a given point within the image (**Stage** radio button selected) or relative to the centre of the image (**View** radio button selected).

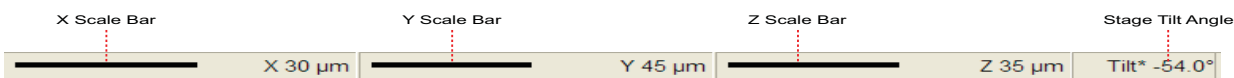


Figure 11: The three different scale bars and the stage tilt angle of the Scale and Position Data bar.

4.1. Changing NPVE Active Beam Control

The Scale and Position Data bar contains two buttons that allow the user to select which beam the NPVE controls.



Figure 12: The SEM and FIB Control Buttons.

In order for the NPVE to control the SEM or the FIB, SmartSEM must set the appropriate beam into external control mode. For the SEM beam, SmartSEM must be in “FIB mode SEM” and the Ext Scan Control for the SEM beam must be turned on. For the FIB beam, SmartSEM must be in “FIB mode EXT” for the NPVE to take control of the FIB beam.

Beam To Be Controlled by the NPVE	SmartSEM State
SEM	FIB mode SEM + SEM Ext Scan Control On
FIB	FIB mode EXT

Table 1 The SmartSEM state for the NPVE to control the SEM or the FIB beam.

The state and the caption colour of the **SEM** and **FIB** control buttons indicates 2 things:

- The active beam (SEM or FIB) for imaging and patterning in the NPVE
- Whether SmartSEM or the NPVE has control of the beam

A down button (with a check mark) indicates the beam that is active in the NPVE. You can change the active beam by clicking on the appropriate beam control button. If you want to scan or pattern with the FIB beam, the **FIB** button must be down and its caption must be black.

When the caption is red, it indicates that SmartSEM has internal control of the beam. In this case, the NPVE does not control the beam and therefore cannot image or pattern.

NPVE can release control of a beam to SmartSEM by clicking the active (black) beam control button.

Hint:

When you start imaging, the NPVE will push the proper state to SmartSEM. If the image looks out of synch, verify that the microscope is in the correct state by ensuring that the appropriate beam control button is down (with a check mark) and its caption is black.

5. General Status Data Bar

The General Status Data Bar is located below the Scale and Position Data Bar and displays information pertaining to the Beam Status, View Status, Last Scan Status, the current imaging status as well as current date and time.

The Beam Status section displays the accelerating voltage of the beam as well as the beam aperture currently being used. It also shows which detector is being used to capture the current frame. These settings can be changed by left clicking the value displayed in the NPVE software and selecting the desired values from a pop-up list.

The View Status section displays the current field of view, the mag range the microscope is in, as well as the stage coordinates at the centre of the image displayed in the Visualization Panel.

Two additional options are available: **Scan Rotate** () , Compucentric Stage Rotate() and **Stage Centre** (, ).

Adjust Image Horizontal

By right-clicking the **Adjust Image Horizontal** button, the user can select between beam scan rotation, compucentric stage rotation, auto-aligning the image, and zeroing the scan rotation. After toggling the **Adjust Image Horizontal** button, the user can click and hold the left mouse button while hovering the mouse cursor over a feature displayed in the Visualization Panel. By dragging the cursor along an image feature and releasing the left mouse button, the rotation will be changed to display the specified line horizontally across the screen.

Centre View

The **Centre View** button, when toggled, allows the user to select a location on the image which will be brought to the centre of the FOV.

The **Centre View** functionality can be accessed by left-clicking the **Centre View** button, and selecting one of three options:

- Beam Shift only
- Stage move only
- Beam Shift and Stage Move in combination

An option to zero the beam shift is also available.

Note:

It is up to the user to ensure that physical movement of the stage does not potentially damage the sample or the microscope.

The Last Scan Status displays the image dimensions in micrometers, the dwell point spacing, the dwell time and the time per frame. The system updates these parameters after a full frame has been scanned.

SEM and FIB Beam State

The current scan status is displayed in the bottom left hand corner alongside the current date and time. During imaging, the word “Scanning” will be displayed in large blue text. If the beam is blanked and no imaging is underway, the word “Stopped” will be displayed in large red text. The date and time cannot be changed in the NPVE software, but will reflect the current date and time set on the NPVE PC.

6. NanoPatterning Parameters Panel

6.1. Overview

Everything related to the execution of a NanoPatterning operation is contained in the NanoPatterning Parameters Panel (Figure 13). This panel allows the user to perform multiple operations such as creating shapes, aligning shapes, changing NanoPatterning conditions, starting/stopping an operation, saving shapes on disk, and much more.

The NanoPatterning Parameters Panel is divided into multiple subpanels each of which is dedicated to a particular function, as well as a toolbar section along the right hand side of the panel which contains groups of tool buttons.

NanoPatterning Subpanels

Progress Display and Control (A)

The Progress Display and Control Panel gives information about the mill or mills in progress.

Shape Identifier / Browser (B)

This subpanel is used to numerically select a specific shape, browse through all the shapes, activate or deactivate shapes and optionally assign names to shapes.

Shape Geometry and Style (C)

Details related to the geometric properties of the selected shapes are displayed in this subpanel. The width, height, angle, position and style of the shape or shapes can all be modified.

Standard Parameters (D)

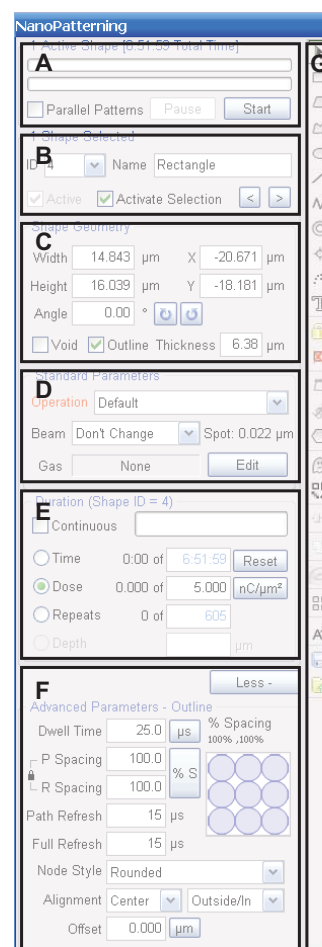
The standard parameters a user needs to accomplish a nanopatterning action are displayed in this subpanel, including any pre-defined Operation parameters (including gas chemistries) which have been applied to the shape as well as the beam conditions and resulting spot size to be used while patterning.

Duration Controls (E)

Adjacent to the standard parameters are the Duration controls which determine whether a shape or selection of shapes is to be patterned continuously until user intervention, or patterned to a specified Dose, Time, Number of Scans or Depth/Height.

Shape List Beam Selection (F)

Independent shape lists exist for both the FIB and the SEM. The **FIB** and **SEM** buttons can be used to change which list (FIB or SEM) the Nanopatterning form edits. Shapes created for the SEM will have the prefix “S” added to the shape ID, while shapes created for the FIB will have the



prefix “F”. When changing the active beam with the **Active Beam Control** buttons, the corresponding shape list will be assigned to the Nanopatterning Form.

Advanced Parameters (G)

The Advanced Parameters subpanel allows the experienced user to control more complex parameters regarding the NanoPatterning of the shape. Parameters such as dwell time, dwell spacing and patterning style, which are automatically applied when a particular Operation is selected, can be modified to allow more precise operations or “on-the-fly” optimization. As the Advanced Parameters subpanel may be unnecessary for the average user, this subpanel can be hidden (or displayed) using the **More/Less** Advanced Parameters button.

Toolbar Buttons (G)

Shape Creation / Selection



A total of ten basic shape types can be created with the NPVE: Rectangles, Trapezoids, Polygons, Ellipses, Lines, Polylines, Spots, Rings, FIBTexts, and Deflection Lists. This section contains all the tools needed to select and create any of the available shape types.

Patterning Direction Buttons



Define the patterning direction either at an arbitrary angle or in the direction of the selected edge of a given shape.

Convert to Polygon



This icon can be used to convert any shape type into a Polygon to increase the capability of editing that shape.

Shape Display Options



Depending on the situation, certain information on the shape should be either displayed or hidden. Six options are available in this menu:

- Show/Hide All
- Toggle Ghost Mode
- Show/Hide Shape Fill
- Show/Hide Patterning Direction
- Show/Hide Shape Controls
- Show/Hide Shape ID

Shape Deletion, Lock / Unlock, Selective Editing



This section can be used to lock or unlock selected shapes, to delete shapes or toggle to a different mode of selective editing.

Save and Load Shapes / Array of Shapes



Shapes can be saved and loaded. The load functionality can also be used to load an array of shapes.

Advanced Options ☒ADV



Advanced Options include the Array and Annular Builders, Set Operations, Bitmap NanoPatterning and the 3D Profiler.

The use of the subpanels and buttons listed above are described in detail in the following sections.

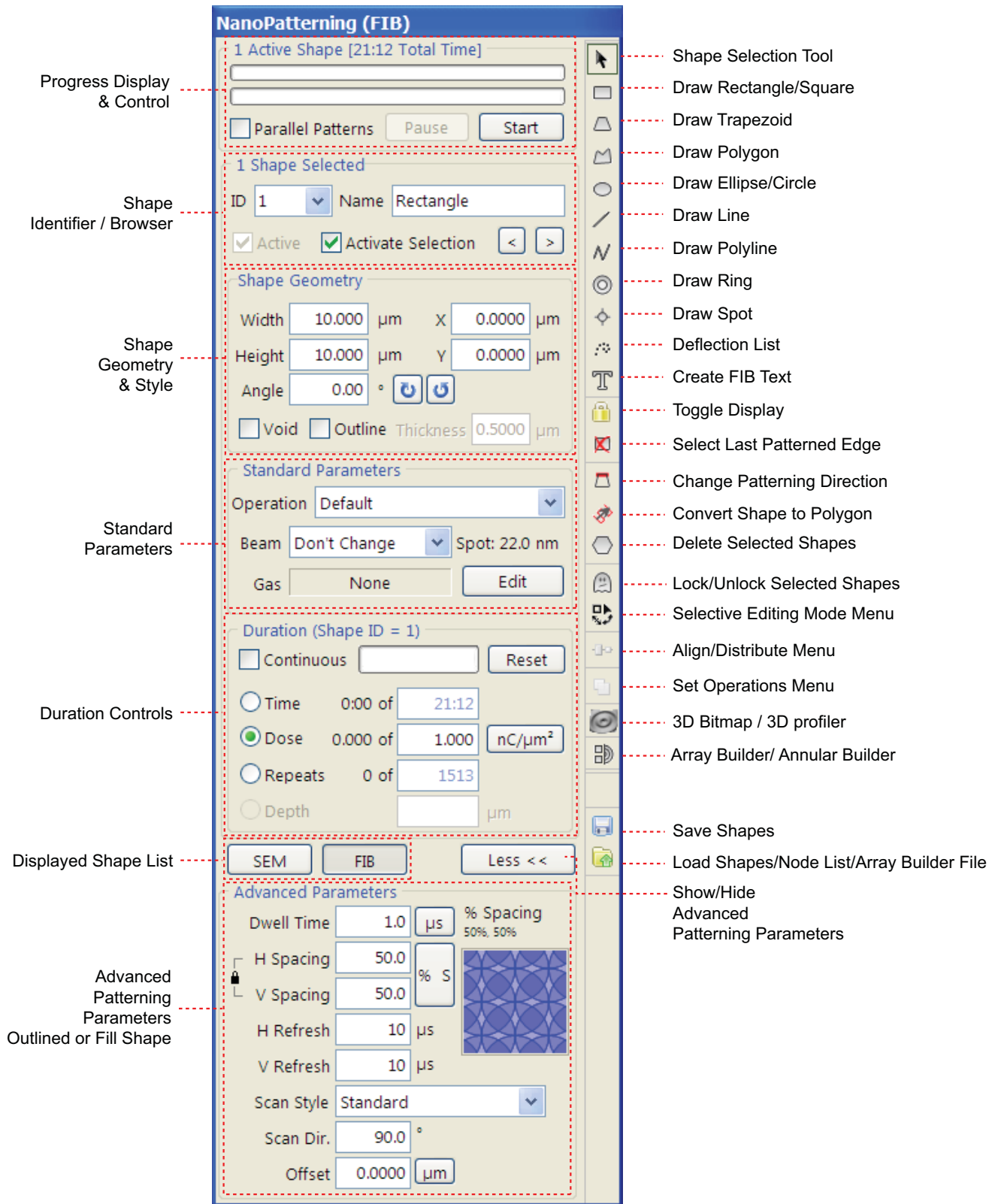


Figure 13: NanoPatterning Parameters Panel.

6.2. Shape Selection and Creation

Before NanoPatterning can take place, one or more shapes must be created using the NPVE graphical user interface.

Shape Creation

The NPVE interface allows the creation of multiple different types of shapes. All of the shape types are created using the shape creation toolbox combined with mouse interaction in the Visualization Panel.

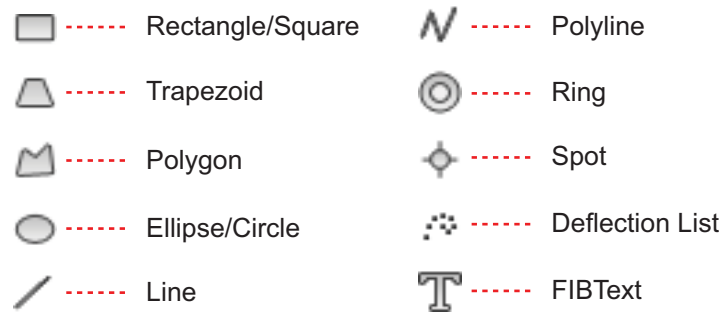


Figure 14: Shape Creation Toolbox.

The user has a total of ten (10) different shape types to choose from:

- Rectangle (Square)
- Trapezoid
- Polygon
- Ellipse (Circle)
- Line
- Polyline
- Ring
- Spot
- Deflection List
- FIBText

Note:

In order to be able to create new shapes, you must be in Patterning Display mode. Refer to Figure 126 for information regarding how to switch modes.

To create a shape, the user has first to click on the shape type of their choice in the toolbox and then click anywhere in the Visualization Panel in order to start the shape creation. Once shape creation has begun, the behaviour of the user interface depends on the type of shape chosen. It is possible to abort the creation of shapes at any time by pressing the **ESC** key.

Note:

To create a shape by importing the coordinates of its nodes, refer to Section 8.5, “Save / Load Shapes”.

The aspect ratio of certain shapes can be changed or constrained at creation time. For example, an Ellipse can be turned into a Circle by pressing and holding the **ALT** key. Shapes can also be created about their centre by holding down the **CTRL** key. These particular behaviours will be explained in more detail below in the sections pertaining to the creation of each type of shape.

Once the drawing process is complete, the NPVE will automatically go back into Shape Selection Mode. While in this mode, a dropdown list of additional shape properties, such as shape geometry, node modification, and transform commands, can be accessed by right-clicking on the shape.

Most shape types have two creation modes: clicking mode and dragging mode (not available for Polygon and Polyline).

Dragging Mode

To create a shape using the Dragging Mode, select the desired shape type then **CLICK** and **HOLD** the left mouse button in the Visualization Panel. Then, by dragging the mouse, the dimensions of the newly created shape will change.

Clicking Mode

In Clicking Mode, click and release the left mouse button on a given point in the Visualization Panel. By moving the mouse the dimensions of the shape will change. To end shape creation, click again in the Visualization Panel. These behaviours will be discussed in more detail below.

Drawing a Rectangle/Square

Use the Rectangle shape creation tool () to create a Rectangle.

1. Click on the **Rectangle** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel to position one corner of the Rectangle.
3. Drag the mouse diagonally until the Rectangle is the desired size.

To create a Square, press and hold the **ALT** key down while dragging diagonally.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard. The Rectangle tool supports both Clicking Mode and Dragging Mode.

Drawing a Trapezoid

Use the Trapezoid shape creation tool () to create a Trapezoid.

1. Click on the Trapezoid creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel to position one corner of the smaller edge of the Trapezoid.
3. Drag the mouse diagonally until the Trapezoid is the desired size.
4. Release the mouse button to enter the base width adjustment step.

5. Move the mouse to adjust the base of the Trapezoid.
6. Click the left mouse button to end the Trapezoid creation.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard. Trapezoid Mode shape creation supports Clicking and Dragging creation modes.

Drawing a Polygon

Use the Polygon shape creation tool () to create a Polygon.

1. Click on the **Polygon** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel to insert the first node of the Polygon.
3. Add as many nodes as desired by clicking at additional desired locations in the Visualization Panel.
4. To end the Polygon creation, double-click in the Visualization Panel. Note: Double-clicking will insert a node.

It is possible to create nodes that are locked relative to one another at multiples of 45 degrees by holding the **ALT** key down at the insertion time.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard.

Drawing an Ellipse

Use the Ellipse shape creation tool () to create an Ellipse.

1. Click on the **Ellipse** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel to position one corner of the square rectangle bounding the Ellipse.
3. Drag the mouse diagonally until the Ellipse is the desired size.

To create a Circle, press and hold the **ALT** key down while dragging diagonally.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard. The Ellipse tool supports both Clicking Mode and Dragging Mode.

Drawing a Line

Use the Line shape creation tool () to create a Line.

1. Click on the **Line** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel in order to insert the first node of the Line.
3. Drag the mouse to move the end point of the Line.
4. Release the mouse button when the end point of the Line is at the desired position, which will insert the second node of the Line.

It is possible to create a Line with its angle locked at multiples of 45 degrees by holding the **ALT** key down when dragging the end point of the Line.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard. The Line tool supports both Clicking Mode and Dragging Mode.

Drawing a Polyline

Use the Polyline shape creation tool () to create a Polyline.

1. Click on the **Polyline** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel in order to insert the first node of the Polyline.
3. Add as many nodes as desired by clicking at additional locations in the Visualization Panel.
4. To end Polyline creation, double-click in the Visualization Panel. Note: Double-clicking will insert a node.

It is possible to create nodes that are locked at multiples of 45 degrees relative to each other by holding down the **ALT** key at node insertion time.

At any time, the drawing process can be aborted by pressing the **ESC** key on the keyboard.

Drawing a Ring

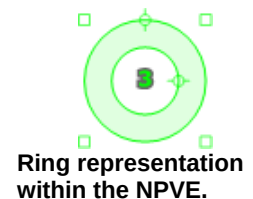
A Ring is a shape which is concentrically patterned. A Ring can only be constrained within a square (as opposed to a rectangle in the case of an Ellipse).

Use the Ring creation tool () to create a Ring.

1. Click on the **Ring** creation tool in the shape toolbox.
2. Click on the desired start point in the Visualization Panel to position one corner of the square bounding the Ring.
3. Drag the mouse diagonally until the Ring is the desired size.

At the time of creation, the thickness of the Ring will be automatically set to 1/8th of the width of the Ring.

The thickness of the Ring can be adjusted after it has been created using the thickness edit box in the Shape Geometry Panel or by using the outline thickness knob directly on the shape using the mouse.



Draw Ring mode shape creation supports clicking and dragging creation mode.

Drawing a Spot

The Spot shape creation tool is used to mill a single point.

Use the Spot creation tool () to create a Spot.

1. Click on the **Spot** creation tool in the shape toolbox
2. Click on the desired location in the Visualization Panel to position the Spot.

It is possible to create several Spots one after the other by holding the **SHIFT** key down on the keyboard. To end the continuous creation mode either double-click at the final position to insert the last Spot or, release the **SHIFT** key, or press the **ESC** key.

Import a Deflection List

The Deflection List creation tool () is used to load an arbitrary list of points to be patterned. Once loaded, the deflection is static which means it cannot be moved, stretched or rotated. The Deflection List can only have the following Patterning Parameters modified:

- Dose/Time/# of Repeats
- Dwell Time
- Full Refresh

Deflection List File Format

The Deflection List file must contain a valid header block and a predefined structure in order for the NPVE to compile it. The file must contain a header which defines the units of the Deflection List (i.e. UNITS=UM or UNITS=%). The NPVE accepts two types of units:

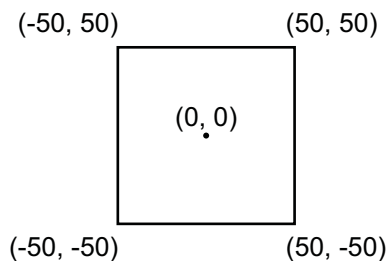
Deflection List Units	Valid Codes
μm	UNITS=UM UNITS=MICRON UNITS=MICRONS
% of the field of view (FOV)	UNITS=% UNITS=FOV% UNITS=FOV

The file header must also contain a tag that tells NPVE where the Deflection List starts (START).

In the case that a percentage of the field of view is used, the values must be between -50 and 50. (0, 0) refers to the centre of the field of view and (50, 50) refers to the top-right corner of the field of view.

The header of a Deflection List file could be as follows:

```
NPVE DEFLECTION LIST
UNITS=FOV
START
```



After the header, the text file must contain a Deflection List composed of at least one XY pair, separated by a comma (.). Each line should be terminated with a Carriage Return / Line Feed combination (CR+LF). At the end of the list of XY pairs, the Deflection List must contain the end tag (END), and once again should be terminated with a Carriage Return / Line Feed combination (CR+LF).

Here is an example of a Deflection List:

```
NPVE DEFLECTION LIST
UNITS=UM
START
-0.3099, 7.5485
-0.3199, 7.5481
-0.3299, 7.5476
-0.3399, 7.5472
-0.3499, 7.5467
-0.3599, 7.5463
-0.3698, 7.5458
-0.3798, 7.5453
-0.3898, 7.5448
-0.3998, 7.5442
-0.4098, 7.5437
END
```

Entering FIBText

FIBText allows the user to enter text to be patterned, which can then be manipulated in the same manner as any other Polygon shape.

Use the FIBText shape creation tool to create text. 

1. Click on the **FIBText** creation tool in the shape toolbox
2. Click on the desired location in the Visualization Panel to position the text.
3. A popup box will appear (Figure 15). Enter text in textbox. Font and Style can be modified and nominal height can be set.
4. Click **OK** to close Text Editor and insert the FIBText shape, or click cancel to abort the operation.

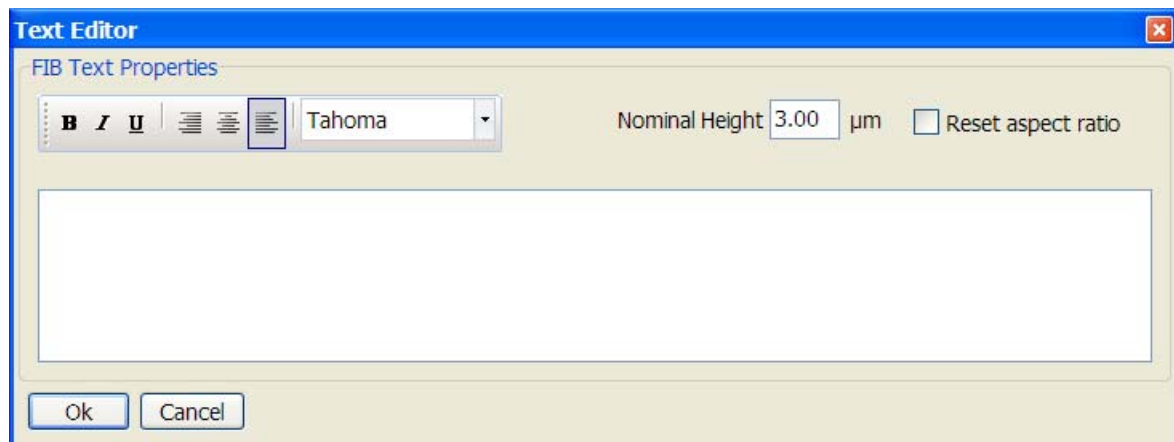


Figure 15 : FIBText Editor.

It is possible to reopen the Text Editor and modify FIBText by double clicking on a previously created FIBText shape or by right-clicking on the FIB Text shape and selecting “Edit Shape”.

Text can be aligned to centre, left, or right using the available text alignment buttons.

During text modification, checking the **Reset Aspect Ratio** checkbox will retain the original aspect ratio of the FIBText from before it was modified. FIBText will also retain its aspect ratio while being resized by holding down the **ALT** key.

Shape Selection

The first icon above the Shape Creation toolbox is a single arrow known as the Shape Selection icon. It is the default tool to manipulate and edit shapes inside the Visualization Panel.



Figure 16 : Shape Selection icon.

Before a shape can be edited, it must be selected. Left-clicking on a shape in the Visualization Panel will select the shape. When the mouse is over a selectable shape, the standard cursor will turn into the Select/Move cursor (Figure 15).

When a single shape is selected, all the fields in the NanoPatterning Parameters Panel will display values pertaining to this shape. Any changes made to any of the fields of the NanoPatterning Parameters Panel will be applied to this single shape.

As will be discussed in more detail in section 2.3, when multiple shapes are selected, the fields of the NanoPatterning Parameters Panel will only display values that are common to all the selected shapes. If a given parameter is different amongst the selected shapes, the corresponding field will be left blank. For example, if two rectangles are selected as follows:

- Rectangle 1: width of 2 μm and height of 3 μm
- Rectangle 2: width of 2 μm and height of 5 μm ,

The Width field in the NanoPatterning Parameters Panel will display 2.000 μm but the Height field will remain blank, reflecting the different height of the two shapes.

It is important to note that any changes made to any fields of the NanoPatterning Parameters Panel will be applied to all the selected shapes. As an example, if three shapes with different dwell times were selected and a value of 5 μs is typed in the (blank) dwell time field, all three shapes will have their dwell times set to 5 μs .



Figure 17 : The Select/Move cursor, indicating the shape at the current mouse position can be selected or moved.

It is possible to select multiple shapes in two ways:

1. Hold down the **SHIFT** key while clicking on the shapes to be selected. This approach can also be used to select additional shapes when one or more shapes have been selected previously.
2. Click and drag on an empty area. This will activate a “rubber band” selection rectangle. On the mouse release, all shapes contained within, or partially within, the area of the selection rectangle will be selected.

To deselect a shape which has already been selected without deselecting other shapes, press the **SHIFT** key and click on the shape to be deselected.

Holding down CTRL + A will select all the shapes.

Note:

In order to be able to select shapes, the software needs to be in Shape Editing Mode. Refer to the section “Annotation Editing Mode vs. Shape Editing Mode”.

6.3. Moving Shapes using the mouse

Shapes can be moved using the mouse in the Visualization Panel. If one shape is selected, clicking and dragging on that particular shape will cause it to be displaced. Similarly, in the case of multiple selections, clicking and dragging on any of the selected shapes will cause all of the selected shapes to be moved.

Moving using the CTRL Key

When the **CTRL** key is down, clicking and dragging anywhere in the Visualization Panel will move all the shapes, including the unselected ones.

Moving using the CTRL + SHIFT Keys

When the **CTRL + SHIFT** keys are down, clicking and dragging anywhere in the Visualization Panel will move all the selected shapes.

6.4. Shape Selection versus Shape Activation

It is important to note that the NPVE system differentiates between the activation state and the selection state of a shape. A shape is Selected to enable editing to be performed whereas it is Activated in order to enable patterning to be performed.

When a shape is Selected, its nodes will be displayed. Any modification made in the fields of the NanoPatterning Panel will affect all of the Selected shapes. Shapes that are not Selected do not display nodes.

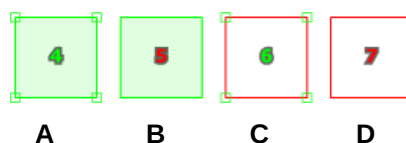


Figure 18 : Four shapes showing the differences between active and selected states. (A) Selected and Active. (B) Not Selected and Active. (C) Selected and not Active. (D) Not Selected and not Active.

An active shape is indicated by its bright green colour. A deactivated shape is indicated by its red colour. By default, Shapes are created in an Active State. To activate or deactivate a shape, use the **Active** checkbox located in the Shape Identifier / Browser. All active shapes will be processed when the **Start** button is pressed to initiate NanoPatterning.

Hint:

For most applications, it is convenient to have the NPVE consider all selected shapes as automatically active, and all unselected shapes as automatically Inactive. This can be accomplished by enabling the **Activate Selection** checkbox in the Shape Identifier / Browser section.

6.5. Shape Geometry and Style Parameters

This subpanel is used to edit the shape geometry properties such as position and dimensions. As mentioned at the beginning of this section, the fields show values for all selected shapes. Modification of any of those fields will be applied to all of the selected shapes, so it is important to only select the shape(s) that is (are) to be modified. These techniques are discussed below.

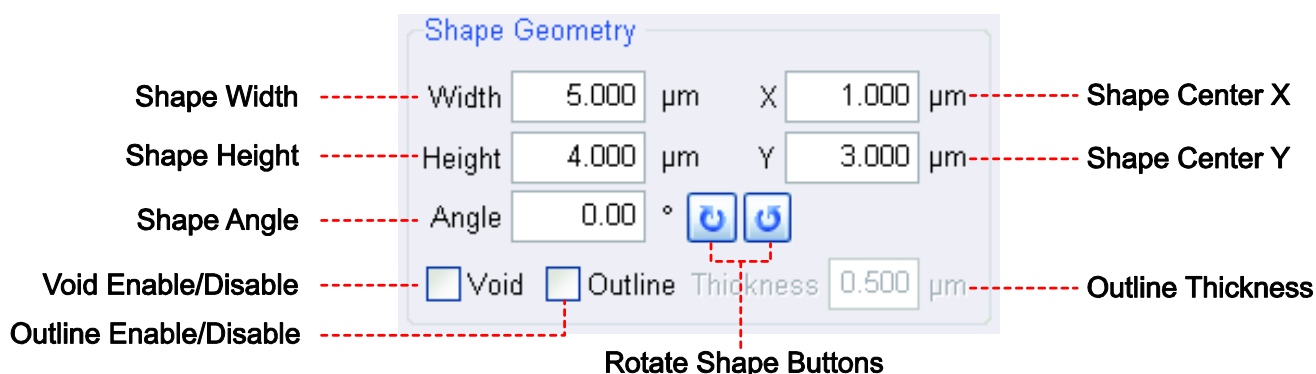


Figure 19 : Shape Geometry and Style subpanel in the NanoPatterning Parameters Panel. Note that Void ☒ADV and Outline ☒ADV styles are optional features in the NPVE Advanced software.

Width and Height

The width and the height of a shape can be entered using the corresponding fields. Again, a change made to the dimension will affect all the selected shapes by applying the value to each individual shape. It is important to note that changing width and height will not stretch or scale the selected group as a whole, but will assign the new value to each of the shapes individually as shown in Figure 10.

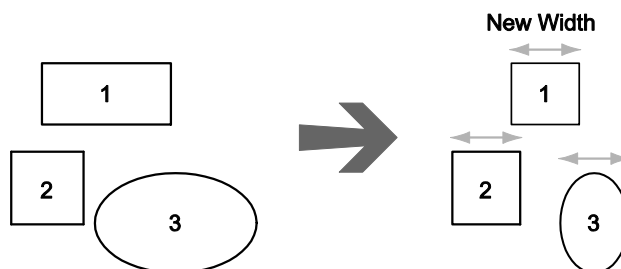


Figure 20: The width of the 3 selected shapes is changed to a common value using the Width field of the NanoPatterning Parameters Panel.

The width and height of a shape have different meanings depending on the type of shape selected. For Rectangle, Ellipse and Trapezoid shapes, these two dimensions correspond to the dimensions of the shape along the angle of the shape as shown on Figure 21. However, for Polygon, Line and Polyline shapes, the width and the height change relative to the shape angle (Figure 22).

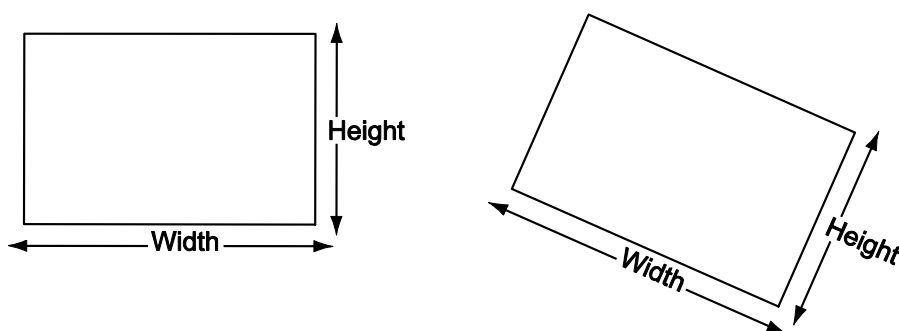


Figure 21: Width and Height for Rectangle (Ellipse and Trapezoid shapes also) are measured according to the actual lengths of the sides, regardless of shape Angle.

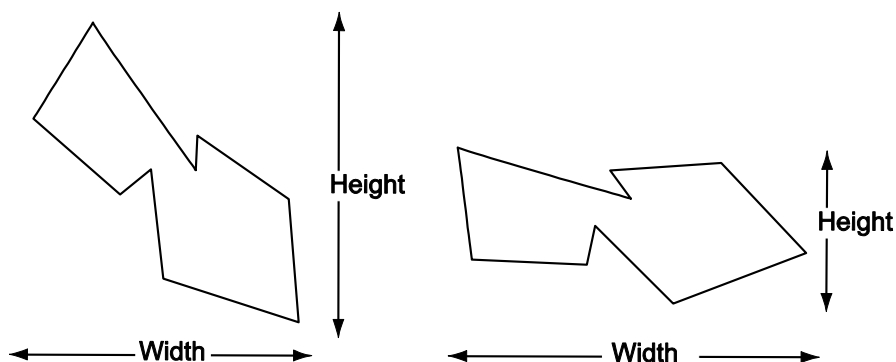


Figure 22: Width and Height for Polygons, Lines and Polylines will vary relative to the Angle of the shape.

Shape Centre X and Y

The XY position of the shape can be edited using the Shape Geometry Panel. The coordinates entered in the X and Y fields correspond to the centre of the shape. The centre of the shape is defined as the centre of the “bounding box” encompassing the shape, as shown in Figure 21. When multiple shapes are selected, the X and Y values correspond to the centre of the bounding box of the selected shapes. Figure 23 illustrates how a modification in X and Y will affect the position of the multiple shapes: the position of each shape relative to one another is not changed but the selection is shifted to the typed-in position.

FOV

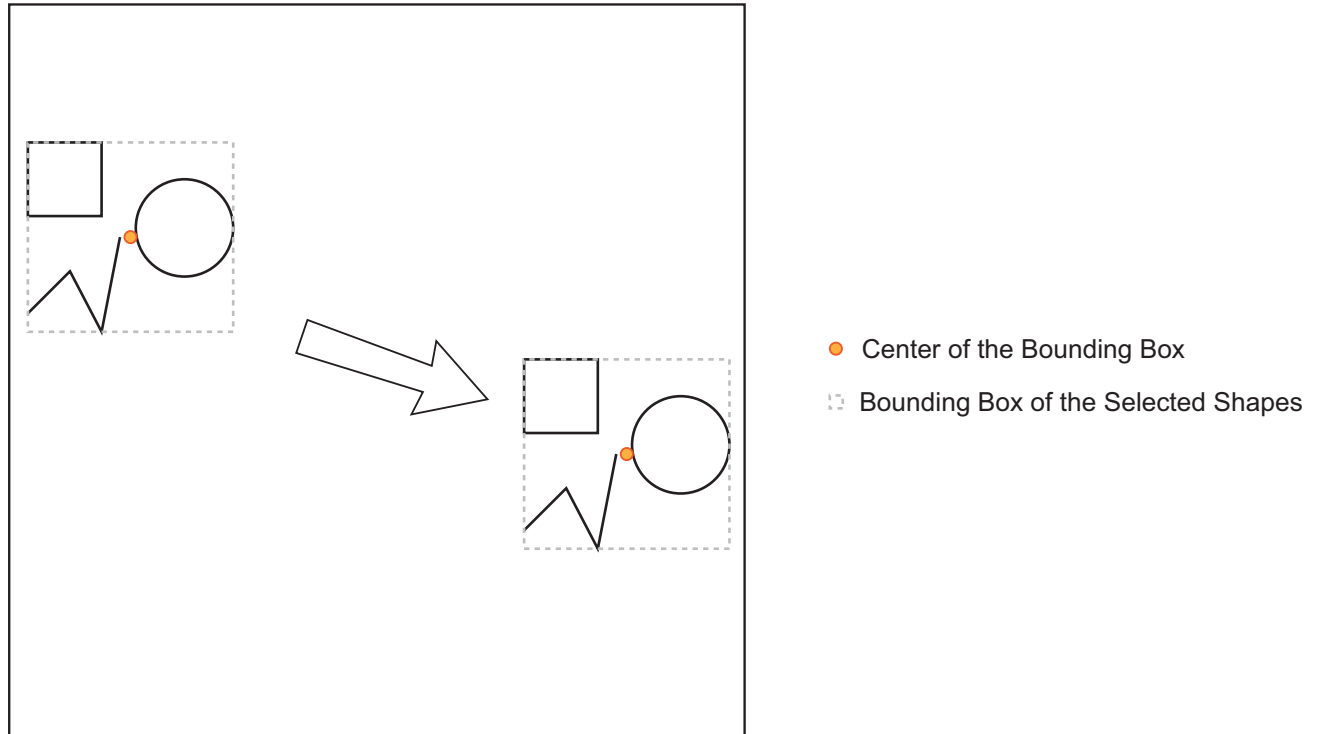


Figure 23: Displacement of selected shapes using the X and Y fields. The relative positions of the shapes are maintained.

Angle

The orientation of the shape is accessible through the Angle field of the Shape Geometry Panel. Changing the Angle of the shape will automatically update the scan angle of the shape. Changing the angle of the shape will rotate the shape according to its rotation centre, which is by default the centre of the shape itself. The section on “Rotation Knobs and Rotation”, found on page 42, discusses how to modify the rotation centre of a shape.

Positive angles rotate the shape counter-clockwise and negative angles rotate the shape clockwise.

Changing the angle of multiple shapes that have been selected will rotate each of the selected shapes individually about its respective rotation centre.

The **Rotate Shape** buttons can be used to increase or decrease by 0.1° the angle of the selected shapes.

Outline ADV

Using the Outline feature, the NPVE has the capability of outlining a shape using a range of parameters, which can be a useful technique. The Outline feature is offered as an option. Please refer to section 0 for more details.

Void ADV

A Void shape is an area that will not be patterned. This feature can be used to easily create more complex shapes or to protect an area of the sample. Please refer to section 9.2 for more details.

6.6. Shape Editing

There are three ways to edit the geometric properties of a shape:

- Using the Shape Geometry section in the NanoPatterning Parameters Panel (see NanoPatterning Parameters Panel section)
- Directly editing the shape using the mouse/keyboard.
- Using the context sensitive edit form.

Since editing using the NanoPatterning Parameters Panel has been covered in the Advanced Parameters Panel section, we will only cover shape editing using the mouse in this section.

Editing Shapes Using the Mouse

Perhaps the most flexible way to edit a shape is to use the controls that are drawn on the selected shapes in the Visualization Panel. Shapes are generally composed of Nodes and Edges, as discussed below. At all times, the software updates the mouse cursor to provide feedback to inform the user as to what action will be performed when the mouse is clicked, as the allowable actions depend on whether the mouse is over the interior of a shape, its nodes, edges, etc. These informative mouse cursors are described below, with each section pertaining to a particular editing mode. A list of the controls that can be used to edit the shape follows. It is important to note that the shape needs to be selected prior to direct editing.

To select a shape, simply left click on it with the mouse.

There are two shape edit modes available: Move/Stretch and Move/Rotate. The following operations are allowed for each mode:

- Move/Stretch Mode:
 - Move the shape
 - Move nodes
 - Move edges
- Stretch the shape
- Move/Rotate Mode:
 - Move the shape
 - Rotate the shape
 - Move the Rotation Centre

Once a shape is selected, clicking on the shape will toggle between both modes.

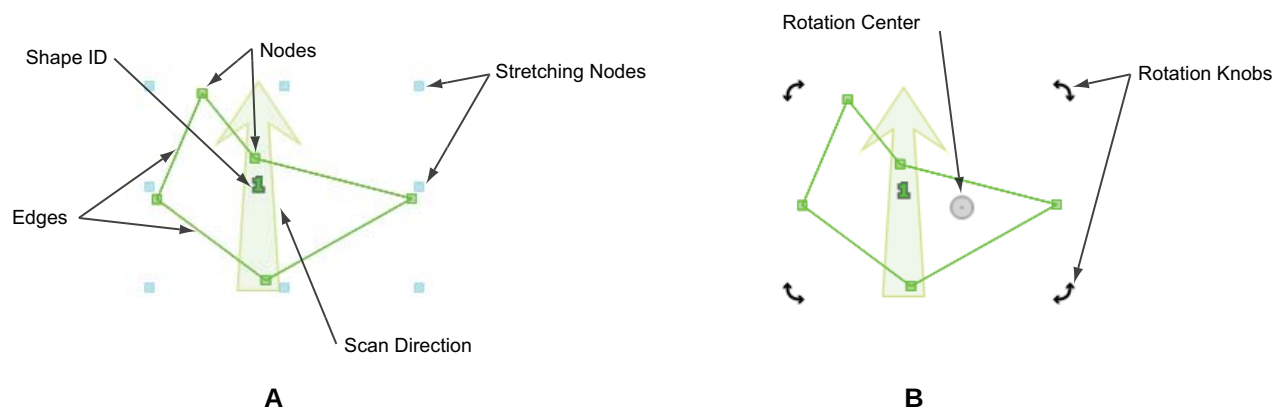


Figure 24: Different controls modes: (A) Move/Stretch Mode. (B) Move/Rotate mode.

Hint:

The arrow keys on the keyboard can be used to move a shape to a particular location. Pressing **SHIFT** + the arrow keys will move the shape by a five times (5X) greater factor.

Shape Nodes

Rectangles, Polylines, Polygons, Trapezoids and Lines are drawn using nodes. A node is represented by a semi-transparent square (Figure 25A), which is green in the case of Nodes along segments of a shape, and light blue in the case of Stretching Nodes which map the bounding boxes of more complex shapes.



Figure 25: (A) Example of a node at the end of two Line segments. (B) The cursor indicating that the mouse is over an editable node.

When the mouse is moved over a node and the node is editable, the mouse cursor will turn into the cursor shown in Figure 25B to inform the user that a node can be moved. To move a node, left-click on it and while holding the mouse button down, move the mouse to relocate the node. Release the mouse button when the node is in the desired position.

Hint:

If a node cannot be selected, the possible causes are:

- The shape is not selected
- The shape is locked
- The node movement has been disabled in the Selective Editing Mode
- The NPVE is in Annotation Editing Mode (see section Annotation Editing Mode vs. Shape Editing Mode)

The node behaviour depends on the shape type. Polygon, Line and Polyline shapes allow free movement of the nodes in any direction. In the case of Rectangle and Trapezoid shapes, node movement is constrained to maintain the fundamental geometry of the edited shape. Moving a node in a Rectangle will cause the two adjacent nodes to move in order to maintain 90° corners. In a Trapezoid, the symmetry of the shape is maintained at all times.

Hint:

It is possible to turn any shape (except Deflection Lists and Spots) into a shape whose nodes are editable shape by using the “Convert to Polygon” tool. Please note that some of the patterning properties will be changed to comply with the patterning of the Polygon. For example, concentric patterning for a Ring will be changed to a standard raster style.

It is possible to deactivate the node movement capabilities from the Selective Editing menu (See section 8.3 for more details about Selective Editing menu).

Node Manipulation

Nodes can also be added, deleted or moved to a specific XY coordinates through the right-click menu.

Note:

Nodes can be added, moved or deleted on a Rectangle, Trapezoid, Polygon, Line and Polyline. If a node is added to a Line, it will automatically be turned into a Polyline. Rectangle and Trapezoid shapes will be turned into Polygons.

Add Node

Nodes can be added to a shape after creation time. In order to add a node, right-click on an existing edge of the shape and select “Add Node” from the pop-up menu.

Delete Node

To delete an existing node, right-click on a node and select “Delete Node” from the popup menu.

Deleting a node from a Rectangle or a Trapezoid will turn the shape into a Polygon.

Set Node Position

To set the position of a particular node, right-click on the node to be moved and select “Set Position” from the popup menu. The position can be entered in nm, μm or mm. To switch units, click on the unit selector button.

Shape Edges

The line segment joining two nodes is called an edge. Edges can be moved just like nodes. Ellipse, Spot, FIBText, Ring and Deflection List do not have any editable edge. When the mouse is over an edge, the mouse cursor will be changed to one of the cursors shown in Figure 26



Figure 26: Cursors indicating the mouse is over an edge.

Moving an edge causes two nodes to be moved at once. The edge behaviour is also shape dependent since the fundamental geometric properties of the shape are preserved at all times. For the Trapezoid and Rectangle shapes, the edges can only be moved at 90° from the edge angle.

The Edge move function can be deactivated from the Selective Editing menu (See section 8.3 for more details about Selective Editing menu).

Stretching Nodes

The size and aspect ratio of certain shapes can be modified by dragging the stretching nodes. The stretching nodes are drawn on the edge of the bounding box encapsulating the shape. They are drawn using a light blue colour. When the mouse is over a stretching node, the cursor will change to one of those shown in Figure 27.

Stretching nodes are not available for Spot, Line, Rectangle and Trapezoid shapes as moving the standard nodes of a Line, Rectangle or a Trapezoid will result in stretching, and by definition a Spot is not stretchable.



Figure 27: (A) Stretching nodes shown in turquoise along the bounding box of the selected shape. (B) Cursors indicating that the mouse is over a stretch node.

Stretching the boundary of a shape over its opposite boundary will flip the shape in the crossover direction.

Hint:

Holding the **CTRL** key while stretching the shape allows stretching from the centre of the shape.

Holding the **ALT** key while stretching the shape will maintain the aspect ratio of the shape at all times.

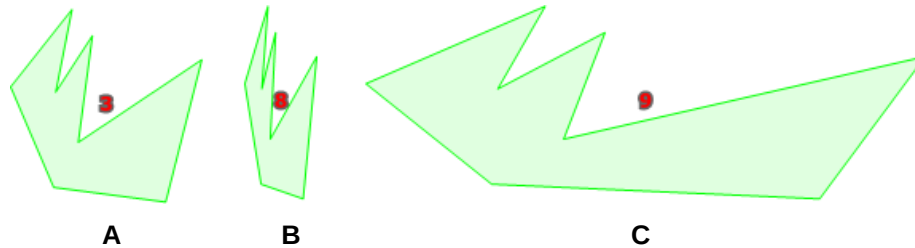


Figure 28: Shapes illustrating stretching. (A) The original shape. (B) The same shape compressed horizontally. (C) The shape extended horizontally.

Rotation Knobs and Rotation Centre

When in the Move/Rotate mode, it is possible to grab the Rotation Knobs of the shape and rotate the shape according to the position of the Rotation Centre (by default the centre of the shape). To access the Move/Rotate mode, click and release the left mouse button on the shape (without moving) when in Move/Stretch mode.

In Move/Rotate mode, it is possible to relocate the Rotation Centre by clicking and dragging the grey circular Rotation Centre icon to the desired location. Double-clicking on the Rotation Centre will reset its position to the centre of the shape.

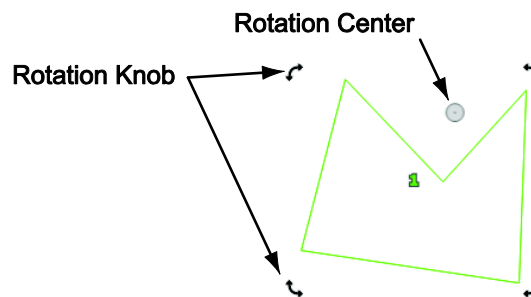


Figure 29: Example of a Polygon placed into Move/Rotate mode. The Rotation Knobs are located in the four corners. The location of the Rotation Centre is indicated by the grey circular icon, which has been moved outside the body of the Polygon in the figure above.

Hint:

Holding the **ALT** key while rotating the shape constrains the rotation to angles which are multiples of 15°.

Outline Thickness Knob

Once a shape is turned into outline mode, the thickness of the outline can be changed using the outline thickness knob illustrated in Figure 30.



Figure 30: Outline shapes showing the outline thickness knob. (A) A Ring showing 2 knobs. (B) A standard shape showing one outline thickness knob.

Dragging the Outline Thickness Knob will set outline thickness to the minimal distance from any edges of the shape and the current mouse position.

6.7. Undo

All of the edit actions (move shape, create shape, move node, move edge, etc.) can be undone. The key combination **CTRL + Z** reverses the last action. Undo is also accessible through the right-click menu. By default, the number of undo levels stored is set to 20. Modifications of the Patterning Parameters can also be undone in a similar manner.

Hint:

The number of undo levels can be modified in the User Preferences Panel.

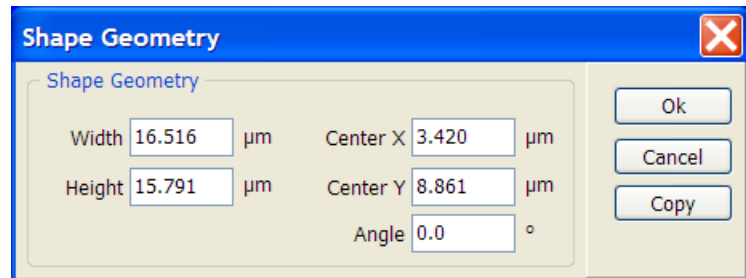
6.8. Context Sensitive Shape Edit

The right-click menu allows the user to edit shape types with context sensitive forms. The context sensitive forms provide a set of useful parameters specific to a shape type.

Rectangles, Ellipses, Polygons and Polylines

When a Rectangle, an Ellipse, a Polygon or a Polyline is selected, the context sensitive form gives the ability to change the following parameters:

- Width
- Height
- Centre XY
- Angle



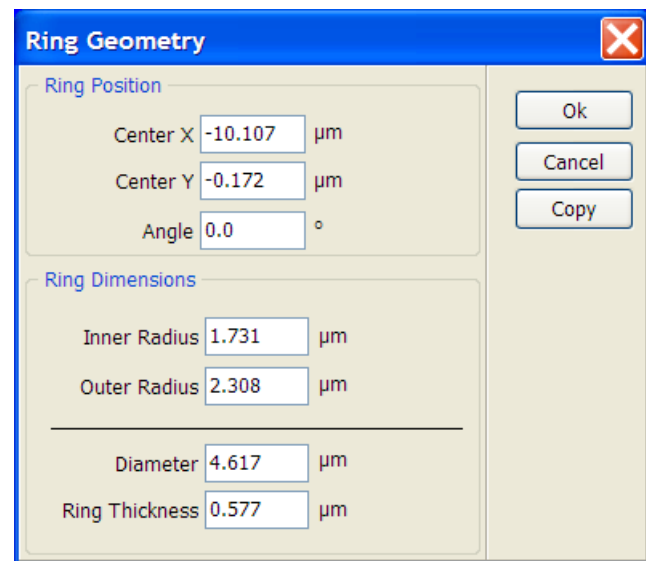
The 'Shape Geometry' dialog box is shown with a blue title bar and a close button (X) in the top right corner. It contains a section titled 'Shape Geometry' with four input fields: 'Width' (16.516 μm), 'Height' (15.791 μm), 'Center X' (3.420 μm), and 'Center Y' (8.861 μm). Below these is an 'Angle' field (0.0 °). On the right side of the dialog are three buttons: 'Ok', 'Cancel', and 'Copy'.

Figure 31 Context Sensitive Edit Form for Rectangles, Ellipses, Polygons and Polylines

Rings

When a Ring is selected, the context sensitive form allows modification of the following parameters:

- Centre XY
- Angle
- Inner Radius
- Outer Radius
- Diameter
- Ring Thickness



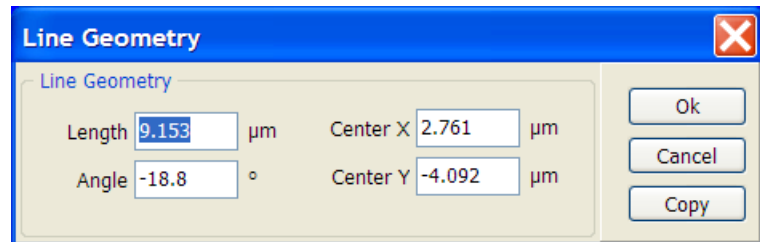
The 'Ring Geometry' dialog box is shown with a blue title bar and a close button (X) in the top right corner. It contains two sections: 'Ring Position' and 'Ring Dimensions'. The 'Ring Position' section has three input fields: 'Center X' (-10.107 μm), 'Center Y' (-0.172 μm), and 'Angle' (0.0 °). The 'Ring Dimensions' section has four input fields: 'Inner Radius' (1.731 μm), 'Outer Radius' (2.308 μm), 'Diameter' (4.617 μm), and 'Ring Thickness' (0.577 μm). On the right side of the dialog are three buttons: 'Ok', 'Cancel', and 'Copy'.

Figure 32 Context Sensitive Edit Form for Rings.

Lines

When a Line is selected, the context sensitive form allows modification of the following parameters:

- Centre XY
- Angle
- Length



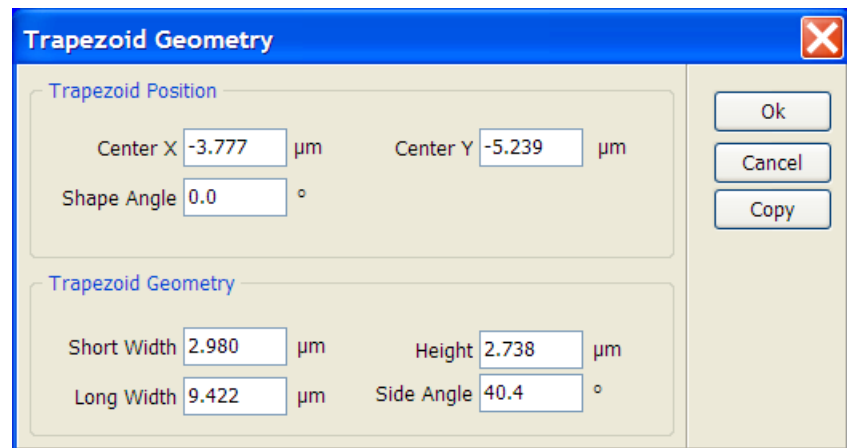
The 'Line Geometry' dialog box has a blue title bar with a close button. It contains two sections: 'Line Geometry' and 'Line Geometry'. The 'Line Geometry' section has input fields for Length (9.153 μm), Angle (-18.8 °), Center X (2.761 μm), and Center Y (-4.092 μm). The 'Line Geometry' section is empty. On the right, there are 'Ok', 'Cancel', and 'Copy' buttons.

Figure 33 Context Sensitive Edit Form for Lines.

Trapezoids

When a Trapezoid is selected, the context sensitive form allows modifying the following parameters:

- Centre XY
- Angle
- Short Width
- Long Width
- Height
- Side Angle



The 'Trapezoid Geometry' dialog box has a blue title bar with a close button. It contains two sections: 'Trapezoid Position' and 'Trapezoid Geometry'. The 'Trapezoid Position' section has input fields for Center X (-3.777 μm), Center Y (-5.239 μm), and Shape Angle (0.0 °). The 'Trapezoid Geometry' section has input fields for Short Width (2.980 μm), Long Width (9.422 μm), Height (2.738 μm), and Side Angle (40.4 °). On the right, there are 'Ok', 'Cancel', and 'Copy' buttons.

Figure 34 Context Sensitive Edit Form for Trapezoids.

6.9. Geometric Shape Transforms

The NPVE offers the capabilities to modify the shapes' geometric properties. Those geometric transforms are accessible under the "Transform" item of the popup menu accessible by right-clicking the main viewport. The transform will be applied to all of the selected shapes. Using those types of transform, the selected shapes are treated as a group.

There are four different transforms:

- Move
- Rotate
- Scale
- Reflect

It is important to note that the geometric transforms behave differently than setting the width, height and angle in the Shape Geometry Panel of the NanoPatterning Form.

When using the Shape Geometry Panel, it sets the values for each shape individually. For example, entering 10 μm in the width field will give all of the shapes a width of 10 μm .

When using the Geometric Shape Transform, the bounding box of the selected shapes is used as the primitive for the transforms. For example, if three shapes are selected, it will use the centre of the bounding box of those 3 shapes as the Rotation Centre for the rotation transform as shown in Figure 36.

For each transform, the user can choose to create a copy of the selected shapes with the transform applied.

Move Shapes

The selected shapes can be moved from their actual location using the Move Shapes Dialog Box. The move can either be specified in x and in y or by using a radius and an angle. One can move to an absolute position or a position relative to the initial location depending on the move type selected.

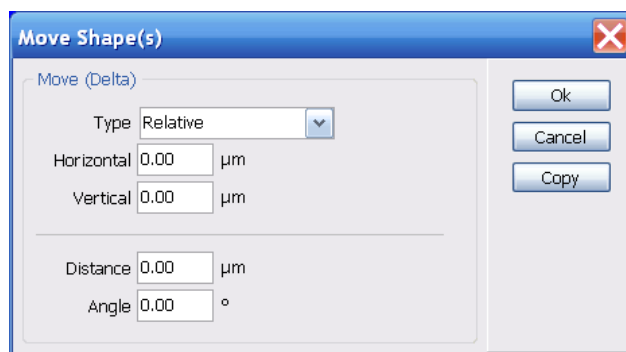


Figure 35 The Move Shapes dialog box.

Rotate Shapes

The selected shapes can be rotated using a common Rotation Centre with the Rotate Shape dialog box.

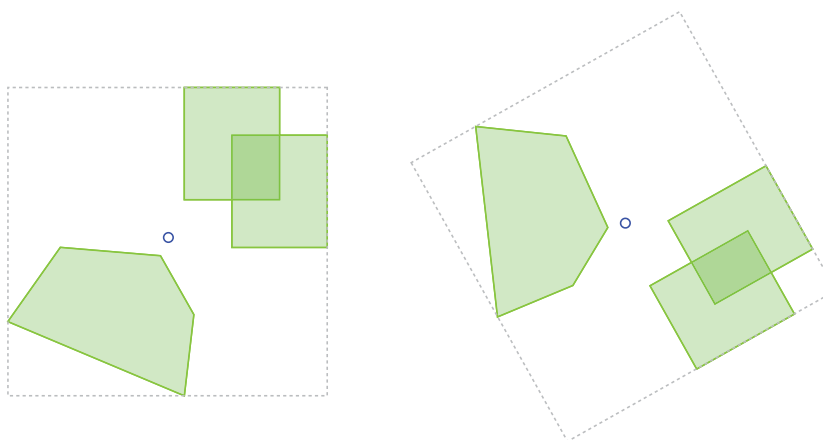


Figure 36 Example of a rotation of 3 shapes using the Transform Rotation tools. Note that the 3 shapes rotated not according to their respective centres, but rather according to the centre of the bounding box encompassing all 3 of them. In the example shown, the shapes were rotated by +30°.

It is important to note that the outcome is different depending on whether the Rotate Shape Transform tool is used or the angle value is changed in the Angle field of the Shape Geometry of the NanoPatterning Form. In the latter case, each shape will be rotated about its own centre. In the former case, the shapes will be rotated about a common Rotation Centre which is the centre of the bounding box of all the selected shapes, as shown in Figure 36. Positive angles are counter-clockwise and negative angles are clockwise.

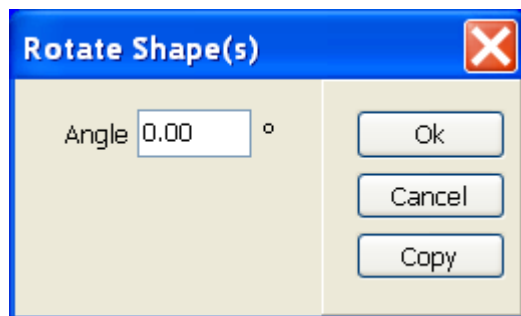


Figure 37 Rotate Shapes dialog box.

Scale Shapes

The dimensions of the selected shapes can be modified using the Scale Shapes dialog box.

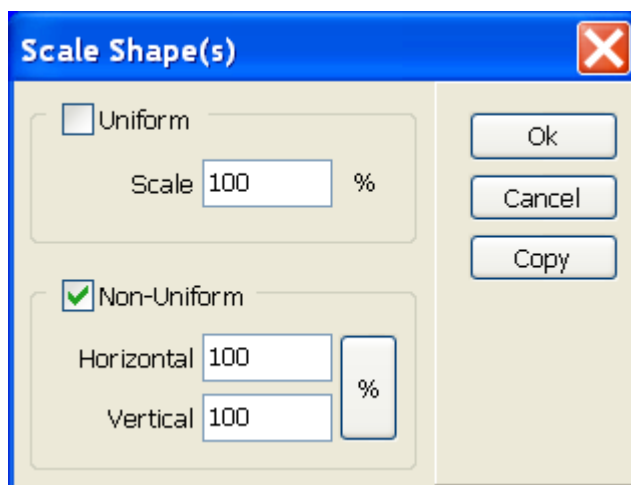


Figure 38: Scale Shapes dialog box.

Shapes can be scaled uniformly or non-uniformly. It is important to note the difference between using the Scale Shape Dialog Box and simply entering values in the Width and Height fields of the Patterning Parameters Form. As shown in Figure 39, the scaling is applied to the bounding box encompassing the selected shapes as opposed to scaling each of them individually.

Please note that in the case of non-uniform scaling, the values can either be entered as a percent or in micrometers.

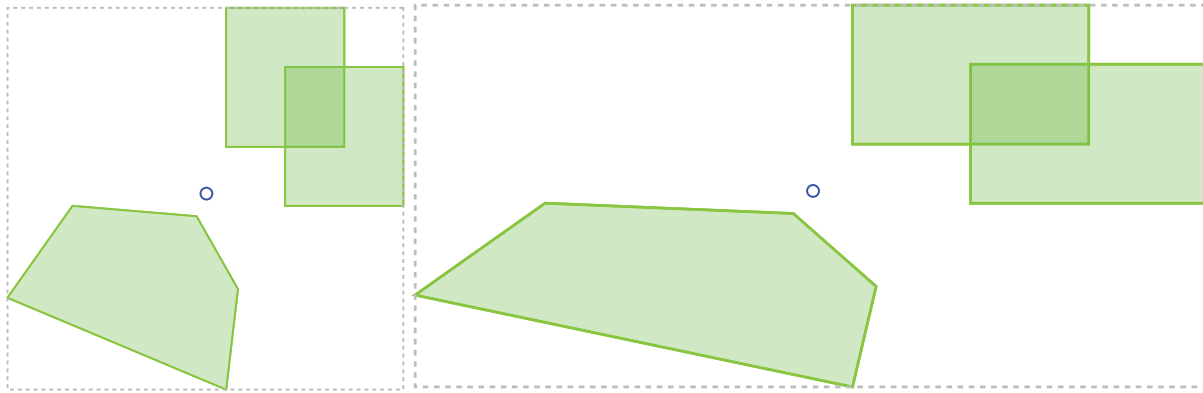


Figure 39 Example of scaling three shapes non-uniformly by 200% in the horizontal direction using the Scale Shapes dialog box. Note that it is the bounding box of all the shapes that is scaled and not each shape individually.

Reflect Shapes

The selected shapes can be reflected using the Reflect Shapes dialog box. The selected shapes can be reflected relative to:

- The horizontal axis
- The vertical axis
- A custom angle

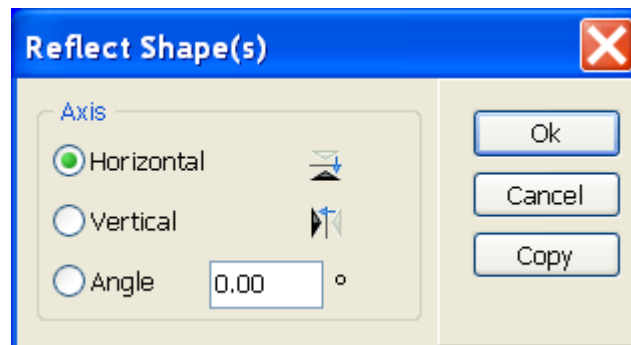


Figure 40 The Reflect Shapes dialog box.

As with other transforms, shape reflection is done according to the centre of the bounding box of the selected shapes.

6.10. Progress Display and Control Subpanel

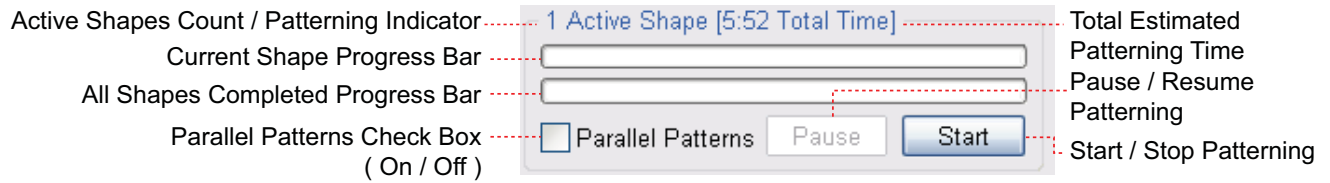


Figure 41: Progress Display and Control Subpanel

Active Shapes Count / Patterning Indicator

This label gives information about the number of active shapes or about the patterning operations in progress. See “Shape Selection versus Shape Activation” for more information regarding Active Shapes.

Patterning can occur serially (consecutively, i.e. one shape at a time) or in parallel (concurrently, i.e. all shapes at once), as discussed in the Parallel Patterning Checkbox section, below.

When patterning is underway the Patterning Indicator will read as follows:

“Executing 1 of N Shapes”	Patterning the first of N shapes serially
“Executing N Shapes”	Patterning N shapes in parallel

For example, if 5 shapes are active and the NPVE is patterning all five shapes in parallel, the Patterning Indicator displays “Executing 5 Shapes”. If Serial Patterning is specified when the second shape is being patterned, the indicator will read “Executing Shape 2 of 5”.

Note:

Parallel Patterning is available only when the active shapes use the same patterning beam or when the gas used is the same. It is also deactivated when an active shape is set to “Line Mill”.

Current Shape and All Shapes Completed Progress Bars

Two progress bars are visible in this subpanel. The upper Current Shape progress bar tracks the status of the shape currently being patterned to its completion. It is updated at the end of each “frame”, which corresponds to the completion of one full patterning scan of the shape being executed. The lower All Shapes Completed progress bar indicates, when performing serial patterning, how many shapes have been completed out of the total number of shapes to be patterned. It is updated every time a shape’s patterning is completed. In parallel patterning mode, both progress bars show identical results.

Parallel Patterning Check Box

This check box toggles between serial patterning and parallel patterning. Figure 42 illustrates the differences between the patterning modes.

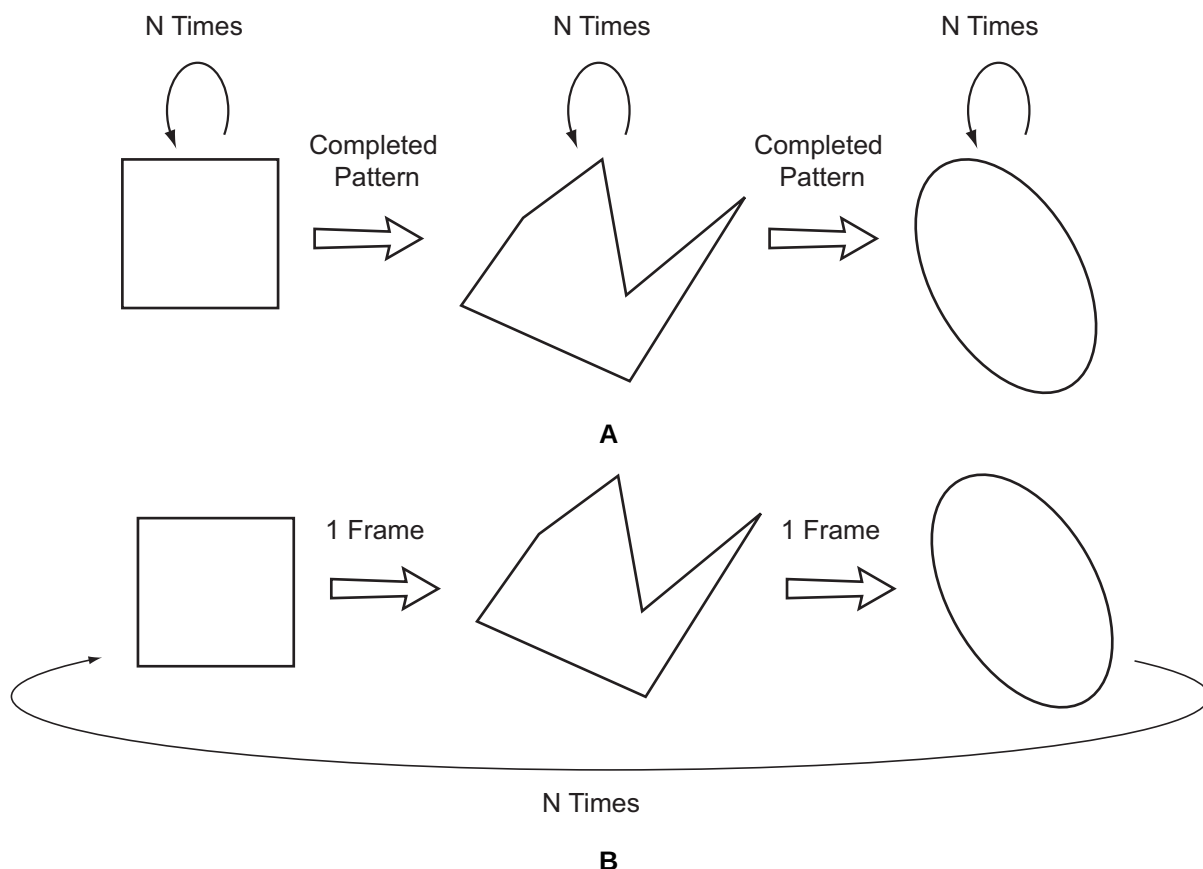


Figure 42: Different types of patterning sequences. (A) Serial patterning where each shape is patterned in a sequence based on their ID. (B) Parallel patterning where all shapes are patterned simultaneously.

Note:

In serial patterning, the shape that is displayed is the only shape currently executing and all the shapes will be sequentially displayed as they are being patterned.

Only shapes defined with the same Operation can be patterned in parallel.

Since "Line" scan style delivers the complete dose for each line at once, it is not possible to perform parallel patterning for shapes using "Line" scan style.

Start/Stop Patterning Button

Pressing the **Start** button will cause all the active shapes to be patterned. It is important to note that it is possible to have as many shapes created on the screen as the user wishes, but only the active ones, shown in light green, will be patterned when the **Start/Stop Patterning** button is pressed.

Hint:

To quickly activate/deactivate shapes, enable the “Activate Selection” option.

Note:

It is possible to pattern shapes with different patterning parameters during a single execution cycle. However, all the shapes must have the same gas (or no gas) and must be milled using the same beam current.

Once patterning is started, the button toggles to “Cancel” while it completes internal calculations.

When the internal calculations are completed, the patterning will start and the button caption will switch to “Stop”. The button can then be pressed to stop patterning whenever the user desires.

Note:

The time taken to complete the internal calculations can vary from a fraction of a second to a few minutes. Outlined shapes usually take longer to process than regular filled shapes. Filled shape with a large number of points can also require a longer period of time to process.

The NPVE keeps track of the time/dose/number of scans delivered to a shape. If the user stops executing the patterning of a shape and then starts patterning that same shape again, the system will not reset the complete time/dose/number of scans delivered to that shape and will continue the patterning from the point at which the **Stop** button was pressed. In order to reset the time/dose/number of scans delivered, the shape must be moved or resized, or the **Reset** button in the Duration Controls subpanel must be pressed.

Stopping and starting a shape using “Line” raster mode will restart patterning at the first line and not at the line that was currently executed when the user pressed **Stop**; in this case, in order to continue patterning at the point where it was interrupted, use the **Pause/Resume** button, discussed below.

Pause/Resume Button

Once patterning has been started, it is possible to pause and then resume. Pausing will suspend the patterning process until the user presses the **Resume** button. Resuming will continue patterning from the point where it was when the **Pause** button was pressed. The behaviour, however, is slightly different for certain patterning operations and styles:

3D NanoPatterning (BMP Patterning)

When a bitmap mill is paused, the system stores the actual frame that is being milled. When resumed, patterning will continue from this frame as opposed to the first frame of the mill, which would occur if the user had pressed the **Stop** button.

Single Shape using line scan style

With line scan style, the complete dose is delivered to each line before moving to the next line. Pausing and resuming a mill under these conditions will cause patterning to be restarted at the line where the mill was paused; this may lead to additional dose being applied to this line. See the Advanced Parameters Panel for further details about the different scan styles.

6.11. Shape Identifier / Browser

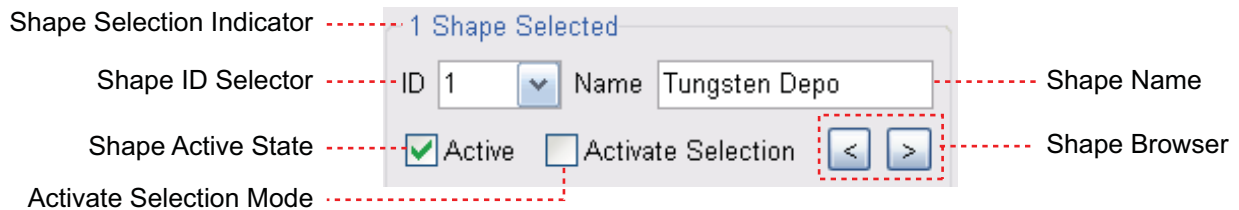


Figure 43: Shape Identifier / Browser annotated.

Shape Selection Indicator

This caption indicates to the user how many shapes are selected.

Shape ID Selector

The Shape ID Selector is used to select a shape by its ID. All shapes have a unique ID so they can be easily identified or selected.

Note:

The NanoPatterning execution order is determined by the Shape ID, from lowest ID to highest ID amongst the selected shapes. The Shape ID also affects the outcome of other operations such as the Set Operations.

The Shape ID Selector can also be used to change the ID of a single shape by entering a new ID number in the Shape ID field. In this case, the selected shape will take the ID the user has entered. If this ID number is already assigned to an existing shape the ID of the now conflicting shape will be incremented by 1. This operation will cascade until there is room to insert the updated ID. Shape ID changes can only occur when there is only one shape selected.

Shape Name

By default, the name of the shape is filled automatically with its type (Rectangle, Ellipse, Line, etc...). However, a custom name can be assigned to a shape to give more insight about the purpose of the shape.

Note:

When the Shape Name is changed, the change is applied to all of the selected shapes.

Shape Browser

It is possible to browse through the list of created shapes by using the “forward” and “backward” buttons of the shape browser, which browses sequentially by Shape ID number.

Note:

Using the **Shape Browser** buttons will cause the currently selected shape(s) to revert to an unselected shape and the next shape to become the sole selected shape.

6.12. Standard Patterning Parameters

Standard Patterning Parameters subpanel is used to set the essential nanopatterning parameters: the beam conditions, the pre-defined Operations and the patterning duration in terms of Time, Dose or Number of Repeats.

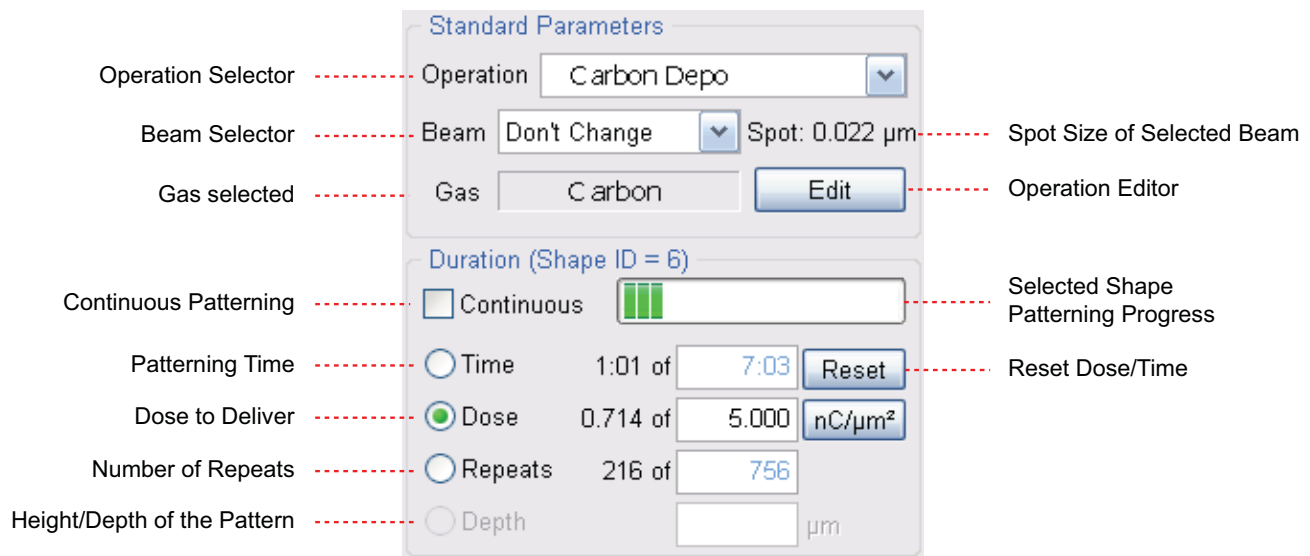


Figure 44: The Standard Patterning Parameters Panel.

Operation Selector

The NPVE introduces the concept of “Operations” in order to improve ease of use and repeatability. Operations can be considered as pre-defined tables of values that set all nanopatterning parameters required to execute a shape. In general, different parameter settings are required to achieve different patterning behaviours. Once the optimal Operation parameters have been determined by an advanced user, a new Operation should be saved. This Operation may then be used in the future by anyone looking to perform a similar task.

The Operation Selector is used to automatically set the Patterning parameters for a particular behaviour (i.e. milling, etching, depositing, etc.) in a straightforward manner that does not require advanced knowledge of the microscope or the particular advanced settings required in order to achieve the desired behaviour. Assigning an Operation to a single shape or to a set of shapes will automatically employ the correct advanced patterning parameters for each shape.

The Concept of “Custom Patterning Parameters”

If the user is experienced and wishes to manually enter shape parameters, they can be accessed in the Advanced Parameters subpanel discussed in section “Advanced Parameters Panel”. Once an Operation is assigned to a shape, it is still possible to modify the Patterning parameters for this particular shape through the Advanced Parameters Panel; this shape would then be considered to have “Custom Patterning Parameters”.

The shape with “Custom Patterning Parameters” will remain in that state up until the Beam or the operation of the shape is reassigned. The NPVE will then ask whether the user wants to keep the custom Patterning Parameters or wants them changed to reflect the new Operation/Beam combination.

Operation Selector Caption Colours

Once an operation has been assigned to a shape, the patterning parameters can be overridden to any value. The patterning parameters for this shape are then considered “Custom Patterning Parameters”. The caption can have 3 different colours indicating whether a shape has custom parameters or not.

- Black** The patterning parameters of the shape are exactly the same as the operation patterning parameters.
- Red** All selected shapes have “Custom Patterning Parameters”.
- Orange** Undefined state: At least one of the selected shapes has “Custom Patterning Parameters” and at least one of the selected shapes has standard Patterning Parameters.

Note:

It is possible to execute shapes with different operations in the same sequence. However, the selected operations must all use the same gas (or no gas at all) and the same beam conditions.

After an expert user defines an Operation, the Operation takes into account the beam current selected to perform the patterning and modifies the Advanced Parameters of the shape accordingly without the need for user intervention.

Beam Selector

When the NPVE is controlling the FIB Beam, it communicates directly with SmartSEM[®] and retrieves information on each of the beams available in the SmartSEM[®] interface. Any of these beams can then be selected to perform a surgery. When “Use Existing” is selected, the NPVE won’t change the beam when the mill is started.

When the NPVE controls the SEM, the Beam Selector displays the current beam state of the SEM column. Any changes related to the SEM Beam can be made by double clicking the displayed beam state and entering the desired values in the SEM Beam Conditions Form. This is described more thoroughly in section 1.2.

The nominal spot size of the current beam is displayed next to the Beam Selector. When “Use Existing” is selected, it is possible to know which beam is selected in SmartSEM[®] by looking into the “General Status Bar”.

Hint:

Changing the beam will assign the optimal patterning parameters for this newly selected beam to all of the selected shapes based on the selected Operation for each shape.

Continuous Patterning

Ticking the **Continuous** patterning checkbox will cause the patterning of active shapes to be performed indefinitely and will disregard any duration entry (Time, Dose, Number of Frames or Depth). Patterning will be stopped only when the user presses the **Stop** button in the Progress Display and Control Panel.

Continuous patterning cannot be selected when “Line” scanning style is selected or when two or more shapes are activated and are not selected to be patterned in parallel.

Duration

The user can choose between 4 different duration types:

- Time
- Dose
- Number of repeats
- Depth/Height

The duration types are mutually exclusive, but the non-selected types are updated when the selected one is changed.

Note:

The Depth/Height field is only activated if the milling rate is defined in the selected operation. To enter a milling rate for a given Operation, click the **Edit** button underneath the beam size.

The time is entered in seconds. However, input of type HH:MM:SS is also accepted.

The dose is entered as nC/μm².

Spot Size of the Selected Beam

The Spot Size label has two different meanings depending on whether the NPVE controls the SEM beam or the FIB beam.

The NPVE Controlling SEM:

When the NPVE controls the SEM, the spot size becomes the beam interaction volume. This interaction volume depends on the physical aperture selected but also on the EHT (Accelerating Voltage) of the probe: the higher the EHT, the greater the interaction will be. This is particularly true when gas chemistry is involved.

The NPVE Controlling FIB:

In this case, the Spot Size corresponds to the value entered in the “Probe Width” field under the FIB Probe Table in SmartSEM®.

7. Advanced Parameters Panel

Located at the bottom of the NanoPatterning Parameters Panel, the Advanced Parameters subpanel is used to customize the more complex parameters of a shape.

Note:

This section refers to a number of advanced concepts and may be safely skipped by the novice user and returned to at a later date.

There are two versions of the Advanced Parameters Panel: the default “filled” shapes and Outlined shapes. The different fields vary between both panels (as illustrated in Figure 45 and Figure 71). For more details about the Outline parameters, see section 0.

Hint:

To avoid confusion for novice users, the Advanced Patterning Parameters subpanel is hidden by default. If the Advanced Patterning Parameters subpanel is not visible, click on the **More +** button and the subpanel will be revealed.

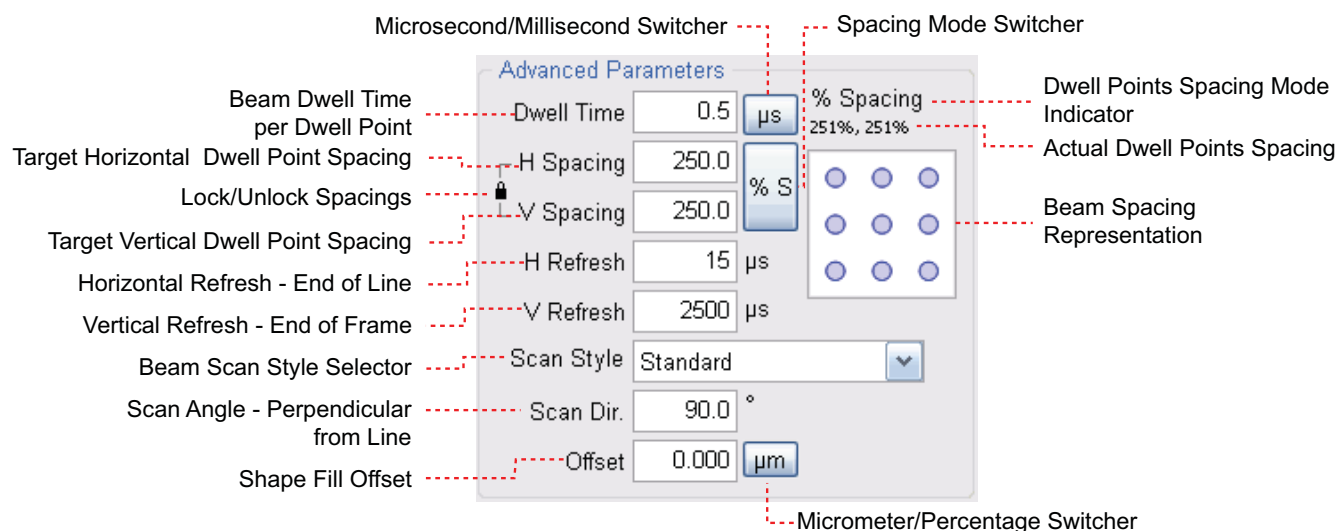


Figure 45: Advanced Parameters Panel for a regular shape.

Dwell Time

The dwell time represents the time in μs (10^{-6} seconds), in ms (10^{-3} seconds) or in seconds that the beam will remain at each dwell point before moving to the next dwell point.

The units of the dwell time (μs , ms or s) can be toggled using the **Microsecond/Millisecond/Second Switcher** button. When toggling between the units, the value in the field is automatically converted into the new unit in order to preserve the same absolute value.

Scientific notation, such as “2.5E5”, can be used to enter values in this field.

Spacing Mode Switcher

For maximum flexibility, and to accommodate users from different instrument backgrounds, it is possible to enter the spacing in three different ways:

3. Relative Spacing (%)
4. Relative Overlap (%)
5. Absolute Spacing (μm)

Clicking the **Spacing Mode Switcher** will toggle between these different modes.

Relative Spacing:

Relative spacing is entered as a percentage and is calculated as:

$$\%_{spacing} = \frac{dwell\ spacing_{\mu m}}{spot\ size_{\mu m}} \times 100\%$$

The minimum relative spacing allowed is 1%. There is effectively no maximum relative spacing.

Relative Overlap:

Relative overlap is entered as a percentage and is calculated as:

$$\%_{\text{overlap}} = \frac{(\text{spot size}_{\mu\text{m}} - \text{dwell spacing}_{\mu\text{m}})}{\text{spot size}_{\mu\text{m}}} \times 100\%$$

The maximum overlap spacing allowed is 99%. A negative overlap value is interpreted as a spacing which is larger than the spot size.

Absolute Spacing:

When absolute spacing is selected, the spacing is entered in micrometers (μm). It represents the physical spacing between two contiguous points.

H Spacing

The horizontal spacing (H Spacing) is the distance between each dwell point at an angle of -90° from the scan direction. Physically, during patterning, the beam will first move in the H Spacing direction. Once a “line” along the H Spacing is completed, the beam moves one step in the V Spacing direction, which is in the same direction as the scan direction. Figure 46 shows a schematic representation of the patterning direction and the meaning of H spacing and V spacing for scan directions of 90° and 135° ; in this figure horizontal spacing is represented as “HS”.

Note:

The spacing parameters (H, V, P and R) are independent of the field of view.

The entered values are considered target values and may be adjusted slightly (to the “Actual Spacing”, as discussed below) in order to fit an integer number of dwell points in the shape. See Figure 49.

V Spacing

The vertical spacing (V spacing) is always the spacing along the scan direction or in between each “line” at 90° to the scan direction. It is represented in Figure 46 as “VS”.

Scan Direction

The scan direction gives the orientation of the lines to be milled by the raster generator. By default it is set to be at 90° from the shape angle. It can be set by entering a value with the keyboard, by selecting the edge of the shape to be patterned last or by using the Scan Direction assignment tool. The Scan Direction is drawn as a light green arrow on the shapes that are selected (see Figure 24).

Hint:

For FIB users performing patterning without gas assistance, in the case of “Line” raster style, the tail of the Scan Direction arrow indicates the side of the shape nearest the starting point of the mill. Normally redeposition will be higher at this point.

The user should note that changing the Scan Direction can be a powerful technique as this direction does not need to be locked to the axes of the shape, and can be set according to features in the sample itself, while still allowing a given shape to be patterned. For example, Figure 47 shows (A) the typical scan direction for a rectangular shape. Sample constraints may dictate that a sample must be patterned with the scan in a specific direction. One way to achieve this is by rotating the shape, as shown in (B). This, however, could lead to milling in unwanted regions of the sample, or the need to create a more complex Polygon in order to achieve the same results. In (C) a more elegant solution is shown, where the Scan Direction itself is rotated while the base rectangular shape remains unchanged, thereby satisfying both requirements.

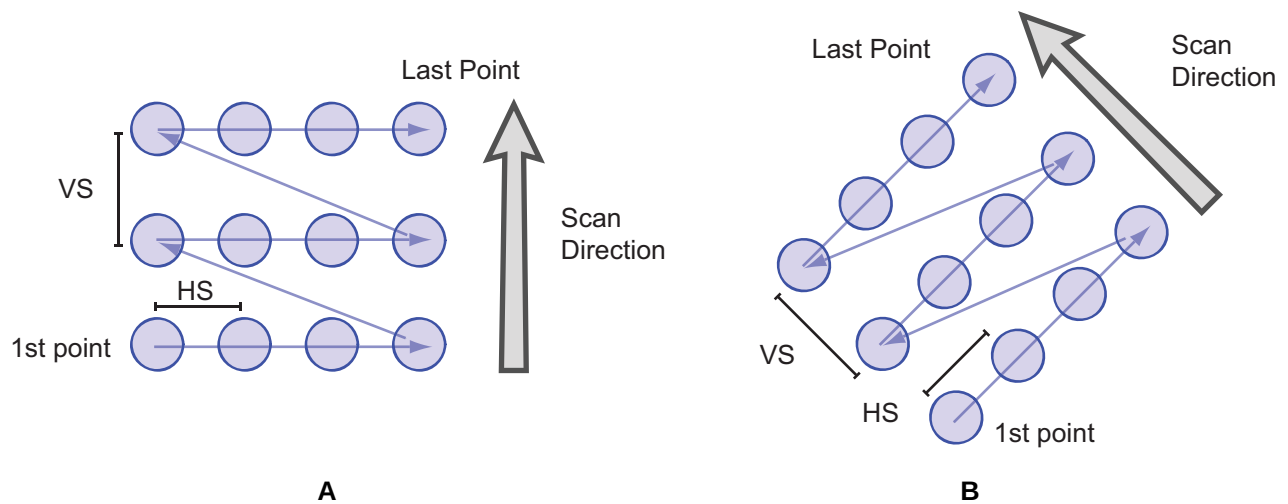


Figure 46: Schematic representation of the dwell spacing (HS : “Horizontal” Spacing, VS : “Vertical” Spacing) relative to the patterning direction. (A) Patterning direction of 90°. (B) Patterning direction of 135°.

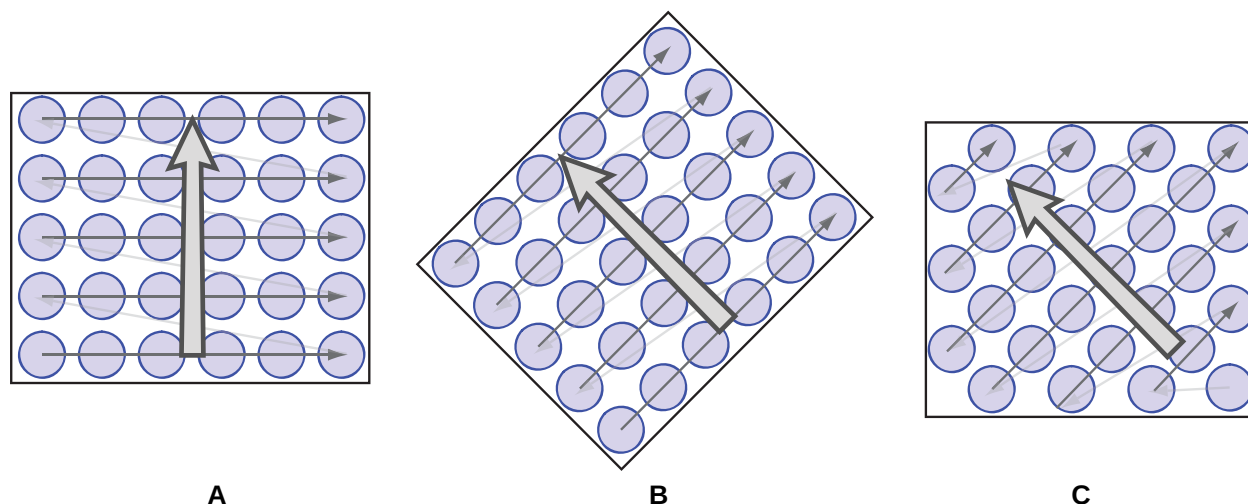


Figure 47: The power of changing the Scan Direction. (A) Illustrates the typical scan direction for a rectangular shape. Sample constraints may dictate that a sample must sometimes be patterned with the scan in a specific direction. One way to achieve this is by rotating the shape, as shown in (B). This, however, could lead to milling in unwanted regions of the sample, or the requirement to create a more complex Polygon in order to achieve the same results. In (C) a more elegant solution is shown, where the Scan Direction itself is rotated while the base rectangular shape remains unchanged, thereby satisfying both requirements

H Refresh

The horizontal refresh (H Refresh) is the period of time that the NPVE will wait before starting a new line. It is bounded by a minimum value in the Preferences Panel of the NPVE. This delay serves two purposes: beam stabilization and/or gas replenishment. The beam is blanked during this H Refresh period.

Note:

When the Scan Style is set to Serpentine, the H refresh is ignored. For more on this, refer to the Scan Style Selector below. When the Scan Style is set to Double Serpentine, both the H refresh and the V refresh are ignored.

V Refresh

Vertical refresh (V Refresh) is the time the NPVE will wait after completing the last dwell point of the shape before starting again at the first dwell point of the shape. The beam is blanked during this V Refresh period.

When a patterning Operation uses Gas processes, choosing an optimal V Refresh is important to ensure gas replenishment. The V Refresh time is enforced even when the Scan Style is set to Serpentine.

Note:

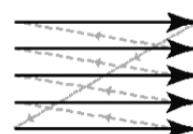
When the Scan Style is set to Double Serpentine, both the H refresh and the V refresh are ignored.

Scan Style Selector

The Scan Style selector modifies the order in which the NPVE executes the dwell points during patterning of a given shape. There are four Scan Styles available: Standard, Serpentine, Double Serpentine and Line.

Standard Scan Style

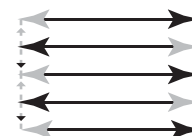
In Standard Scan style the beam always moves in the same direction relative to the scan direction: “left” to “right” and “bottom” to “top”. “Standard” Scan Style is sometimes referred to as a “television style” scanning method. Using the television analogy, in this mode the NPVE will wait for a period of time equal to the H Refresh between each “line” and V Refresh between each “frame” in the patterning of a shape.

**Serpentine Scan Style**

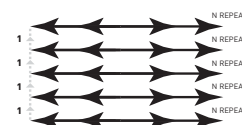
In Serpentine Scan style the direction patterning of the “line” alternates from left to right then from right to left. However, the “frame” direction is always bottom to top (Figure 48B). Serpentine Scan Style may be more efficient (time wise) than the Standard Scan Style since the H Refresh delay is ignored, however this can lead to issues that the user must be aware of when defining gas-based Operations.

**Double Serpentine Scan Style**

The double serpentine is the most efficient (time wise) way to pattern a shape: both the H Refresh and the V Refresh are ignored. The first frame is scan just like the Serpentine Scan Style. However, instead of going back to the first dwell point once the first frame is completed, it walks the same pattern but in the reverse direction. Once again, this can lead to issues that the advanced user must be aware of when defining gas-based Operations.

**Line Scan Style**

When Line Scan style is selected, the NPVE processes each line separately: it will deliver the complete dose of a single line before moving to the next line. The horizontal refresh is ignored.



Hint:

Choosing Line Scan style of “Line” is often the most efficient approach for “rough cutting” many materials in a FIB system, but the user is encouraged to experiment with various scan styles when investigating new materials.

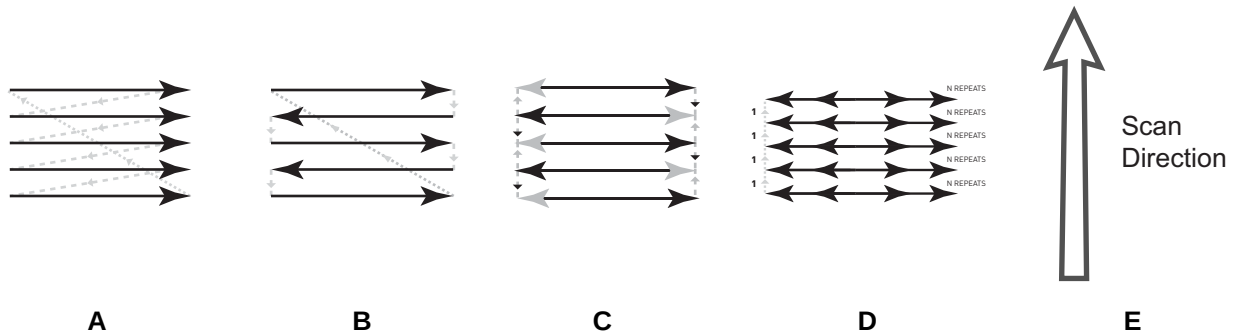
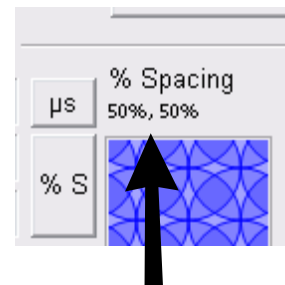


Figure 48: Different Scan Styles shown according to the Scan direction. (A). Normal Raster. (B). Serpentine Raster. (C). Double Serpentine Raster. (D) Line Raster. (E) The Scan Direction arrow indicates the ultimate direction of advancement for all scans (A) to (D), above.

Actual Spacing vs. Target Spacing

To be consistent, NPVE has the notion of Actual and Target dwell spacing. In most cases, both values will be close enough that the user will be unlikely to notice the difference between the two. However when dwell spacings become significant fractions of shape dimensions, differences between the Actual and Target dwell spacings may become apparent.

From a user interface standpoint, the Target dwell spacing is the value in the fields (H Spacing, V Spacing, P Spacing, and R Spacing) in the Advanced Parameters Panel. The Actual dwell spacing is the spacing that will be used to create the mill and its value is displayed beneath the beam spacing representation image in the Advanced Parameters Panel. An arrow points to the Actual Spacing values in the figure on the right.



To understand these concepts, consider a non-rotated rectangular shape. The need for both Actual and Target spacings arises because the NPVE algorithms must fit an integer number of dwell points on each “horizontal line” that makes up the scanned pattern of the rectangle. Since it is not guaranteed that a given Target dwell spacing will fit to an integer number of dwell point for a given length, the Target spacing needs to be adjusted (typically only slightly) in order to perform an integer number of dwell points along the horizontal line of the rectangle. The alternative is to modify the size of the shape slightly, which is often much more problematic, so this process is not performed. For maximum flexibility, the Target spacing is maintained in the user interface at the desired values, and the Actual spacing is recalculated each time the dimensions of the shape are changed.

Note:

The actual spacing takes also into account the maximum number of dwell points the NPVE system can load. The actual spacing will always be updated so that the total number of dwell points for a pattern does not go over the maximum number of points allowed.

Figure 49 illustrates that the Actual spacing is the Target spacing adjusted to fit the closest integer number of points on each line (H spacing) and the closest integer number of lines along the scan direction (V Spacing). As is shown below, the Actual spacing is selected to be as near to the Target spacing as possible, which results in a value larger or smaller than the Target value, depending on which modification results in the smallest difference between Target and Actual value.

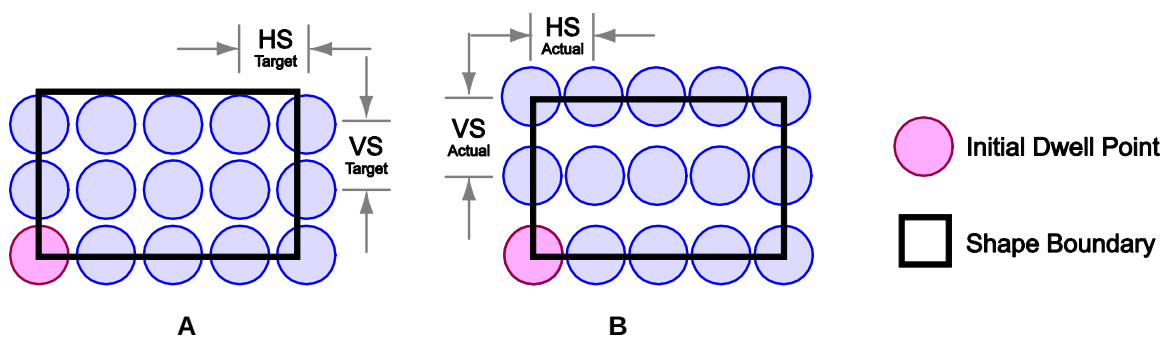


Figure 49: Target spacing compared to Actual spacing for a Rectangle. Note that for illustrative purposes both the spot size and spacings shown are much larger than would be typical for accurate NanoPatterning of this Rectangle. While this leads to an informative illustration, it overemphasises the degree to which the Target and Actual target spacings would typically vary. In this example, $VS_{\text{target}} < VS_{\text{actual}}$ and $HS_{\text{target}} > HS_{\text{actual}}$. (A) Target spacings desired for the Rectangle shown. (B) Readjusted Actual spacings to fit an integer number of dwell points with the minimum difference between Actual and Target spacings possible.

Beam Spacing Representation

The Beam Spacing Representation gives a graphical representation of the physical beam dwell spacing requested, based on the relative size of the beam spot and the spacings. As is shown in Figure 50, a degree of transparency is employed to better illustrate the effect of overlapping spot spacings. As a result, overlapping spots become progressively bluer as the spacing becomes smaller and smaller relative to the beam spot size.

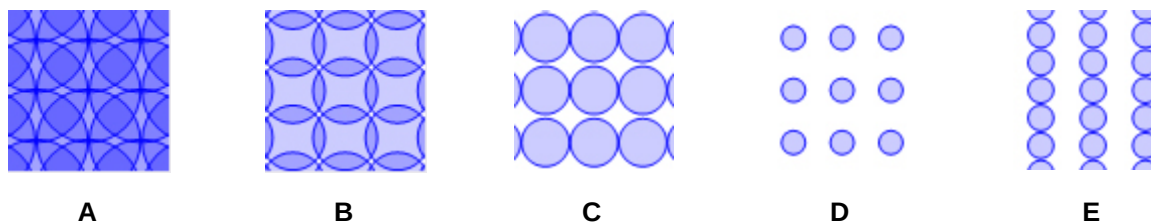


Figure 50: Different representations of the dwell spacing versus spot size as shown in the NPVE Beam Spacing Representation. (A) Spacing of 50%. (B) Spacing of 75%. (C) Spacing of 100%. (D) Spacing of 200%. (E) Spacing of 100% in the vertical direction and 200% in the horizontal direction.

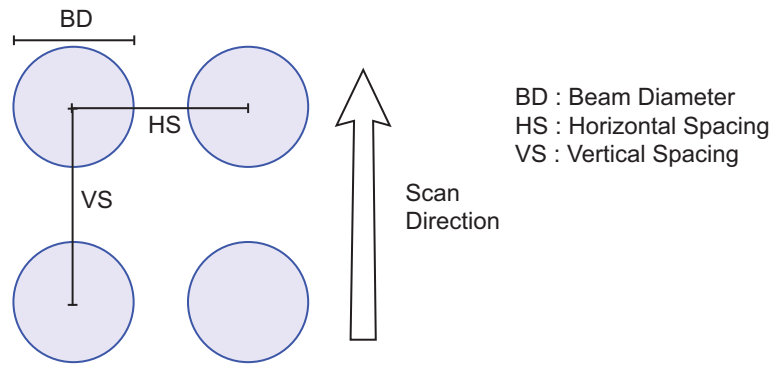


Figure 51: Beam spacing representation.

Offset

The NPVE allows the possibility to expand or contract the area of a shape by shifting its perimeter inward or outward. This functionality is called Offset and it is illustrated by Figure 52.

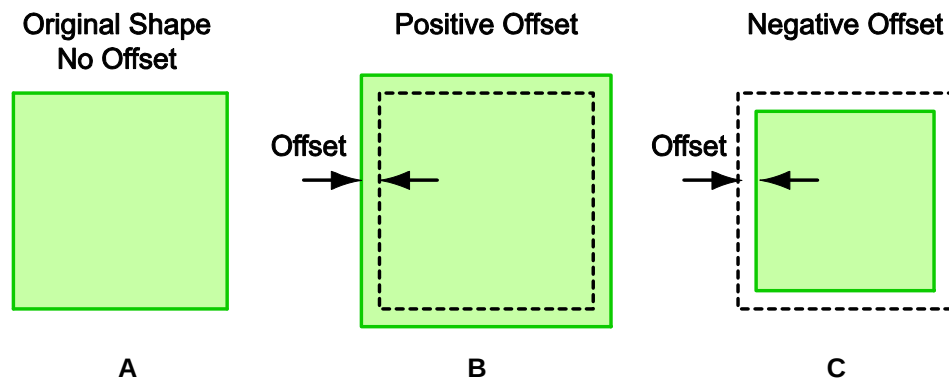


Figure 52: Representation of a shape with and without Offset. (A) The original shape. (B) The same shape with a positive offset (outset). (C) Same shape with a negative Offset (inset).

Most of the filled shapes, except the FIBText, can have an Offset applied to them.

The offset can be typed in as:

- absolute value in μm
- percentage of the spot size

To toggle between modes, press the Micrometer/Percentage Switch next to the Offset input field, as indicated at right.



Application of the Offset

As seen in Figure 49, the dwell points are distributed on the very edge of the perimeter of a shape when patterned as a filled shape. In the case of a large interaction volume several times the spot size, i.e.: in

gas assisted deposition, the outcome of the patterning may be a shape whose boundaries extend much further out of the defined patterned area due to this increased interaction volume.

This problem can be readily solved if a negative Offset (i.e. an inset) is applied to the shape. The user must measure or estimate the interaction area of the probe for a given operation and use half of this value as the negative offset. As Figure 53 shows, when applying a negative Offset, the area of a shape that is scanned is reduced by shifting its perimeter inward. If the estimate used for the Offset is correct, this will result in deposition precisely within the area defined by the original shape perimeter..

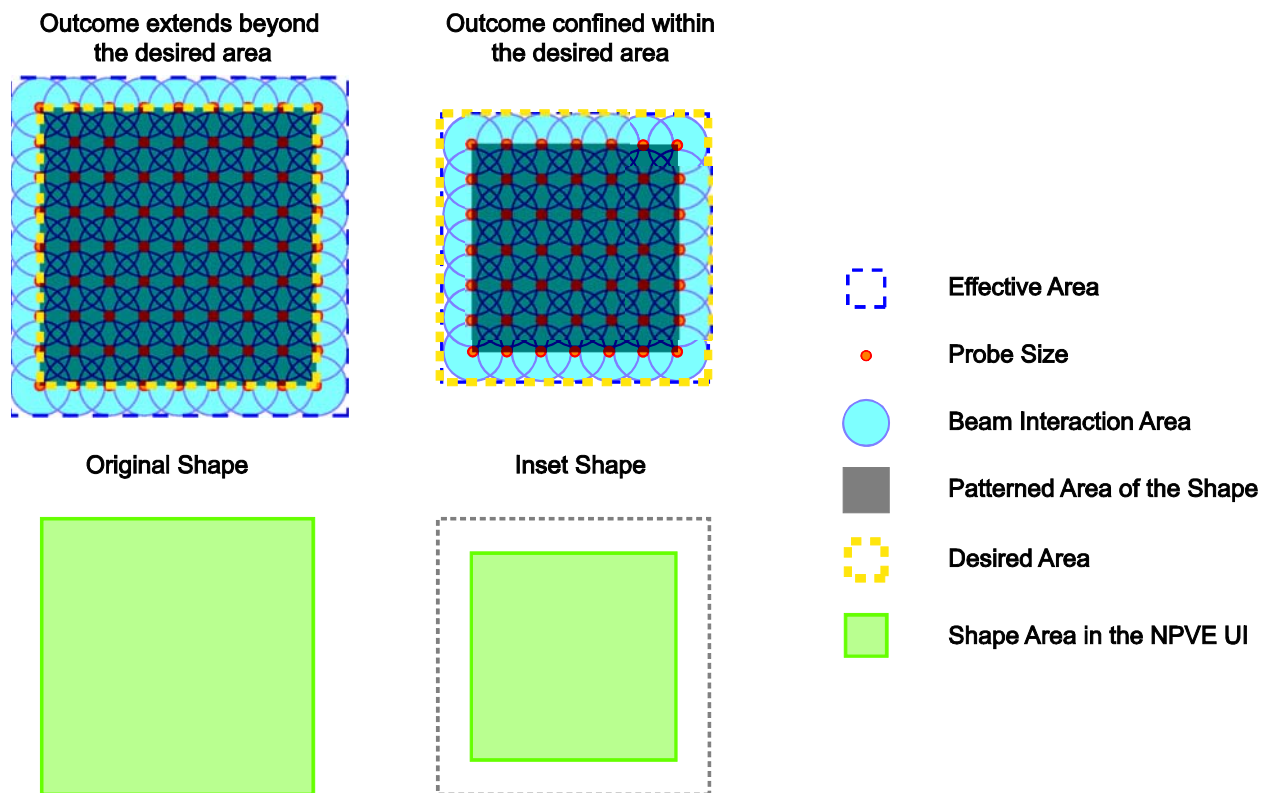


Figure 53: Application of the Offset Parameter (inset) where the patterned area of the shape needs to be reduced to take into account the beam interaction volume/area.

8. Tool Bar

A toolbar containing several buttons grouped by category can be found on the right hand side of the NanoPatterning Parameters Panel. The state (Enabled or Disabled) of the different buttons in the tool bar depends on the shapes selected, and in some cases on which Options have been purchased for a particular NPVE installation licence.



8.1. Display Properties Menu

Sometimes the graphical representation of a shape can hide valuable information in the FIB/SEM images. For this reason, it is possible to toggle different display mode for the shapes. These display modes can be toggled using the following Display Properties menu items:

- Show/Hide All
- Ghost Mode
- Show/Hide Milling Direction
- Show/Hide Shape ID
- Show/Hide Shape Controls
- Show/Hide Shape Fill

The icon of the menu will blink when some of the display properties are turned on.

Show/Hide All

It is possible to hide all the shapes and the annotations in the Viewport. This functionality can be useful to save images without having to delete any of the unwanted overlays. **Show/Hide All** can be toggled using the **F10** key.

Ghost Mode

A display feature known as “Ghost” Mode () is available to partially hide shapes blocking some portion of the FIB/SEM image. When **Ghost Mode** is enabled (Figure 54B), all the existing shapes on the screen become semi-transparent; only the activate shape is drawn normally. To ease shape selection in this mode, shapes are highlighted when the mouse moves over them (Figure 54C).

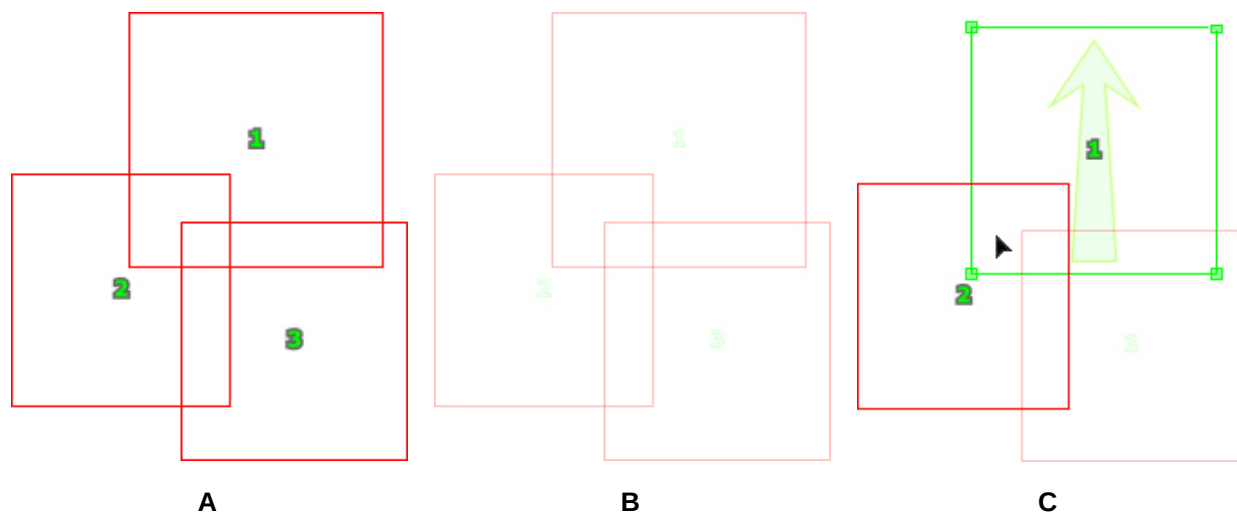


Figure 54: Three shapes showing Ghost Mode. (A) Ghost Mode disabled (normal condition). (B) Ghost Mode enabled. (C) The Ghost Mode display when one shape (with ID “1”) is activated and the mouse is over another shape (with ID “2”).

Note that during patterning, Ghost Mode is automatically activated to reveal the FIB/SEM image below.

Show/Hide Scan Direction

By default, when a shape is selected, the Scan direction is shown as a transparent arrow centred on the shape. The display of the Scan Direction can be turned off by selecting the “Hide Scan Direction” in the menu ().

Show/Hide Shape ID

If the shape is small relative to the field of view, the shape ID can be obtrusive and prevent the user from manipulating the shape as desired. The Show/Hide Shape ID menu item () toggles the display of the shape ID.

Hint:

Hiding the Shape ID when several hundreds of shapes are displayed will improve the user interaction responsiveness.

Show/Hide Shape Controls

This function hides all knobs and nodes used to edit the shapes. It also prevents them from being edited.

Show/Hide Shape Fill

The semi-transparent fill of the shape can be hidden using this function. It also hides any bitmap fills assigned to a shape.

8.2. Setting the Patterning Scan Direction



Select Edge

Once a shape is activated, clicking the **Select Edge** button allows the user to choose the edge which will be patterned last. The selected edge will be drawn red and the scan direction will be oriented toward the selected edge. Physically, the Scan Direction indicates the direction in which the beam advances in V Spacing increments after the desired series of H Spacing increments have occurred, as shown in Figure 55.

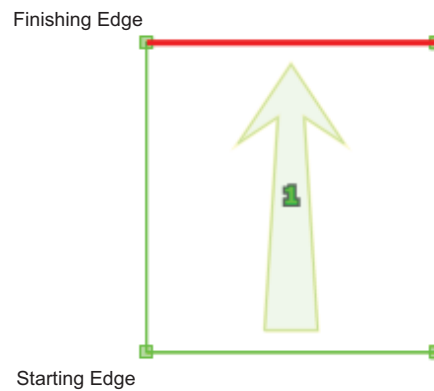


Figure 55: A Rectangle shape with its top Edge selected, and the Scan Direction arrow also displayed.

In Select Edge mode, the mouse cursor will be changed to one of the two cursors shown in Figure 56 depending on whether or not the mouse is over a valid edge.



Figure 56: Mouse cursor in edge selection mode: (A) not over a valid edge, (B) over a valid edge.

Assign Scan Direction

It is possible to set an arbitrary scan direction for patterning by using the Assign Scan Direction tool. In the Assign Scan Direction mode, the scan direction is calculated using the centre of the shape and position of the mouse on the screen: moving the mouse around the selected shape will cause the scan direction to rotate, as shown by the Scan Direction arrow. Once the desired scan direction is obtained, click anywhere to exit the Assign Scan Direction mode.

Note:

When entering the Assign Scan Direction mode, the cursor will be changed to .

Convert Shape to Polygon

Polygons offer fewer constraints on how nodes and edges can be moved around. For this reason, it is sometimes convenient to be able to convert an existing shape into a Polygon to benefit from this ease of movement. When a single shape is selected, it is possible to convert it to a Polygon by using the “Convert to Polygon” icon (.

Hint:

Converting a shape to a Polygon can allow the user to create complex shapes much more easily. An example is shown in Figure 57.

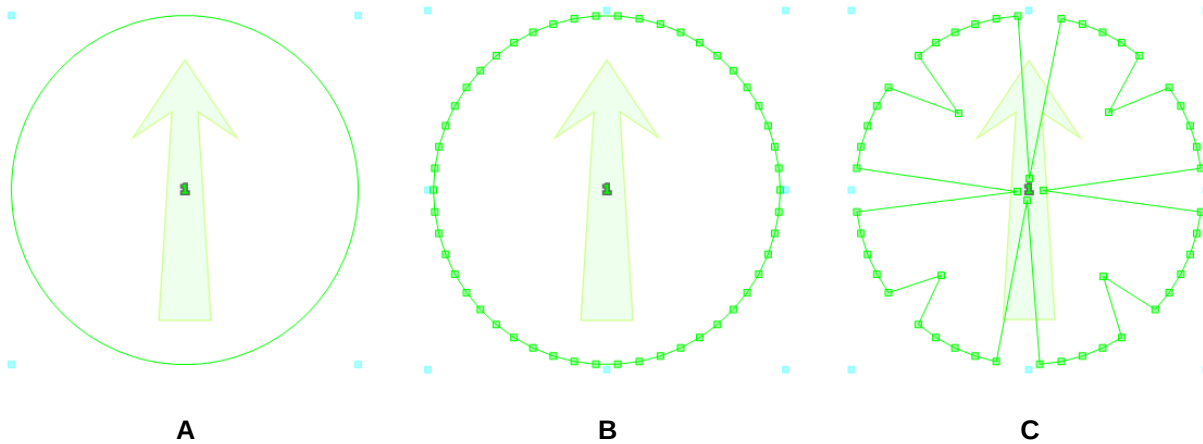


Figure 57: The process of converting a shape to Polygon using 50 nodes and moving its nodes freely. (A) Original Ellipse. (B) The Ellipse converted to Polygon with 50 nodes. (C) Eight nodes moved from their original position to create a complex shape.

Most of the non-polygon shapes are automatically converted to polygons when this button is pressed. For ellipses, however, the user must specify the number of points the final Polygon will have: the higher the number of points, the closer to the original shape the Polygon will be, but a large number of segments is not always as easy to manipulate.

Note:

Lines, Polylines and Spot shapes cannot be converted to Polygons as they are unfilled shapes.

Converting to a Polygon may cause the patterning parameters to be changed. For example, a Ring is patterned as an outlined shape: concentrically. However, converting a Ring to a Polygon will cause the shape to be milled like a regular filled shape.

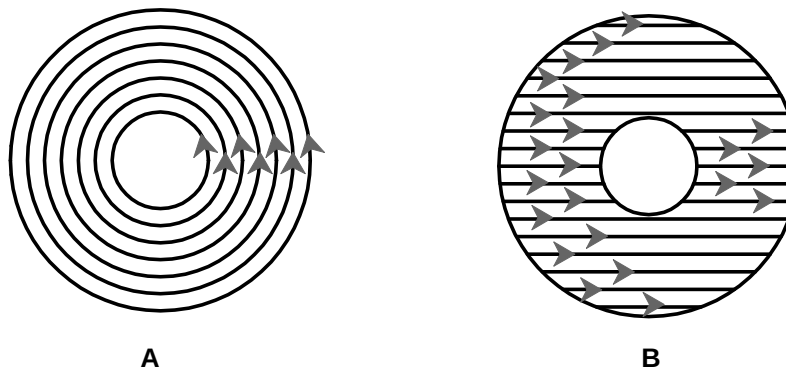


Figure 58: The difference between the way a Ring and a Ring converted to Polygon are patterned. (A) Regular Ring shows the concentric patterning style. (B) A Ring converted into a Polygon is patterned in the same manner as a filled shape.

8.3. Shape Deletion, Lock/Unlock and Selective Editing



Shape Deletion

The **Shape Deletion** button () allows the user to delete the selected shapes. It is also possible to delete the selected shapes by pressing the **DELETE** key on the keyboard.

Hint:

CTRL + DELETE will delete all the shapes, not just those that are selected.

Lock/Unlock Shape

Any shape can be locked to protect it from being moved, edited or having its patterning parameters inadvertently changed. A locked shape will be drawn with a red ID and orange edges if it is locked and deactivated.

The Lock Shape icon indicates the action that will be performed.

When a shape is locked and the mouse moves over it, the cursor changes to the move shape cursor, with an added lock to indicate moving the shape is not possible while it is locked.



A



B

Figure 59: (A) All possible Lock/Unlock shape icons. In order from left to right: Lock Shape, Unlock shape, Lock Shapes (but there are already some locked shapes in the selection), Disabled. (B) The mouse cursor when over a locked shape.

Selective Editing Mode

The NPVE can restrict the Editing Mode of all shapes when using the mouse. This can be very useful when editing small shapes or when certain operations (such as accidental node or edge movement) would cause unwanted altering of the shape itself. For more on editing shapes, refer to section 6.6 on Shape Editing. There are four different restriction options available in Selective Editing Mode which ultimately enable or disable mouse actions for the following:

- Move Shape
- Rotate Shape
- Move Edge
- Move Node

Clicking on the **Selective Editing Mode** button () will pop up a menu where the user can select what can be edited and how the shapes are editable. This icon on this button gives feedback on the state of the different Editing Modes since the icon is divided in four quadrants as represented in Figure 60.

Move Node	Move Shape
Move Edge	Rotate Shape

Figure 60: Four quadrants of the Selective Editing Mode icon.

When one of the four options is disabled, the background of the deactivated quadrant turns red.



Figure 61: Example of the different mode displayed by the Selective Editing Mode button. (A) Move Shape Off, Rotate Shape Off. (B) Move Shape Off. (C) Move Node Off, Rotate Shape Off.

8.4. Align and Distribute Shapes

The NPVE comes standard with shape alignment and distribution tools. There are a total of seven different alignment actions:

Concentric align	
Horizontal align right	
Horizontal align left	
Horizontal align centre	
Vertical align top	
Vertical align bottom	
Vertical align centre	

The NPVE offers eight different shape distribution algorithms.

Vertical distribute top	
Vertical distribute bottom	
Vertical distribute centre	
Horizontal distribute left	
Horizontal distribute right	
Horizontal distribute centre	
Space equally vertically	
Space equally horizontally	

Horizontal Distribution

Horizontal Distribution is used to arrange a group of shapes at equal distance along the X axis. The point used (left, right or mid-point of each shape) to calculate the distances depends on the type of distribution.

When distributing shapes horizontally, the leftmost shape and the rightmost shape within the selection group remain at the same position; however, the shapes located in between those two shapes will be arranged according to the desired distribution, as discussed below.

Horizontal Distribute Centre: The centre of each shape will be spaced by an equal distance from one another.

Horizontal Distribute Left: the leftmost feature on each shape will be spaced by an equal distance from one another.

Horizontal Distribute Right: the rightmost feature on each shape will be spaced by an equal distance from one another. .

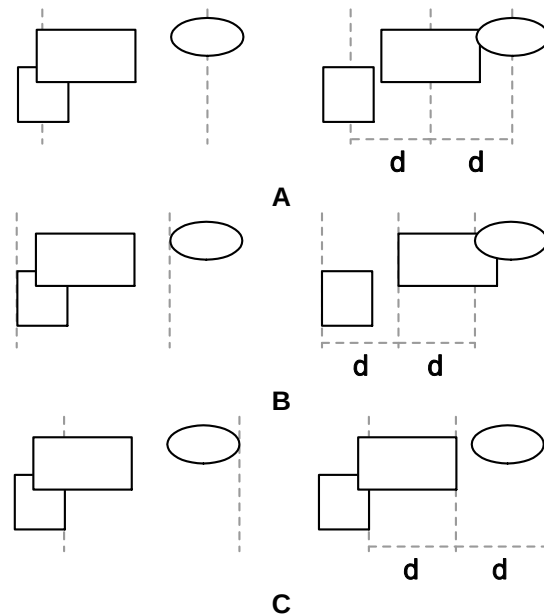


Figure 62: Three types of Horizontal Distribution. (A) Horizontal Distribute Centre - Spaces centres equally. (B) Horizontal Distribute Left - Space leftmost edges equally. (C) Horizontal Distribute Right - Space rightmost edges equally.

Note:

Horizontal distribution does not guarantee equal spacing between shapes, especially if the shapes in the set have different widths. Use the Space Equally functionalities of the NPVE for that requirement.

Vertical Distribution

Vertical Distribution works in the same manner as the Horizontal Distribution except that the distribution occurs along the Y axis. The topmost shape and the bottommost shape will not be moved.

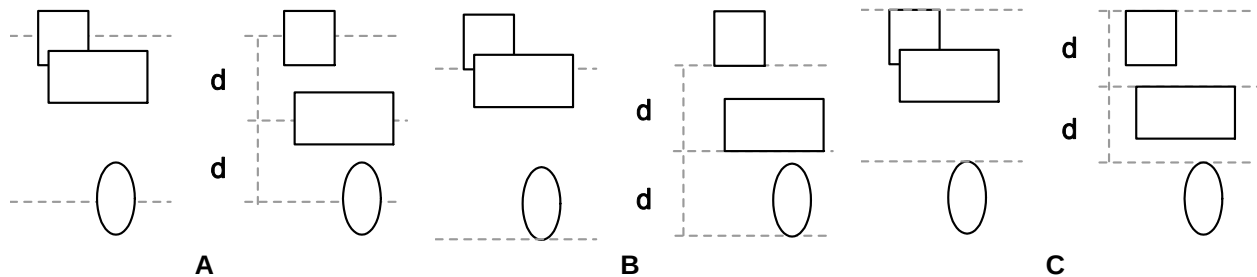


Figure 63: Representation of various Vertical Distributions. (A) Distribute the centres of the shape equally. (B) Space the bottom of each shape equally. (C) Space the top of each shape equally.

Vertical Alignment

Vertical Alignment will arrange a set of shapes using their top, bottom or middle parts along a horizontal line (constant Y value). After a vertical align centre, all shapes within the set will have the same centre Y value.

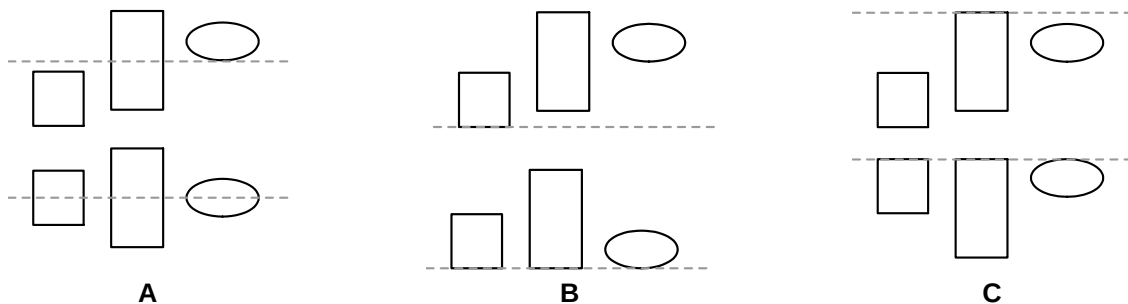


Figure 64: Set of three shapes vertically aligned. (A) Vertically aligned using the centre of the group of shapes. (B) Vertically aligned using the bottom most part of the shapes. (C) Vertically aligned using the top most part of the shapes.

Horizontal Align

Horizontal Alignment will arrange a set of shapes using their left, right or middle parts along a straight vertical line (constant X value).

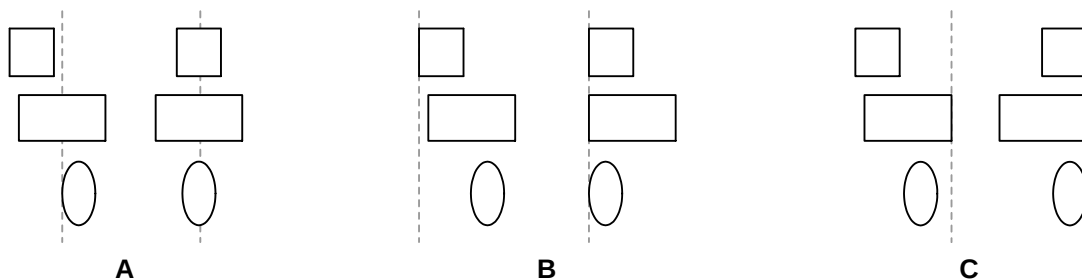


Figure 65: A set of three shapes horizontally aligned. (A) Horizontally aligned using the centre of the group of shapes. (B) Horizontally aligned on the left most part of the bounding box encompassing the 3 shapes. (C) Horizontally aligned on the right most part of the bounding box encompassing the 3 shapes.

Concentric Alignment

Concentric Alignment essentially combines the Horizontal Align Centre and Vertical Align Centre in the same function. It will position each selected shape on the centre of the bounding box encompassing all of the shapes.

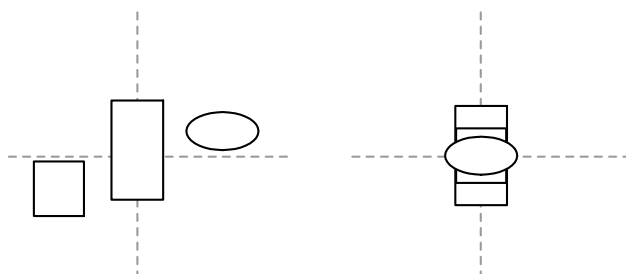


Figure 66: Concentric alignment of a set of 3 shapes.

Space Equally Horizontally and Space Equally Vertically

Once Space Equally is applied to a set of shapes, the selected shapes will all have equal distance between them.

If the selected shapes overlap, spacing them equally will result in equal overlap (in length, not area) for all the selected shapes.

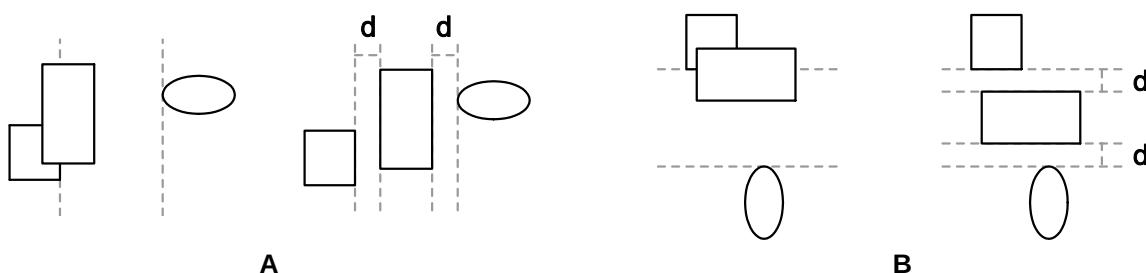


Figure 67: Sets of three shapes spaced equally (A) along the horizontal axis and (B) along the vertical axis.

8.5. Save/Load Shapes

With the NPVE it is possible to save or load any shape or set of shapes. The save (💾) and load (📁) icons are located at the very bottom of the Toolbar. The NPVE saves the files under the .nps (NanoPattern Shape) extension. All shapes and their associated parameters are saved, regardless of whether a shape is visible at the time of saving.

Parameters of an array can also be loaded using the **Load** button. Loading the Array Builder parameters using the **Load Shape** button will use the embedded shape in the Array Builder File (.ab). If the user wants to load the parameters with some other shapes, the saved file needs to be loaded within the Array Builder itself.

Node Lists and the Node List Format

Through the Load icon, it is also possible to load a Node List for patterning. A Node List is a simple text file consisting mostly of (X,Y) pairs in micrometers which represent the positions of each node in a Shape. Importing a Node List can be useful when the user wishes to generate a Shape with precise or complex node placement, or other cases such as for generating Shapes based on mathematical functions. When importing a Node List file, the NPVE interprets each line in sequence according to the format described below. Any errors in the file format cause the interpreter to stop and identify the offending line.

The Node List file format is not case-sensitive (the interpreter does not distinguish between keyword and KEYWORD). However, for clarity, keywords are given here in upper case characters.

Commas “,” hash symbols “#” and equal signs “=” have special meaning to the interpreter, and every line statement must end with an end of line character sequence (Carriage Return + Line Feed). Note that Microsoft Windows based systems will typically insert the necessary end of line sequence automatically.

The # character indicates the beginning of a comment. Any text after the # character and before the end of a line will be ignored by the importer. The # character can be used anywhere inside of a line. For example:

```
# THIS LINE IS A COMMENT
START #THE REST OF THE LINE IS A COMMENT
```

When a node list file is read, the statements are interpreted in the order in which they appear in the file.

A node list file must start with the following identification statement:

```
NPVE NODE LIST
```

The identification statement can be followed by the version of the file. This statement is optional and if ignored, the interpreter will use the latest version available.

```
VER = 1
```

The Node List can contain multiple shapes. The Shapes definition must be encapsulated between BEGIN and END statements.

```
BEGIN
...
END
```

Each Shape definition must begin with the STARTSHAPE statement and be closed with the ENDSHAPE statement.

```
STARTSHAPE
...
ENDSHAPE
```

Note that Shape IDs are created in the order the Shapes are defined in the file.

Each node consists of an (X,Y) pair. There should be one (X,Y) pair (one node) per line. X and Y values use floating point representation.

The decimal symbol separating the integer and fraction portions of the number can be a period (.) or a comma (,) and the interpreter will correctly interpret the information. Similarly the X and the Y values must be separated by a comma (,) or a semicolon (;). However if the decimal symbol is a comma, a semi-colon must be used to separate the X,Y pair value.

```
1.34, 32.49  
-102, 18.93
```

or

```
1,34; 32,49  
-102; 18,93
```

or

```
1.34; 32.49  
-102; 18.93
```

When multiple values are required on a single line (such as an (X, Y) pair of coordinates shown above) with commas “,” or semicolons “;” interpreted as separating the values, this format is commonly known as the Comma Separated Value or CSV format that is output by many programs including Spreadsheets such as Microsoft Excel – please refer to the manual for your spreadsheet program to determine how to export a CSV file.

Note also that multiple trailing commas or semicolons are ignored, i.e.

```
1.34, 32.49,,,,,
```

Is treated by the interpreter as:

```
1.34, 32.49
```

Example of a Node List File

In the following example, a sine wave is created using an Excel spreadsheet and then exported as a CSV file. By choosing appropriate column headers and layout that incorporated the necessary file header statements as text in the spreadsheet ("NPVE Node List", "VER = 1", "BEGIN", "STARTSHAPE", etc.) and appropriate use of the "#" character to separate columns not required to be input, no editing of the CSV file is required after export.

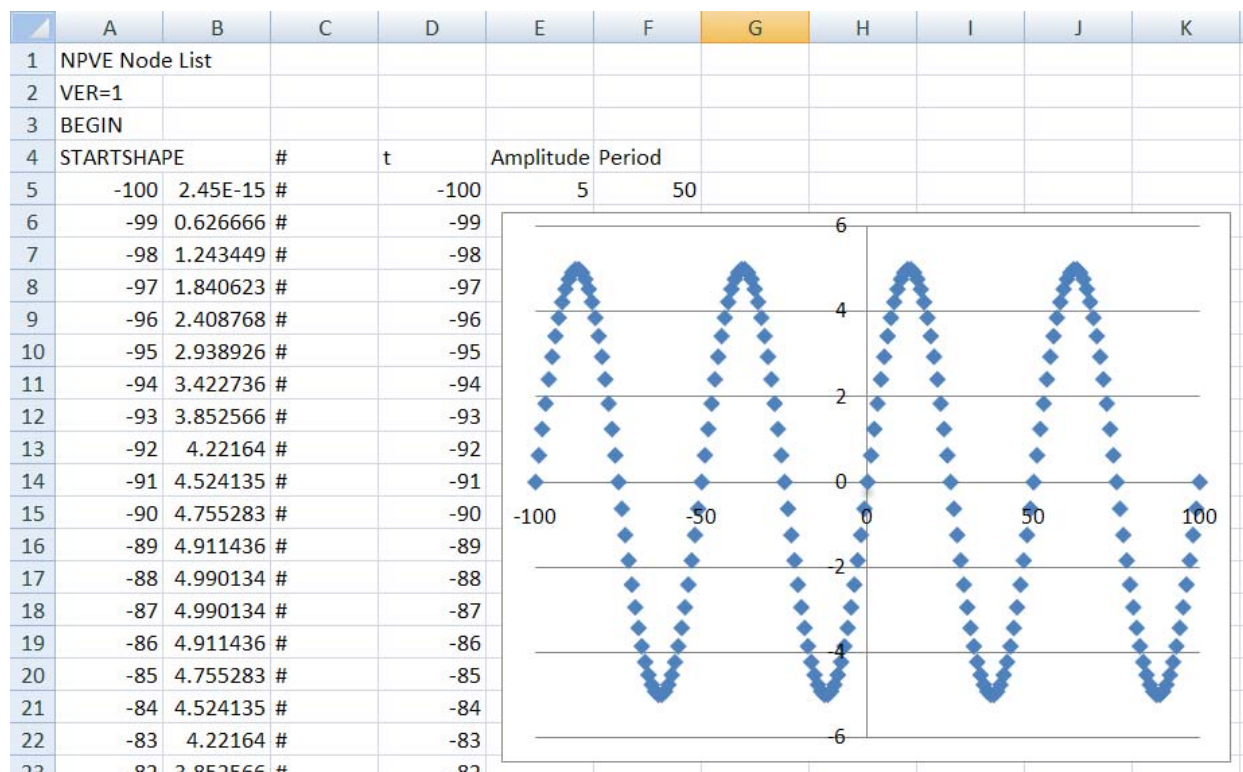


Figure 68 Screenshot of the Excel Worksheet used to create an NPVE Node List.

As shown here, the first four rows of the spreadsheet are used to enter the required header information. Below these rows, columns A and B contain the (X, Y) location pairs required to be imported by the interpreter. Column C contains a # character, making the interpreter treat the rest of the line (row) as a comment. Thus in the spreadsheet the (X, Y) pair can be calculated as a function of other parameters (in this example: Time "t", the Amplitude and Period) which are ignored by the interpreter when inputting the Node List file.

Note that when exporting as a CSV file, Excel will add trailing commas to make sure that each row has the same number of cells. Those trailing commas will be ignored by the NPVE Node List interpreter. By default, the file will be saved with a ".CSV" file extension.

The exported file will be as follows:

```
NPVE NODE LIST,,,,,
VER=1,,,,,
BEGIN,,,,,
STARTSHAPE,,#,T,AMPLITUDE,PERIOD
-100,2.4503E-15,#,-100,5,50
-99,0.626666168,#,-99,,
-98,1.243449436,#,-98,,
-97,1.840622763,#,-97,,
-96,2.408768371,#,-96,,
-95,2.938926261,#,-95,,
-94,3.42273553,#,-94,,
.
.
.
148,-1.243449436,#,148,,
149,-0.626666168,#,149,,
150,-3.67545E-15,#,150,,
ENDSHAPE,,,,,
END,,,,,
```

To import this Node List file, choose the **Load Shape** button on the Toolbar and select Files of Type: Node List, "*.csv, *.txt", then select and Open the appropriate file. If the file has no errors, the interpreter will indicate the file is valid, and pressing OK will create the shape(s). If the interpreter detects an error, the valid lines will be displayed up to and including the line with an error, and the user will be informed of the error at the appropriate line number.

Node List Codes

Code	Explanation	Mandatory
NPVE NODE LIST	File header.	Yes
VER=X	Version indicators.	Yes
BEGIN	Indicates the beginning of the list of shapes.	Yes
STARTSHAPE	Indicates the beginning of the list of points that will form the shape. Several shapes can be defined in a single file. Make sure ENDSHAPE is inserted before adding another STARTSHAPE	Yes
X,Y	Pair of coordinates in μm of a node. Must be included between a STARTSHAPE and ENDSHAPE statement.	YES
ENDSHAPE	Indicates the end of the nodes of a single shape.	YES
END	Indicates the end of the file.	YES
CLOSED = TRUE/FALSE	Tells NPVE if the current shape will be created as a Polygon or as a Polyline. Default is Polyline.	NO

9. Additional Features Available in the NPVE Advanced System

The Standard NPVE system comes with a rich set of features and tools, enabling a wide range of NanoPatterning procedures. However, several optional modules are available in the “Advanced” version of the NPVE that extend the capabilities and solutions provided by the NPVE for specific requirements. These optional modules are discussed in the following sections.

9.1. Outline Patterning ☒ADV

Normally, when a shape is patterned, the area defined “inside” this shape is filled using the desired patterning parameters. There are many applications, however, where it is desirable to perform patterning following the vector path of the shape itself. For these applications, the NPVE Advanced system can pattern the outline of a shape and leave the interior of the shape intact. Figure 69 shows the difference between the original shape and its outline version. Outlining a shape is the equivalent of taking a “pen” of a certain width and drawing the contour of the shape with this pen.

The width of the Outline to be applied can be entered manually using the Outline Thickness field in the Shape Geometry subpanel. The Advanced Parameters Panel offers more parameters to control how Outlines are patterned (see Figure 71).

By adjusting these parameters (discussed below), it is also possible to completely fill a shape in a scan manner very different to the standard Scan Styles otherwise available for non-Outlined shapes.

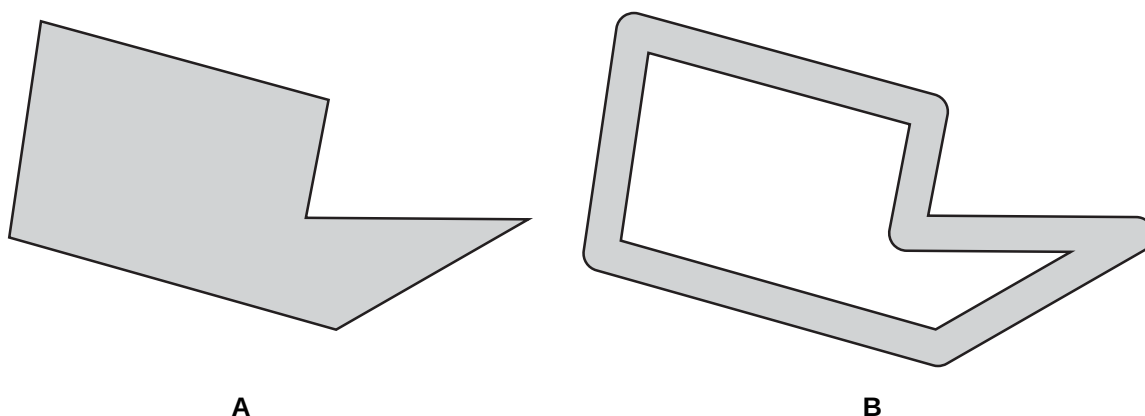


Figure 69: The difference between the same shape is shown with: (A) Outline turned off (regular fill). (B) Outline turned on with rounded node style.

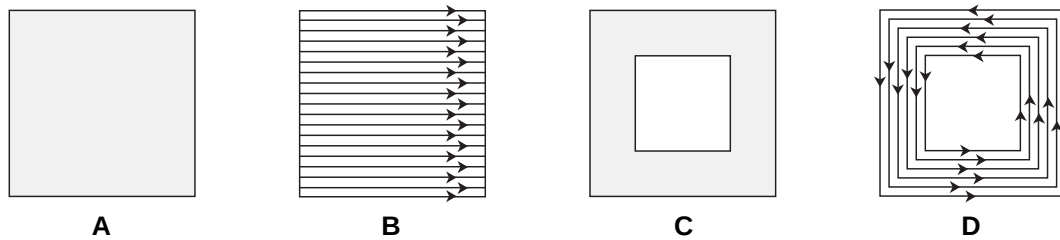


Figure 70: The difference between a Square patterned as a filled shape and patterned as an outline shape. (A). The Square filled. (B)The patterning lines of the filled Square. (C) The Square outline. (D) The patterning lines of the outline Square.

Advanced Parameters Panel for Outlined Shapes

The Advanced Parameters of the NanoPatterning Parameters Panel will be different if the shape selected is outlined. Figure 71 shows the Advanced Parameters Panel for an Outlined shape.

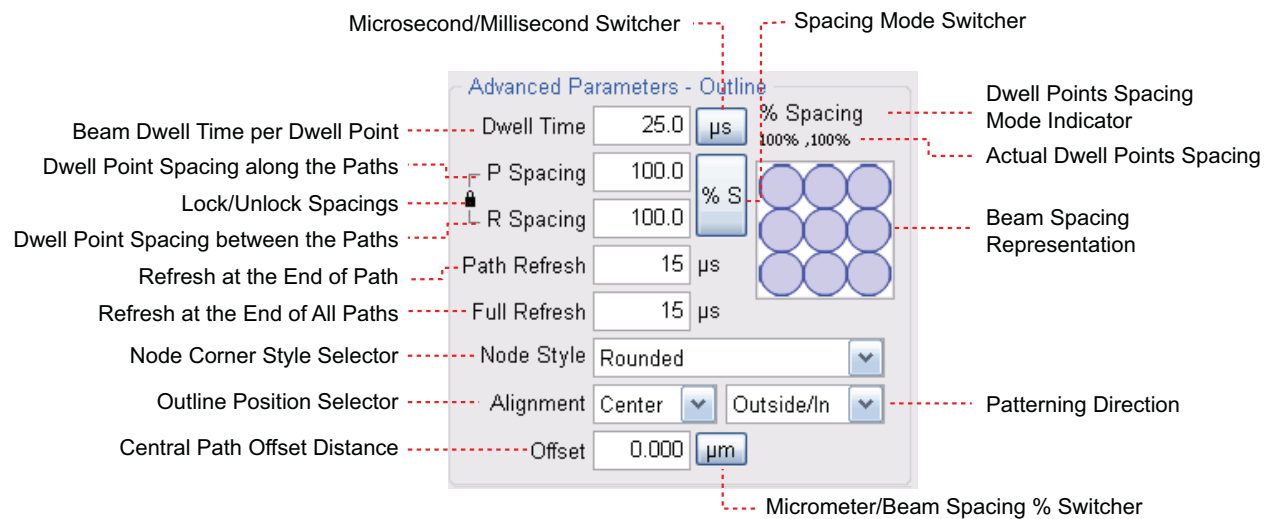


Figure 71: Advanced Parameters Panel when editing an Outline shape.

Spacing

Path spacing (P spacing) and Radius spacing (R Spacing) are used when the shape is outlined. When the shape is outlined, the Scan Direction is no longer used; rather, patterning follows the direction of the outlined path in a concentric manner.

P Spacing

Path Spacing (P Spacing) is the spacing of the dwell points along these outlined paths. It is represented in Figure 72 as DP.

R Spacing

Radius spacing (R spacing) is the distance between each Outline path. Each path of the Outline will have equal concentric space.

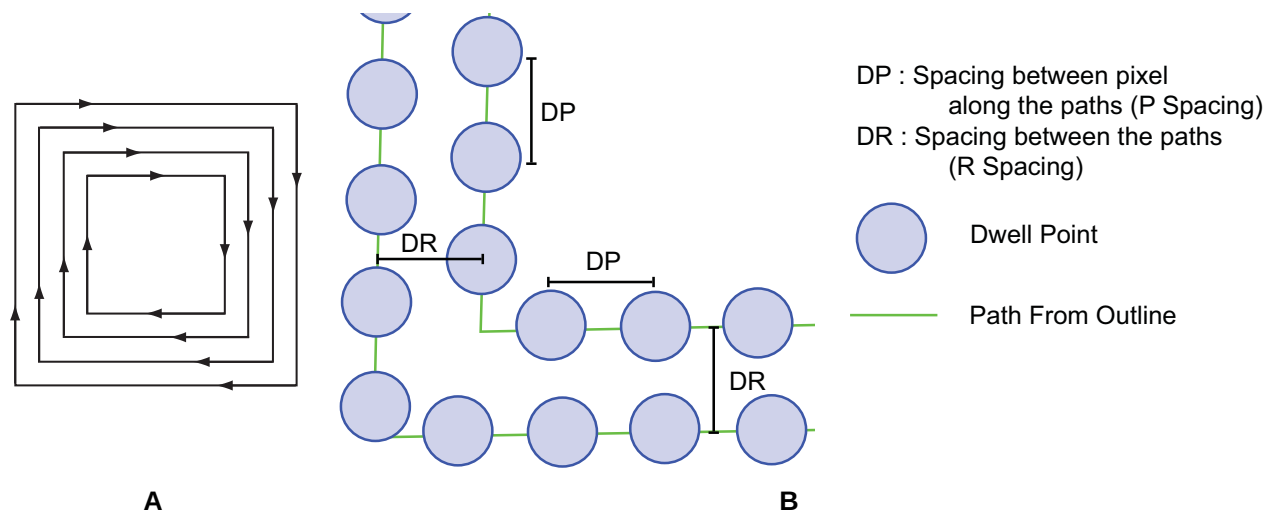


Figure 72: Explanation of Outline patterning. As shown in this schematic for a Square that has been outlined, the pattering direction, represented by the arrows, is constantly changing.

In Outline mode, proprietary algorithms ensure the dwell point distribution is optimized in order to maintain a constant spacing over the entire shape in order to maintain the desired DP and DR despite any crossing edges and sharp geometries.

P Refresh

The path refresh (P Refresh) is equivalent to the horizontal refresh (H refresh): Once one path of the outlined paths is finished patterning, the NPVE will wait for the P Refresh before starting the mill of the next path. The beam is blanked during this time.

F Refresh

The same analogy can be made with the full refresh (F Refresh) and the vertical refresh (V Refresh): once the last path of the outline paths is patterned, the beam is blanked and will ramp during the F Refresh from the last point of the last path to the first point of the first path and wait for the F Refresh time period before commencing the next patterning pass.

Node Style Selector

In Outline mode, the node style tells the NPVE how to render the corner of each node. There are two different types of nodes:

- Rounded
- Mitered

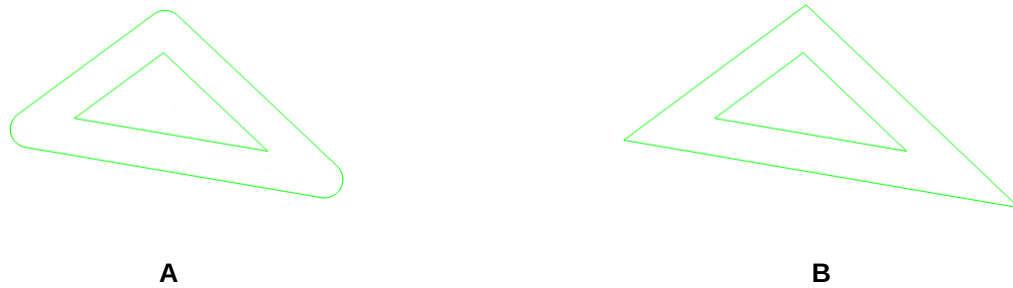


Figure 73: Node styles for Outline patterning. (A) Rounded corners. (B) Mitered corners.

Outline Alignment Selector

By default, the Outline is Centred on the edges of the shape. However, it is possible to Inset the Outline fully to the interior of the shape or Outset the Outline fully to the exterior of the shape. The three different possibilities are demonstrated in Figure 74. An inset shape can be particularly useful if it is desired to completely or partially fill the shape using a scan style that follows the vector paths of the shape edges.

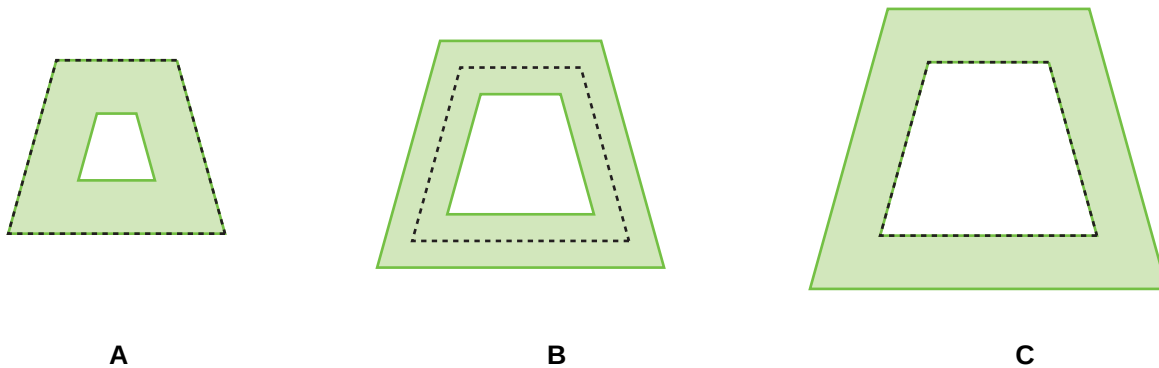


Figure 74: All three possible Outline alignments. (A) Inset, (B) Centred, the default state, (C) Outset.

Outline Offset

It is possible to Offset the position of the Outline relative to the edges of the shape. Entering a non-zero value in the Offset field of the Advanced Parameters Panel will Offset the Outline to the interior or exterior of the shape depending on the sign of the Offset entered.

A negative Offset moves the Outline of the shape towards the interior of the shape. A positive Offset moves the Outline towards the exterior of the shape.

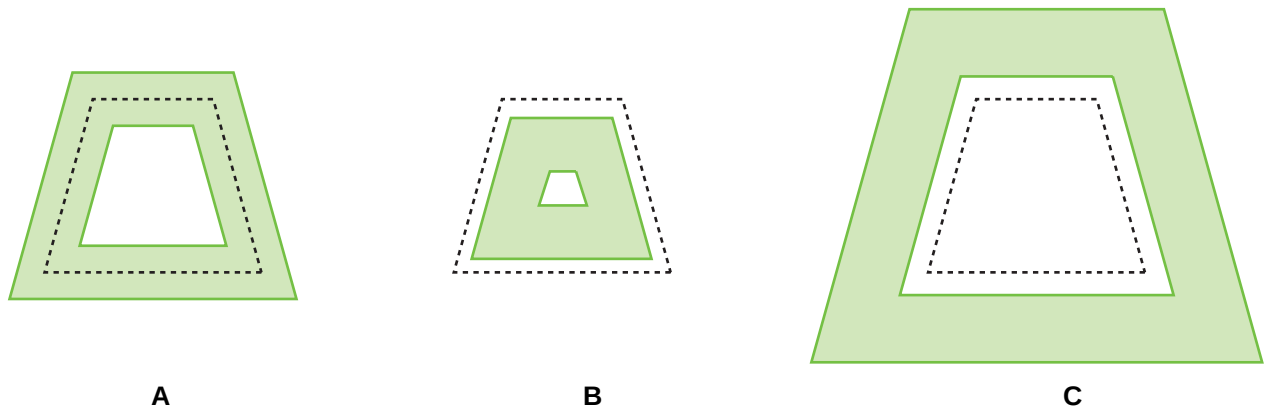


Figure 75: An Outlined shape with different values of Offset. (A) No Offset is applied. (B) A negative Offset moves the outline of the shape towards the interior of the original shape. (C) A positive offset moves the Outline towards the exterior of the original shape.

9.2. Void Shapes ☒ADV

Most shapes, with the exception of Lines, Polylines and Spots, can be configured as a void using the **Void** checkbox on the Shape Geometry subpanel (Figure 19). A Void shape is a region that is excluded from all patterning. It can be used to protect a specific area from accidental patterning or to simplify the creation of a more complex shape.

A Void shape is displayed in blue with a diagonal line fill pattern as shown in Figure 76. As with regular shapes, Void shapes show transparent backgrounds to ease alignment to features in the microscope image. Void shapes are always active.

When Voids are present, shape boundaries are updated in real time as the user moves the shapes or the Voids around in the user interface. This dynamic boundary updating provides a visual representation of the intended end result.

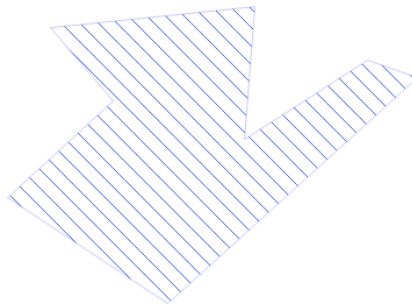


Figure 76: Void shapes appear with a blue diagonal fill pattern.

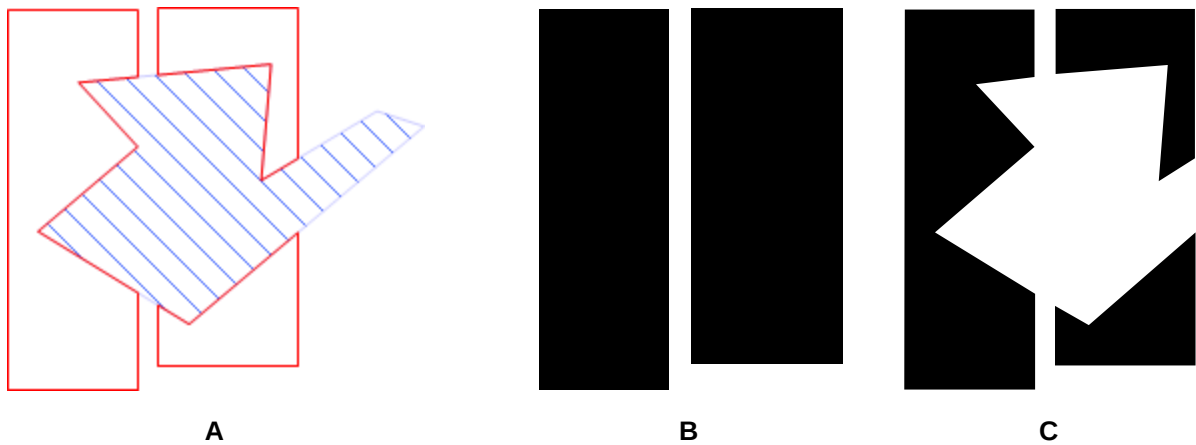


Figure 77: (A) Two rectangular shapes (red) are modified by the presence of a Void shape (blue with diagonal fill). (B) The patterned area of the original rectangular shapes if the Void is not Active. (C) The resulting patterned area when the Void is Active.

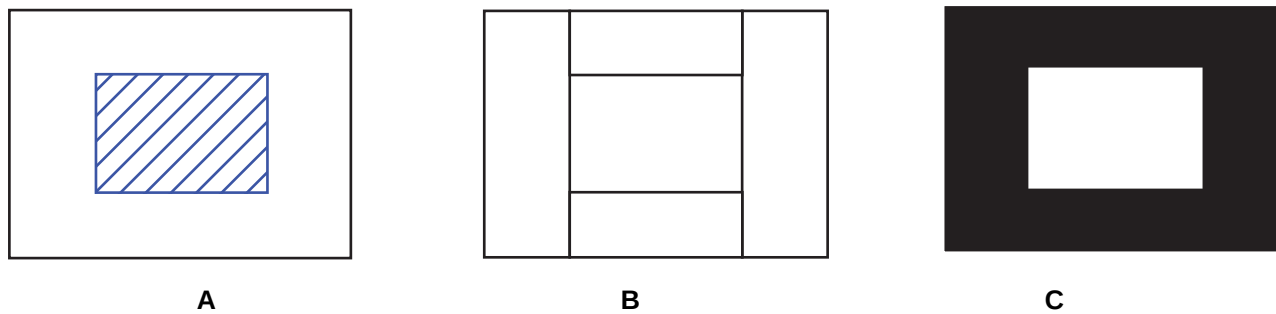


Figure 78: The patterned area shown at right can be created either using four Rectangles, as shown at centre, or a single Rectangle and a Void, as shown at left. As the complexity of the regions to be patterned and to be excluded from patterning increases, the usefulness of Voids also increases.

Note:

Voided shapes are not applied to Outline shapes. Even if a Void covers the complete area of an Outlined shape, the Outline patterning will still proceed as if the Void were not there.

Deflection Lists and Spots are not affected by any Void shape.

9.3. Set Operations ☒ADV

It is possible to combine shapes in different fashions using Set Operations. This results in the creation of new shapes, potentially with different nodes. The NPVE offers a number of different Set Operations:

- Union
- Difference
- Intersection
- Exclusive Or (XOR)
- Clip (Overlap resolver)

Set Operations are applied to all Selected shapes. Set Operations can be accessed using the **Set Operations** button ().

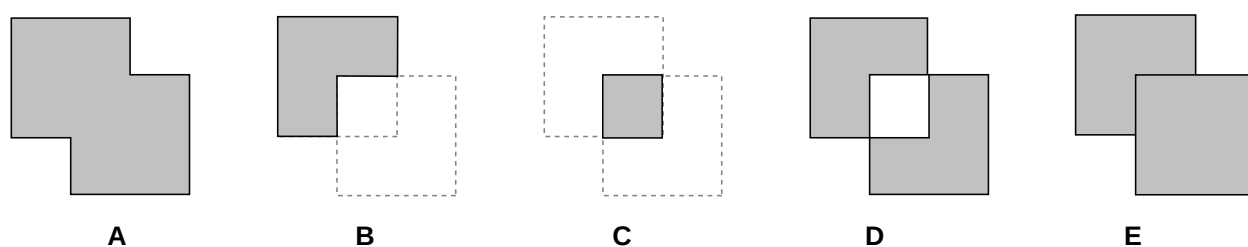


Figure 79: All available Set Operations shown using 2 squares. (A) Union (B) Difference (C) Intersection (D) XOR and (E) Clip (Overlap Resolver).

Set Operations can be used to create shapes that would otherwise be much more complicated to draw (Figure 80). Set Operations can also prevent undesirable "double dosing" for areas where shapes inadvertently overlap (Figure 81).

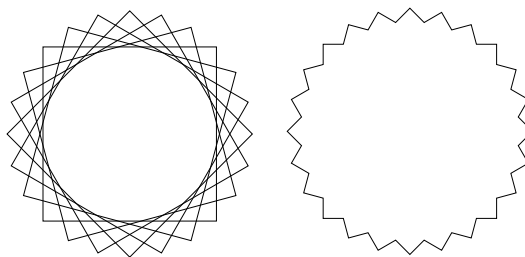


Figure 80: A complex shape (right) obtained from the Union of multiple simple shapes (left).

Note:

Using the set operations will first convert the selected shapes into a Polygon. For example, outline shapes will preserve their geometric aspect, but will not be patterned as an outline.

For some Set Operations, the order of operation does not matter. This is the case for Union, Intersection and XOR Set Operations. However for the Difference Operation and the Clip Set Operation, the order in which the user applies the Difference is important. The NPVE uses the ID of the shape to order the Set Operation. It orders the shapes in ascending order from lower ID to higher ID.

In the case of the Difference Operation, it will iteratively perform the Difference between the first two shapes and then apply the results of that first difference to the third shape, etc.

$$S_{diff} = (((S_i - S_{i+1}) - S_{i+2}) - S_{i+3}) - \dots - S_{i+n}$$

For the Clip Set Operation, the algorithm will clip the shapes with the higher ID using the shapes with lower ID. In other words, lower ID shapes have priority over the higher ID shapes. The shape with the lowest ID will never be clipped.

For this reason, it is important to number the shape IDs in a way that will produce the desired result.

Eliminating Double Dosed Area

It is sometimes difficult to position two closely adjacent shapes in a manner to eliminate the possible overlap between both areas. Set Operations are a powerful tool to prevent double-dosing when two shapes are to be patterned in a continuous fashion.

Using Union

Before performing NanoPatterning, creating the Union of two adjacent shapes will create one shape out of the two original shapes and consequently eliminate the possible overlap between the two shapes. Figure 81 illustrates this useful feature.

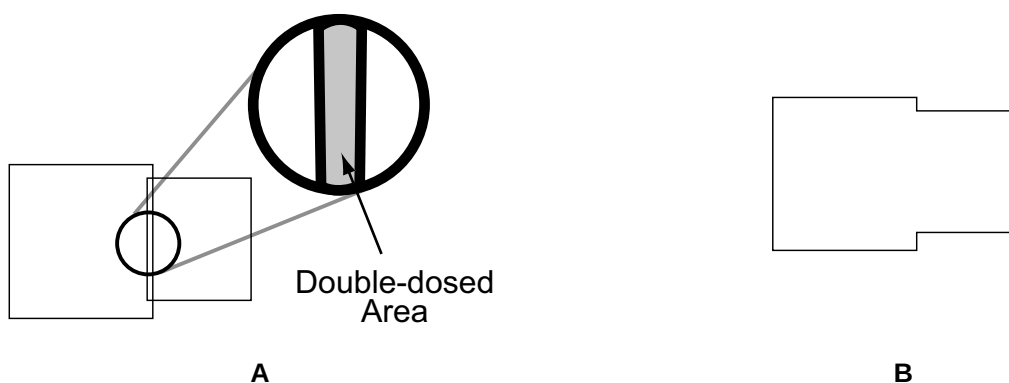


Figure 81: One benefit of the Union Set Operation: preventing double dosing of overlapping areas. (A) Two shapes showing an overlapping region and consequently double dosing in the overlapping region. (B) The same shapes as in A after applying the Union Set Operation to eliminate the overlapping area.

Using Clip

Using the Union Operation combines the selected shapes into one single shape. This is appropriate if the user wants to use the same Operation/Patterning Parameters for both shapes, but can be problematic if the user wants to maintain the Parameters of each shape. If this is a concern, the Clip Operation should be used. The Clip Operation will keep every single shape as a single entity.

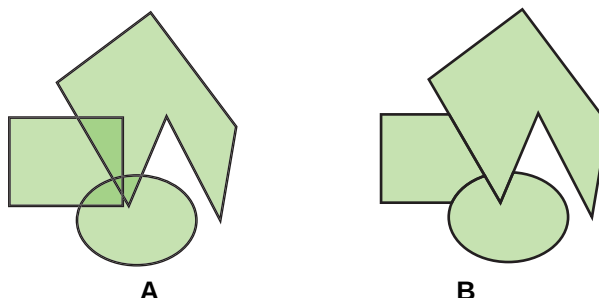


Figure 82: Resolving overlap using the Clip Operation. The overlapping area is shown as darker green. (A) Overlapping shapes. (B) Resulting shapes after using the Clip Operation. Note that the resulting shapes are still editable separately.

9.4. Greyscale Bitmap 3D NanoPatterning ☒ADV

The NPVE is capable of NanoPatterning using greyscale bitmaps (colour bitmaps, which are automatically converted), which can be used to produce three dimensional structures. The selected bitmap is loaded, then resized and clipped to the size of the shape according to user preferences.

Greyscale Bitmap 3D NanoPatterning can be accessed by clicking the  icon in the toolbar.

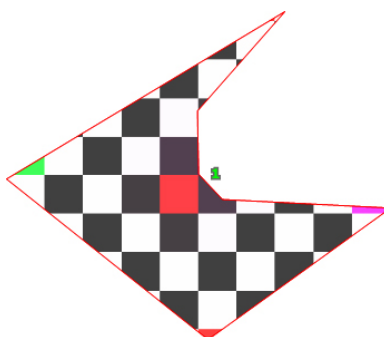


Figure 83: A bitmap applied to a shape with clipping and resizing according to user selected Stretch Modes.

At a basic level, assigning a bitmap to a shape is equivalent to varying the dose pixel by pixel using the greyscale intensity as a weight, although which patterning parameters and algorithms are implemented to deliver the dose can have significant impact on the final result.

A colour bitmap assigned to the shape is converted to greyscale using the human eye lumen response. The dose is then distributed over the shape as a function of the brightness of the pixels and the Advanced

Parameters (dwell, spacing, etc.) selected. The brightest (white) pixels will have the full dose (100%) delivered and the darkest (black) pixels will have no dose delivered. Black and white bitmaps are treated as one might expect – either 0% or 100% on a pixel by pixel basis.



Figure 84: Distribution of the dose according to the brightness. Pure black pixels will not receive any dose. Pure white pixel will receive 100% of the dose.

Once the bitmap to pattern has been chosen, the dialog box shown in Figure 85 appears, allowing the user to select several options.

Note:

To optimize display performance of shapes with an applied bitmap, the bitmap used for display is re-sampled to 512 by 512 pixels. Nevertheless, the original full resolution of the bitmap will be used when NanoPatterning.

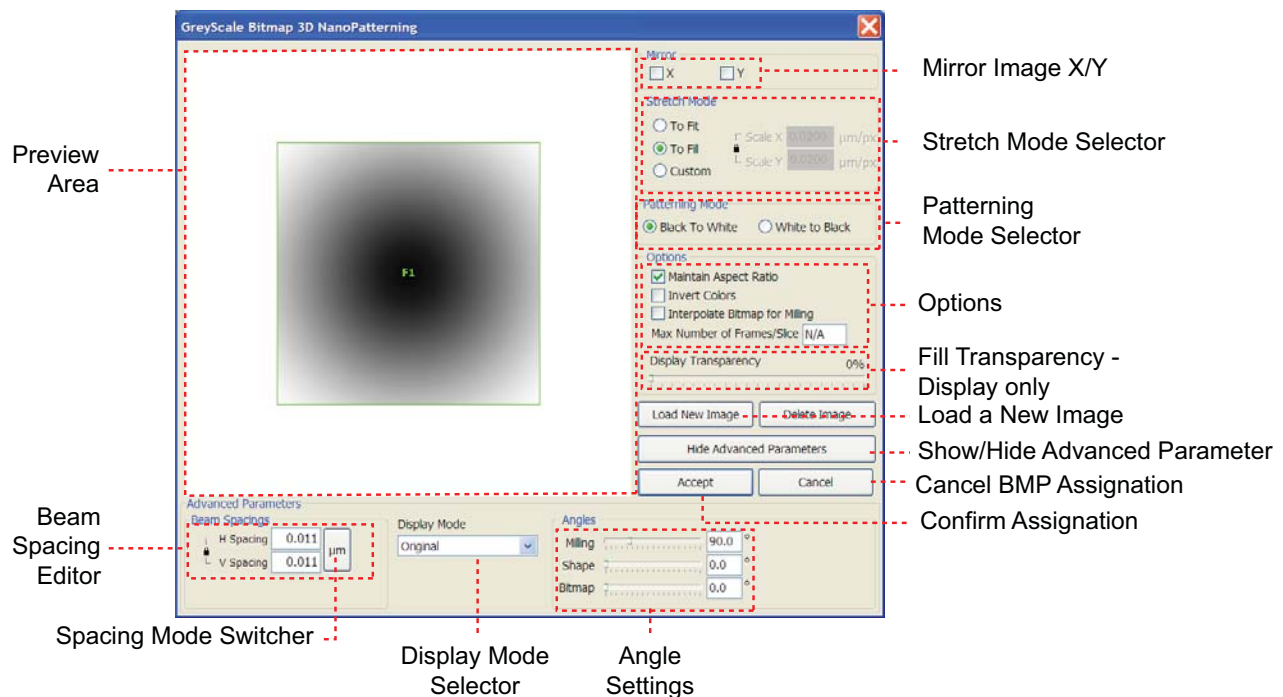


Figure 85: Greyscale Bitmap 3D NanoPatterning dialog box.

The following options are available to customize how the bitmap is processed:

Mirror

Mirror (flips) the bitmap in x or in y.

Stretch Mode

Stretch mode determines how the bitmap should be fit to the shape. Stretch To Fit will use the smallest ratio between the bitmap dimensions and the dimensions of the shape to be milled as the scaling factor. This ensures the bitmap fully fits within the shape (Figure 86A). Stretch To Fill will maintain the aspect ratio of the bitmap while stretching it to cover the entire shape (Figure 86B).

If the option “Maintain Aspect Ratio” is enabled, the scale factor for the width and height will be held equal. However, if “Maintain Aspect Ratio” is not enabled, the bitmap will be stretched along its X and Y axes independently to cover the entire shape. When “Maintain Aspect Ratio” is not enabled, stretching “To Fill” and “To Fit” yield the same result (Figure 86C).



Figure 86: Stretch modes demonstrated. (A) Stretch To Fit with Maintain Aspect Ratio enabled. (B) Stretch to Fill with Maintain Aspect Ratio enabled (note the grey portion outside the shape will not be patterned). (C) Stretch To Fit or Stretch To Fill with aspect ratio turned off.

Patterning Mode

The Patterning Mode determines whether patterning will proceed from black to white, lowest to highest dose, or from white to black, highest to lowest dose. When NanoPatterning with a FIB system in a pure milling environment, the “White to Black” option is frequently more useful in order to have sharper, shallower pixels since shallower pixels will be milled at the end and consequently will not suffer from any redeposition that could be induced from the milling of the deeper areas.

Invert Colour

It is possible to invert the colours of the bitmap. This is equivalent to inverting the dose distribution over the greyscales.

Interpolate Bitmap for Patterning

Most of the time, the bitmap used for patterning will not have the same number of pixels in width and height as the shape will have “horizontal” and “vertical” dwell points in the patterning space (the number of dwell points per row and the number of dwell points per column in a patterned Rectangle, for example). For this reason, it is important to specify how the software will resample the image when the shape is patterned.

When interpolation is enabled, the image may be re-sampled, resulting in transitions that are smoother. However, if the interpolation is disabled, transitions may be sharper and under certain

combinations of patterning parameters, the results may appear “blocky”. The optimal settings will need to be determined on a case by case basis depending on the application. Final results are highly dependent on both the initial bitmap resolution and the patterning parameters chosen for the application.

Figure 87 shows a 10 x 10 pixel bitmap with interpolation turned off and on. Figure A is much sharper than Figure B. In general it is good practice to ensure the bitmap chosen has a sufficiently high number of pixels for the nanopatterning task at hand, regardless of whether Interpolation is enabled or not.

Hint:

It is suggested to use Interpolation for 3D NanoPatterning, however, interpolation should in many cases, be turned off for less complex “binary” or “mask” patterning.

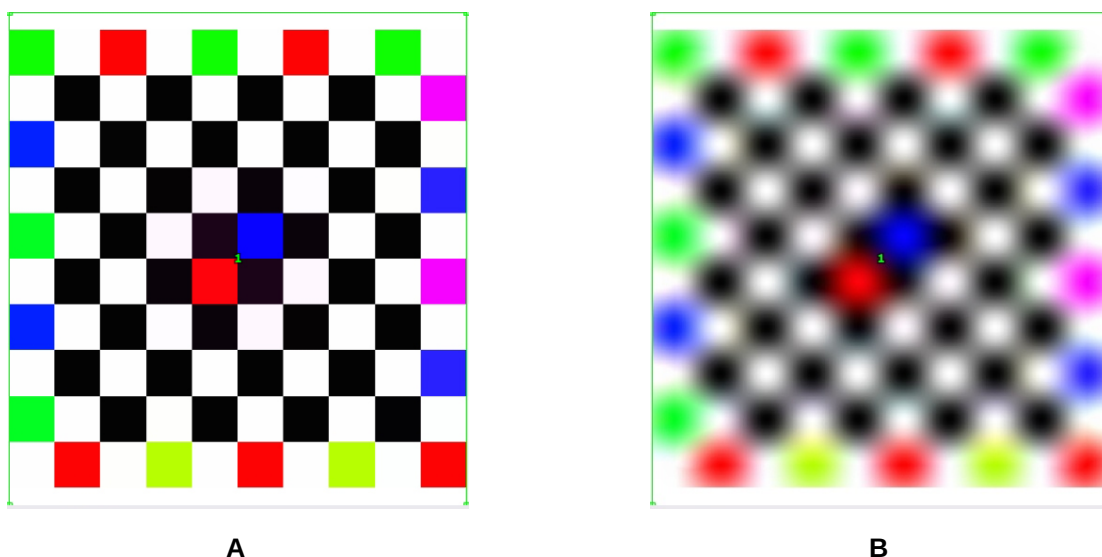


Figure 87: A 10 x 10 pixel bitmap used for patterning. (A) Interpolation disabled (B) Interpolation enabled.

Transparency

Transparency is applied strictly for display purposes and allows the user to align the shape in the Visualization Panel more readily to features in the microscope image. By default, a Transparency of 25% is applied. It is possible to increase or decrease this value by using the Transparency slider.

Note:

Transparency is applied strictly for display purposes and does not have any effect on the patterning of the bitmap applied to the shape.

Advanced Parameters

Additional advanced parameters that can assist the user in optimising the patterning specifications are available, including Beam Spacing, Display Modes, Milling Angles, Shape Angles and Bitmap

Angles. As mentioned above, for optimized display speed, bitmaps are re-sampled to 512 by 512 pixels for display purposes only.

Display Modes

Original Display the bitmap as-is.

Greyscale Display a colour bitmap in greyscale in order to better perceive the patterning levels.

Alpha Display the bitmap with intensity relating to transparency in order to better perceive the patterning levels. Pixels where more dose is to be delivered appear more opaque than the pixels where less dose is delivered.

Patterning Display the bitmap in patterning space. This mode represents more accurately the eventual patterning as it takes into account the dwell spacing parameters to be applied during patterning. See Figure 88.

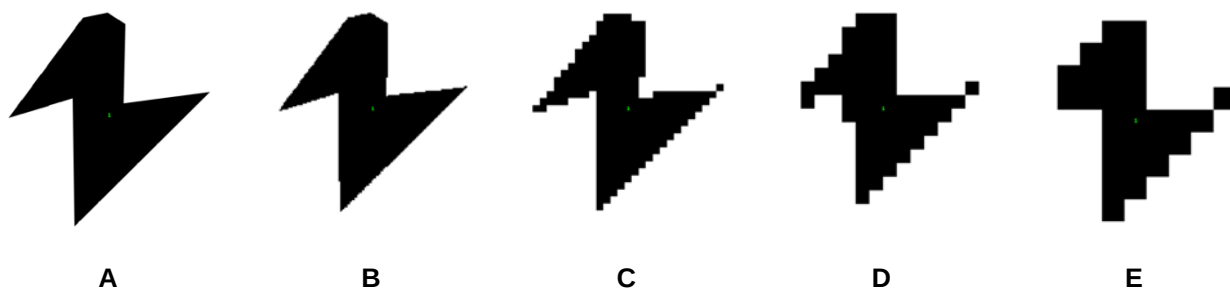


Figure 88: The effect of dwell spacing on the NanoPatterning of a monochrome bitmap illustrated with the Display mode set to Patterning: (A) Original Image (equivalent to setting the Display mode to Original) (B) 10% Spacing (C) 25% Spacing (D) 50% Pixel Spacing (E) 75% Pixel Spacing. These arbitrary spacing values are for illustrative purposes only, as in practice they depend on the spot size and shape size chosen.

Note:

For ease of positioning, the bitmap applied to a shape is not displayed when the user performs mouse operations on the shape.

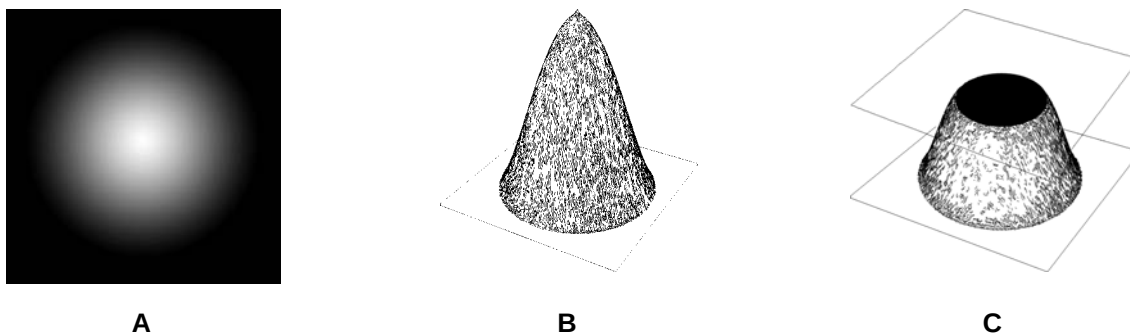


Figure 89: Example of a BMP patterning. (A) The original greyscale bitmap. (B) 3D Simulation of the milling. (C) Representation of a frame from the patterning corresponding to a grayscale value.

9.5. The 3D Profiler

The 3D Profiler allows the user to produce a structure with variations in depth or height. To access the 3D profiler, select a single shape (Rectangle, Trapezoid, Ellipse or Ring) and select the 3D profiler icon located in the toolbar of the NanoPatterning Parameters Panel. This icon is located in a drop down menu that appears after clicking the **Bitmap Milling / 3D Profiler** button in the tools section of the NanoPatterning Form.

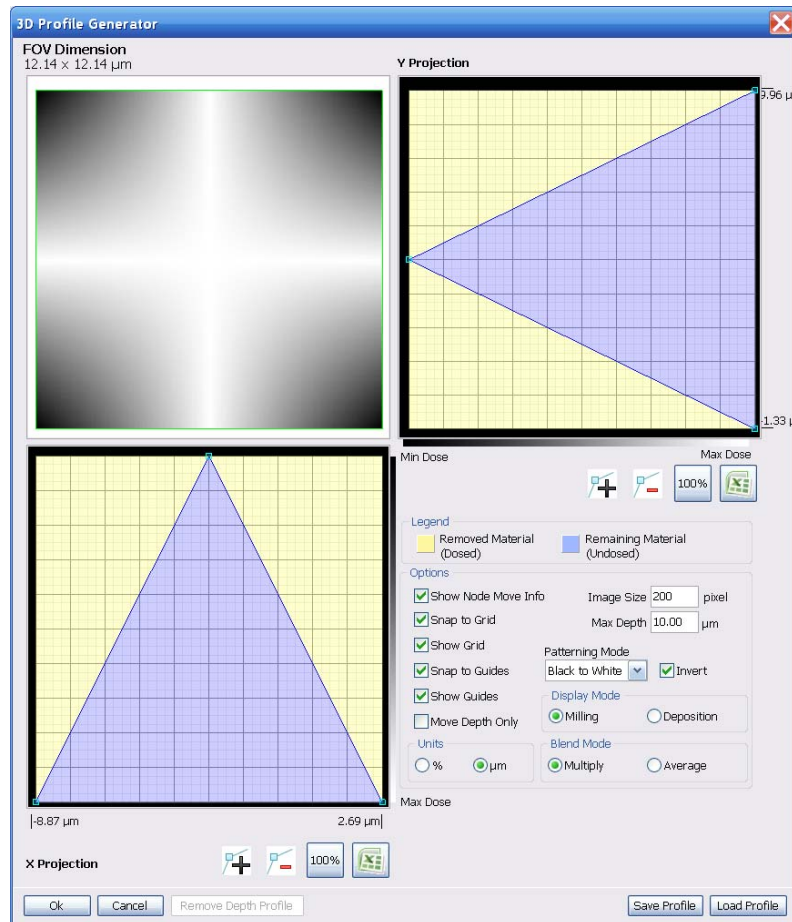


Figure 90: Screenshot of the 3D profiler applied to a rectangular shape.

The 3D profiler consists of two graphs which correspond to the projection of each of the orthogonal axes of the milling space, hereafter referred to as the X Projection and Y Projection. In other words, the 3D profile is relative to the milling angle: rotating the milling angle by 45° will rotate the 3D profile by 45° as shown in Figure 91.

In a manner similar to 3D bitmap milling, the 3D profiler modulates the dose as a function of the greyscale level of the combined profiles of the shape: Black will receive no dose and white will receive the full dose.

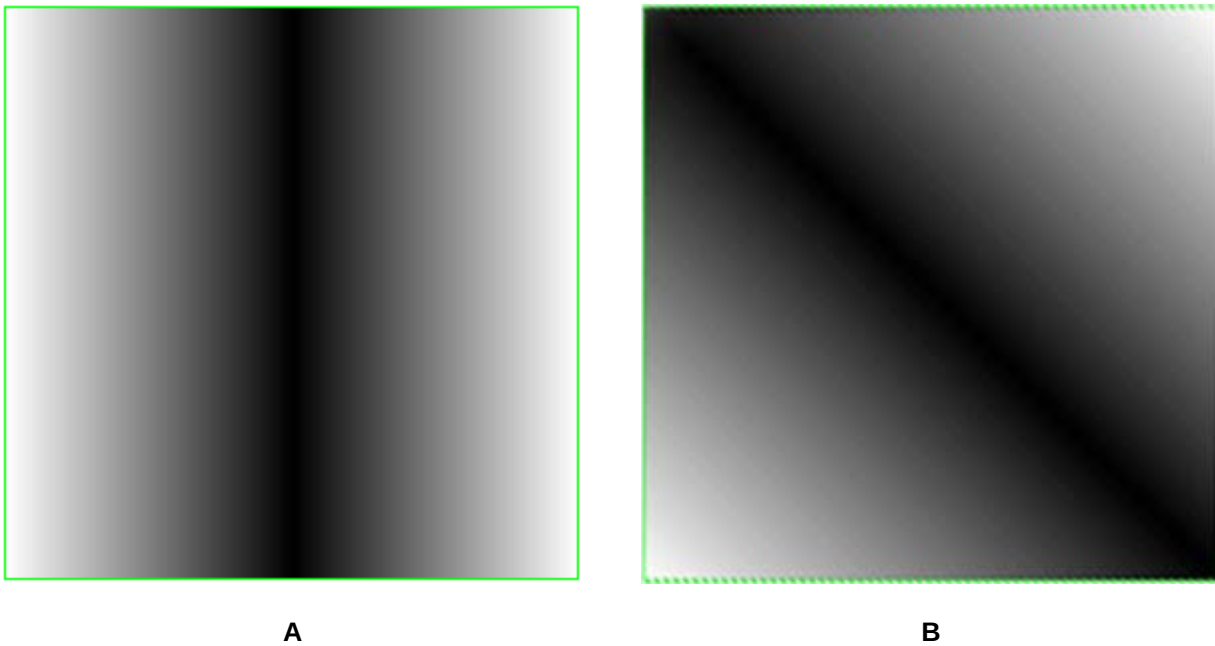


Figure 91: Two rectangular shapes with the same profile but different milling directions. A) The milling direction is 90°. B) the milling direction is 45°.

The 3D profile results in the spatial combination of both dose projections. The resulting final profile can be the average or the multiplied value of both orthogonal projections. The 3D profile can also be inverted, as shown in Figure 92C.

For the Ring shape type, the 3D profiler will have a single profile that varies along the Ring radius.

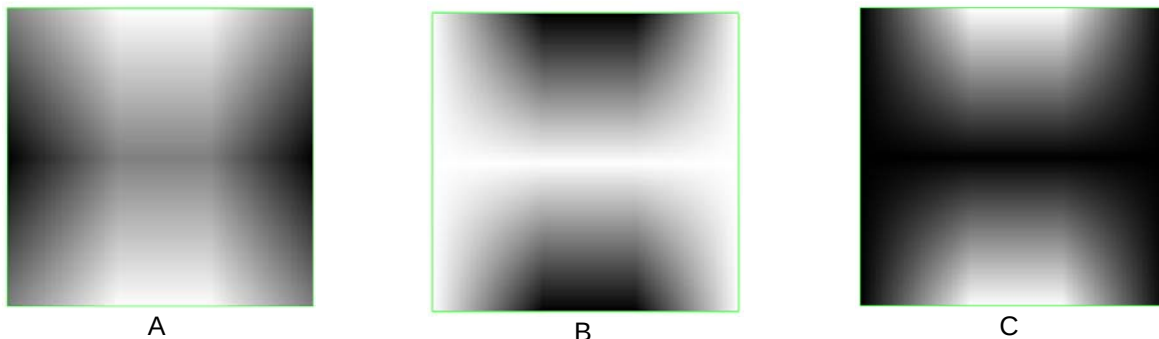


Figure 92 A Rectangle where the same profiles are applied. A) The profiles are averaged. B) The profiles are multiplied. C) The profiles are multiplied and inverted.

Editing Profile

Using the mouse, the user can modify the profile of the shape by adding/removing/moving the nodes along either of the projections. To add nodes, the user can double-click on a given projection profile, right-click and select “Add Node”, or use the “Add Node” button. Removing nodes can also be done by right-clicking on the node to be removed.

Level Nodes

Level nodes are a set of four nodes that are added at the same time. They use the horizontal or vertical guides to position themselves on the profile. They are used to quickly define plateau or ramp structures.

Guides

Guides can be added to help position the nodes in the projection profiles. The guides can be added in two ways: right-click in the projection profile viewport or right-click on the shape itself. If turned on, the nodes will snap to the guides when moved around. The guides can be displayed or hidden using the related checkbox.

Display mode

The 3D profiler can be used both to remove material or to deposit material (using gas assisted deposition). The display of the 3D profiler can be changed from “Milling” to “Deposition”. In both modes, the yellow area will be the “dosed” area. In Milling mode, the area which receives a dose has material removed, whereas in Deposition mode the area which receives a dose has material added (deposited).

Colour	Deposition Mode	Milling Mode
Yellow	The deposited material	The removed material
Blue	-	The remaining material
White	Zone with no deposition	-

Table 2: Signification of the colours used in the 3D profiler display.

Save/Load 3D profiles

Once generated, a 3D profile can be saved to be used at a later time. To save a profile click the “Save” button located in the bottom right of the 3D Profiler window. This will generate an XML file containing all of the information to regenerate a given profile.

Import a profile from Microsoft Excel

A profile can be generated using Excel and then imported into the 3D Profiler. The Excel file must be saved in the Excel 97-2003 format. The file is simply a list of the normalized positions (%) in the first column and of normalized depths (%) in the second column, as shown in Table 3.

0	0
25	100
50	0
75	100
100	0

Table 3: Example of an Excel file used to generate a 3D profile. This file will generate 2 peaks centred at 25% and 75% of the bounding box of the shape.

Functions can be used to generate more complex projection profiles such as sinusoidal or second-degree polynomial profiles as shown in Figure 93.

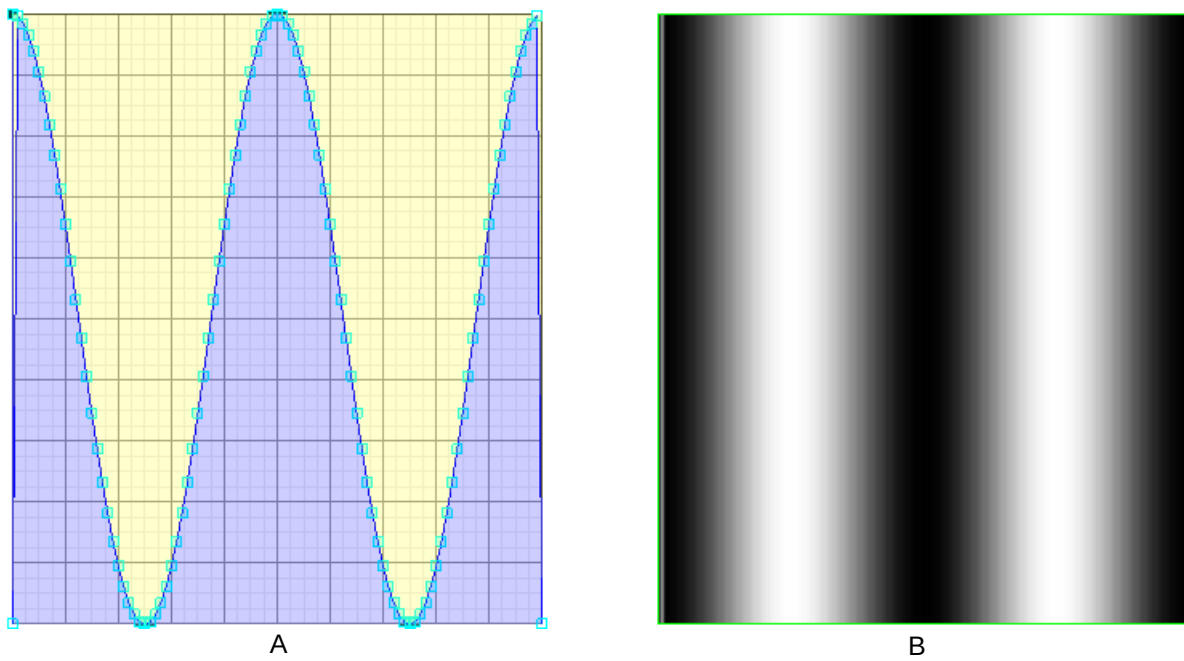


Figure 93: A) A sinusoidal profile generated in Excel then imported into the X Projection of the 3D Profiler. B) The resulting pattern in a rectangular shape.

Below are two examples of the use of the 3D Profiler to remove material in a user determined way from within a shape.

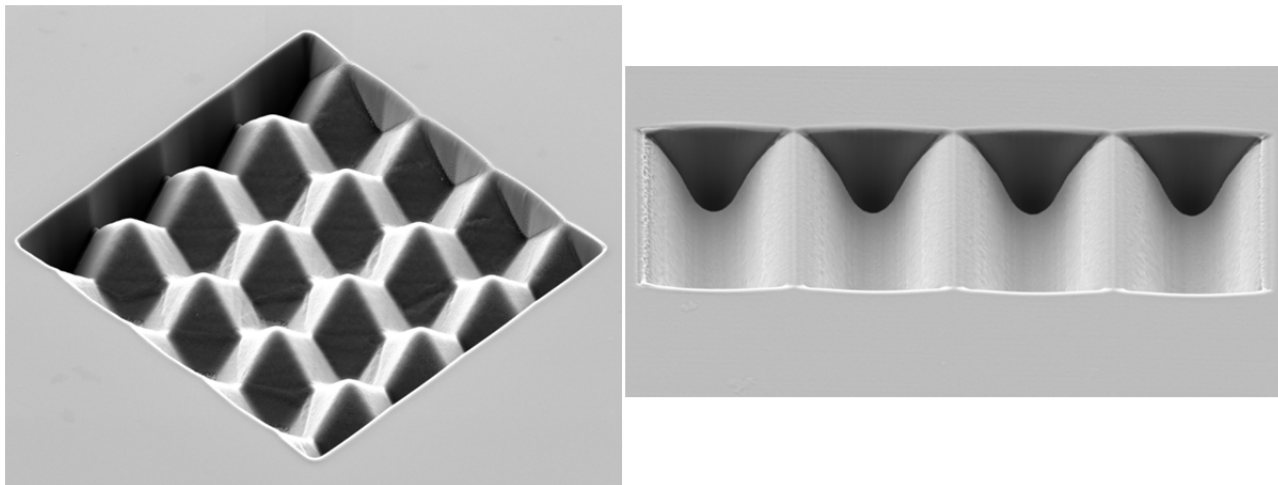


Figure 94: Example of shapes patterned with the 3D Profiler.

9.6. The Annular Builder ☒ADV

The Annular Builder is used to produce pillars in a step-by-step manner. By setting the inner radius, outer radius and the number of steps, concentric rings are generated to help the user efficiently create a pillar.

Hint:

Since multiple FIB apertures / probe conditions are involved, it is recommended that a FIB Daily Adjust be run prior to using the Annular builder. Additionally, the drift correction tool can be used to assure that the shapes in the different steps are correctly aligned.

Predefined general parameters are used to generate a list of shapes. Once generated, the shapes can be individually modified using the table at the bottom of the form.

Warning:

Changing the general parameters will override any custom entries made in the Ring list table.

The “Scale Width” option is used to scale the area of each Ring. For example, if a value of 3 is used, it means that the radius of the next Ring will be a third of the current Ring. This is useful for reducing the patterning time since the probe used normally needs to be scaled down when patterning the innermost rings.

The Annular Builder will also attempt to suggest what aperture should be used to pattern each Ring of the set. If the estimate is not what the user desires, the probe can be changed once the rings have been generated.

The current state of the Annular Builder can be saved and reload as needed.

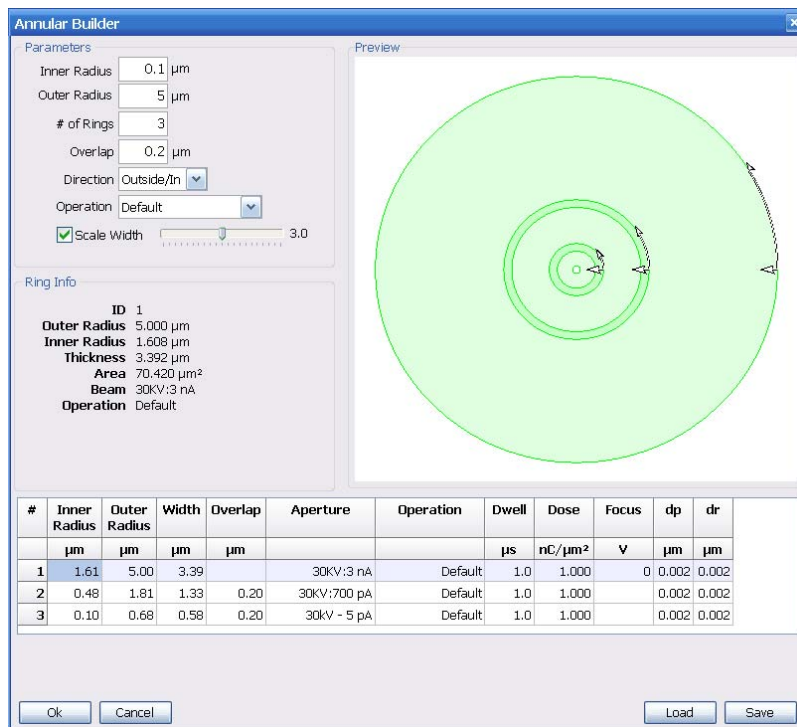


Figure 95: Screen shot of the Annular builder.

9.7. Array Builder ☒ADV

The Array Builder has many uses including systematic investigation of patterning parameters, process development and optimization, and even convenient shape duplication. It allows the rapid creation of duplicates of a single shape, with or without systematic modification of any or all of a wide range of parameters. Clicking the **Array Builder** button () will pop up a panel as shown in Figure 96.

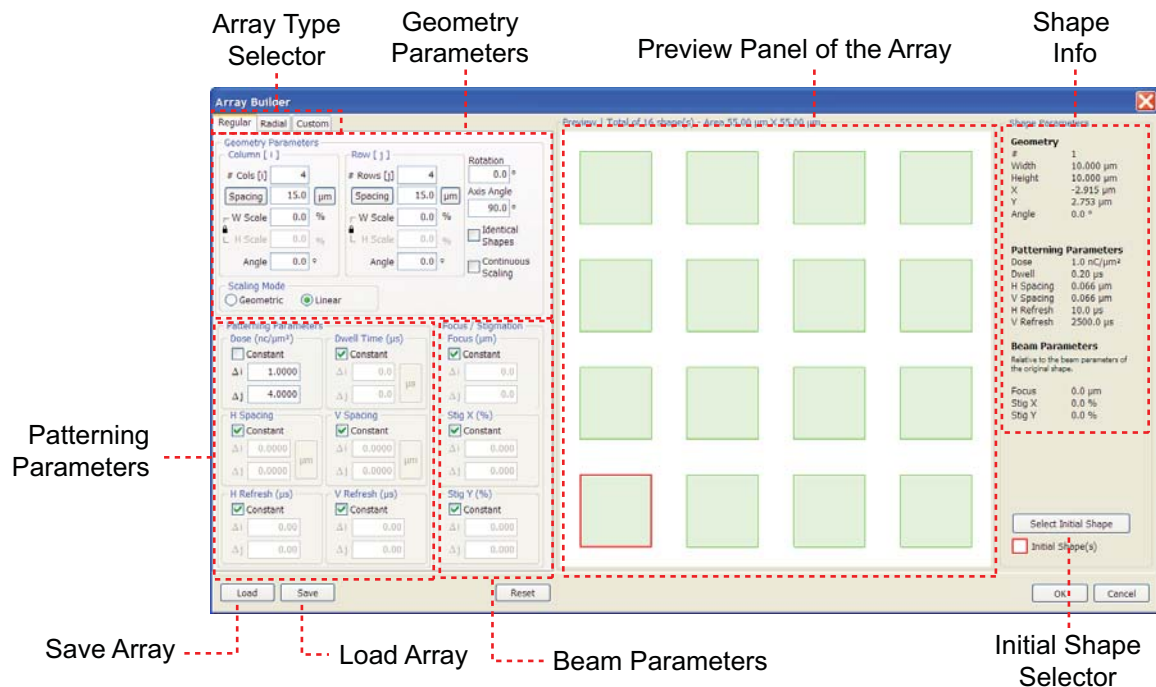


Figure 96: Overview of the Array Builder Panel.

The Array Builder offers three variations: a Regular Array, a Radial Array and a Custom Array. Regular Arrays are used to create multiple shapes distributed along rows and columns. Radial Arrays are used to create shapes in circular or ring-type distribution. Custom Arrays can be used to generate a series of shapes with user-defined positional distribution.

The Array Builder Panel consists of three main sections:

1. The Editing section, which is located on the left-hand side of the panel.
2. The centrally located Preview section to assist the user while the different geometry parameters are being entered or modified.
3. The Shape Info section, which is found on the right hand side of the panel.

In all three types of arrays, the initial shape is drawn in red. In the Regular Array, the initial shape(s) can be specified. The Initial shape(s) corresponds to the (0, 0) index. The index increases towards the right and up and decreases towards the left and down.

In the case of the Regular Array and the Radial Array, multiple shapes can be used as the initial shapes. In those cases, the shapes are considered a group and the shapes will be distributed in the fashion shown in the figure to the right.

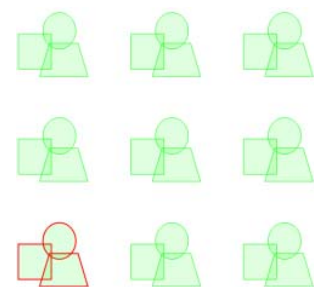


Figure 97: A regular array with the initial group of shapes highlighted in red.

Shape Info Panel

A panel showing most of the essential information about a particular shape can be found to the right of each of the three types of arrays. By hovering the mouse cursor over a shape in the preview viewport, the Info Panel will display the information of that particular shape.

Regular Array

The Regular Array tool can be used to create a regular array of shapes distributed along rows and columns. The initial shape is located at the bottom left corner of the array and is displayed using a thicker red outline. The shape designated as the Initial Shape (0,0) can be switched to any element in the array by clicking the **Initial Shape** button then selecting the desired array element. The editing section of the Array Builder is separated into three sections: Geometry Parameters, Patterning Parameters and Beam Parameters.

The regular array utilises row and column indices, as shown in the Figure 98.

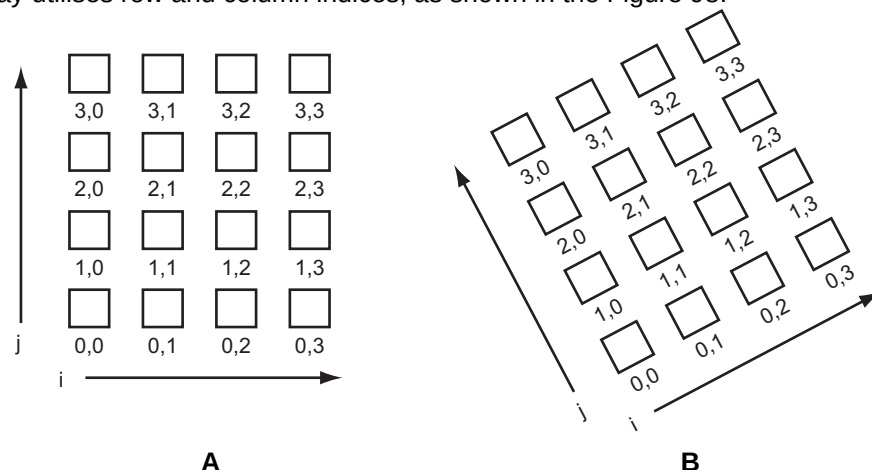


Figure 98: Row and column conventions of the Regular Array. (A) Regular Array without any global rotation. (B) Regular array where a global rotation of 30° has been applied.

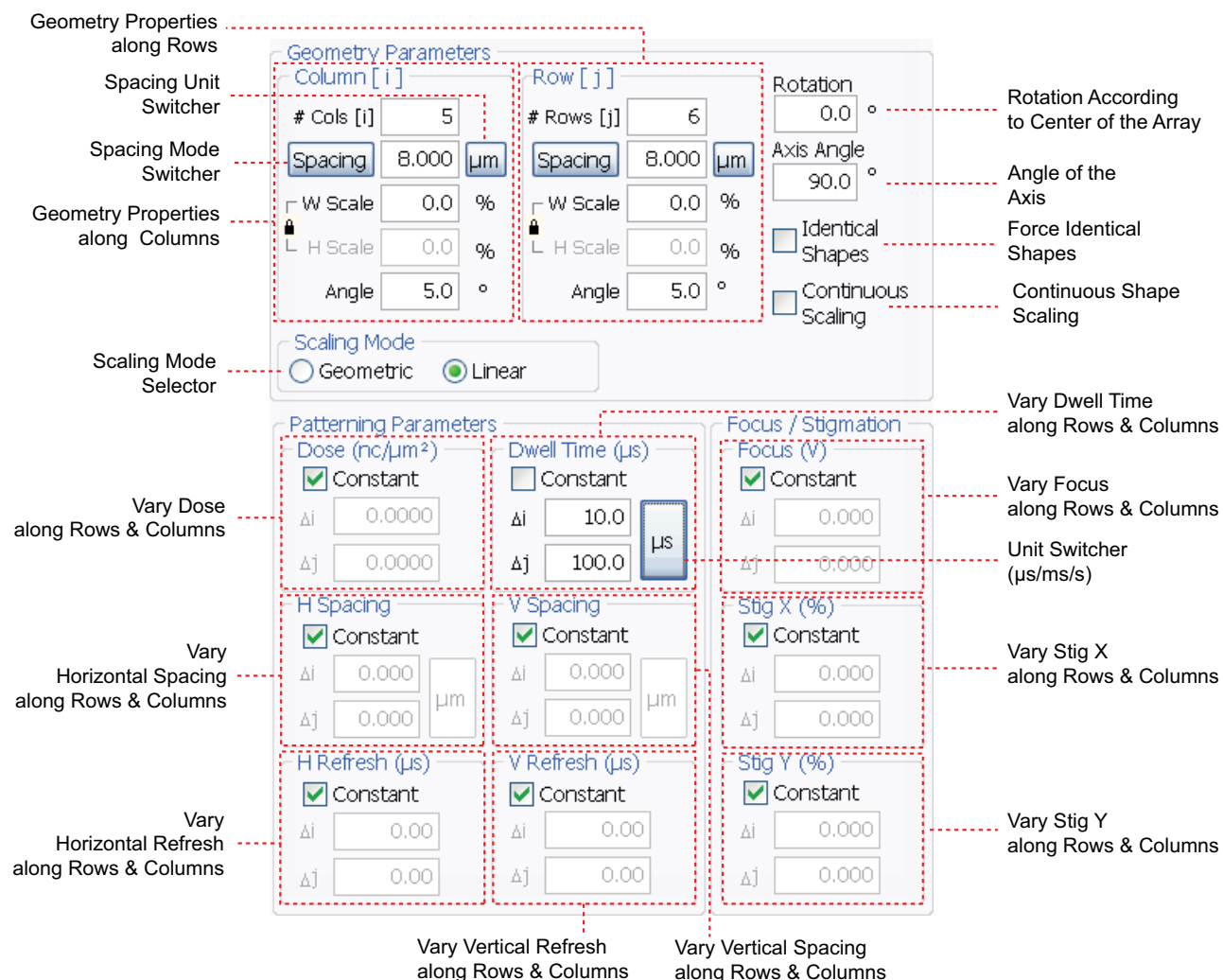


Figure 99: The Parameters section of the Regular Array Builder Panel.

Geometry Parameters

Columns and Rows

Use these parameters to specify the dimensions of the array of shapes. The number of elements (shapes) in the array will be equal to the value of the number of Columns multiplied by the value of the number of Rows.

Spacing Mode

The spacing between elements of a row or a column can be set in two ways: spacing centre-to-centre or fixed gap between the shapes.



Figure 100: The two spacing modes of the Regular Array Builder. (A) Spacing mode, where the entered value is the spacing between the centre of adjacent shapes. (B) Gap mode, where the entered value is the gap between the left edge and the right edge of adjacent shapes.

Scaling

The Scaling is a factor by which a shape in the array will be scaled relative to the previous shape element in a Row or Column. The scaling is applied along the repeating Row or Column (i or j) and is applied according to the Scaling Mode. Width and height of the shape can be scaled separately or together. Turning **Identical Shapes** on will prevent any scaling from occurring.

Scaling Mode

Two Scaling Modes exist: Geometric and Linear.

Geometric Scaling:

Geometric scaling applies the Scaling value geometrically by Row or Column element. For example, under Geometric Scaling with a value of +10% for Row Scaling, and a single Column of shapes, the first shape (element 0) will have no scaling applied, the second (element 1) will have +10% scaling applied, the third (element 2) will have +21% scaling applied ($10\% \times 10\%$), etc.

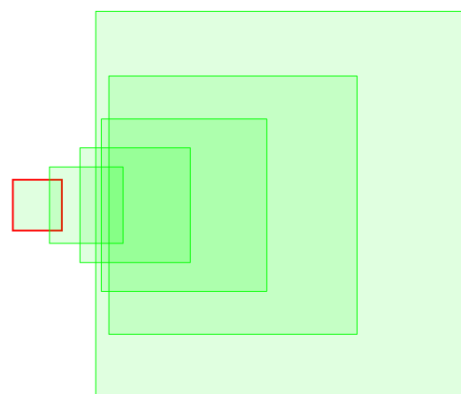
Under Geometric Scaling, a scaling of 0% means “no scaling”.

Linear Scaling:

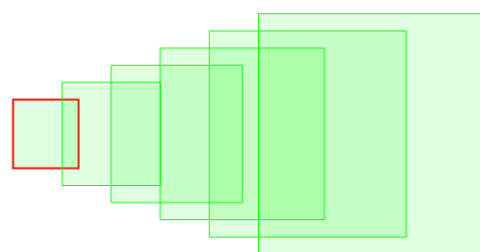
Linear Scaling applies the Scaling value linearly by Row or Column element. For example, under Linear Scaling with a value of +10% for Row Scaling, and a single Column of shapes, the first shape (element 0) will have no scaling applied, the second (element 1) will have +10% scaling applied ($0\% + (1 \times 10\%)$), the third (element 2) will have +20% scaling applied ($0\% + (2 \times 10\%)$), etc.

Under Linear Scaling, a scaling of 0% means “no scaling”.

Equations for the Scaling process under Geometric and Linear Scaling Modes are discussed below.



Geometric Scaling



Linear Scaling

Hint:

Under the Linear Scaling Mode, if the resulting scaling factor becomes negative, the resultant shapes will be flipped about their respective axis, allowing interesting symmetric patterns to be created.

Angle

The amount by which each Shape element will be rotated relative to the previous element. (Figure 101C)

Rotation

The rotation applied to the array as a whole. (Figure 101B)

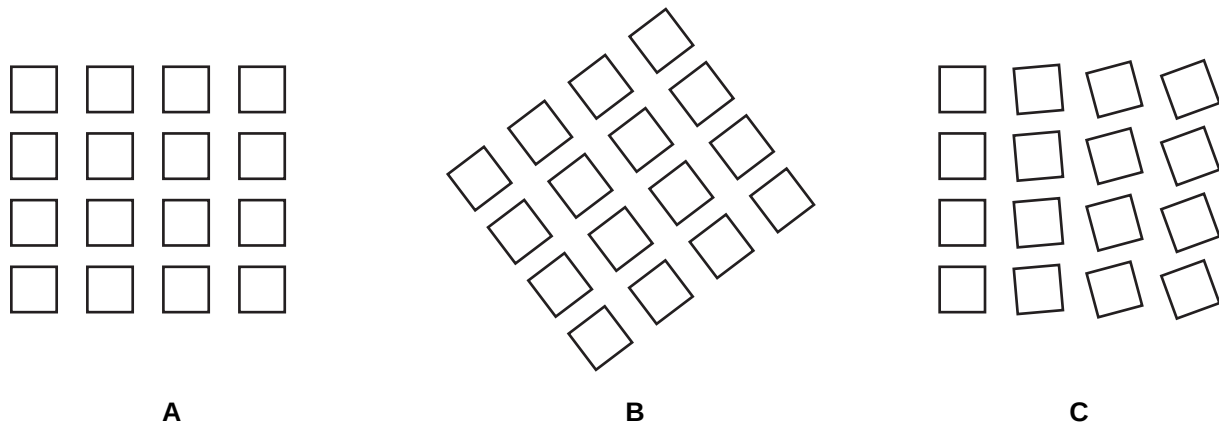
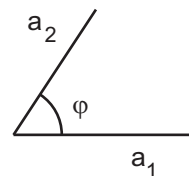


Figure 101: A 4x4 array with different Angle and Rotation values. (A) No rotation applied. (B) Global Rotation of 30°. (C) Incremental rotation in X of 5° achieved with the Angle parameter.

Axis Angle

The rows and columns of the array form a system of 2 vectors a_1 and a_2 separated by an angle of φ . The length of a_1 and a_2 are given by the spacing in x and spacing in y, respectively. The angle φ corresponds to the axis angle in the parameters section. An array using the same spacing in x and y but three different axis angles is shown in Figure 102. The row spacing corresponds to the length of the vector a_2 and the column spacing corresponds to the length of the vector a_1 .

**Hint:**

The position of the initial shape in the Regular Array can be changed from bottom-left corner to top-left by inputting -90° in the Axis Angle field.

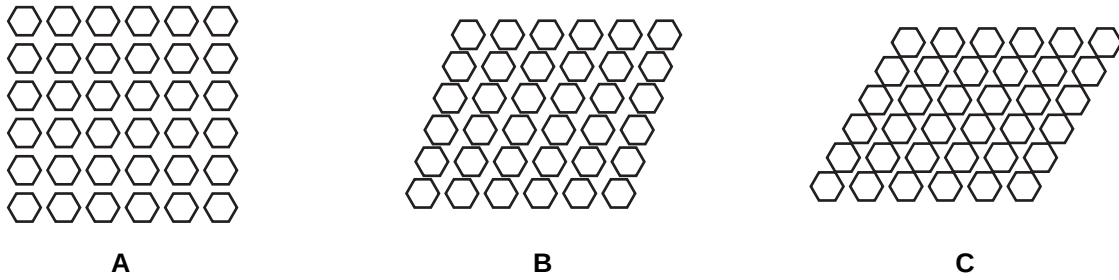


Figure 102: A 6 by 6 array build with (A) 90° (B) 75°, and (C) 60° rotation.

Shape Position

The position of the shape_{ij}, where i is the column index and j is the row index, is given by the following equation:

$$x_{ij} = x_0 + i \times spacing_i + j \times spacing_j \times \cos(\phi)$$

$$y_{ij} = y_0 + j \times spacing_j \times \sin(\phi)$$

where x_0, y_0 is the position of the original shape, i.e. the shape used to create the array.

When the axis angle is 90°, the position of the shape_{ij} becomes:

$$x_{ij} = x_0 + i \times spacing_i$$

$$y_{ij} = y_0 + j \times spacing_j$$

Please note that if a global rotation is applied to the array, the centre of the shape_{ij} will be different than the results of the equation.

Shape Size

When the Scaling Mode is set to Geometric, the size of each shape_{ij} is an element in a geometric sequence based on the original shape size modified by the factor entered in W Scale and H Scale of the row and columns. The width and height of the ijth shape are given by:

$$w_{ij} = w_0 \times sw_i^i \times sw_j^j$$

$$h_{ij} = h_0 \times sh_i^i \times sh_j^j$$

where:

sw_i and sw_j are the width scaling factor along the columns and the width scaling factor along the lines, respectively.

sh_i and sh_j are the height scaling factor along the columns and the height scaling factor along the lines, respectively.

w_0 and h_0 is the width and height of the original shape.

When the Scaling Mode is set to Linear, the size of each shape_{ij} is scaled by a multiple of the factor entered in Width Scale and Height Scale. The width and height of the ijth shape are given by:

$$w_{ij} = w_0 \times (1 + sw_i \times i + sw_j \times j)$$

$$h_{ij} = h_0 \times (1 + sh_i \times i + sh_j \times j)$$

When Continuous Scaling is enabled, the dimensions of the shapes linearly increase. Under Continuous Scaling, the scaling of the shape_{ij} is given by:

$$s_{ij} = 1 + (i + (RowCount + 1) \times j) \times Scaling$$

Under Geometric Scaling, the scaling factor when Uniform Scaling is enabled is:

$$s_{ij} = Scaling^{i + (RowCount + 1) \times j}$$

The width and the height of the ijth shape are then given by:

$$w_{ij} = w_0 \times s_{ij}$$

$$h_{ij} = h_0 \times s_{ij}$$

The angle of the shape_{ij} is given by:

$$\theta_{ij} = \theta_0 + i \times \Delta\theta_i + j \times \Delta\theta_j + \theta_{rotation},$$

where:

θ_{ij}	Angle of <i>shape_{ij}</i>
θ_0	Angle of the original shape
$\Delta\theta_i$	Angle increment along the rows
$\Delta\theta_j$	Angle increment along the columns
$\theta_{rotation}$	Global rotation

Note:

The Angle and the Rotation parameters have different effects. The Angle property is applied to each shape individually with the centre of each shape used as the Rotation Centre. The Rotation property is applied to all shapes as a group: its Rotation Centre is the centre of the bounding box encompassing all the shapes (Figure 101).

By default, the spacing of the shapes is automatically calculated to be 1.5 times the dimensions (width or height) of the shape used to create the array, however the user may modify these values. For example, changing the spacing to zero in x and y will allow the user to create multiple overlapped patterns, such as those shown in Figure 104.

Patterning Parameters

- Dose
- Dwell Time
- Horizontal Spacing
- Vertical Spacing
- Horizontal Refresh
- Vertical Refresh
- Beam Parameters
 - o Focus
 - o Stigmation in X
 - o Stigmation in Y

The Patterning Parameters of the shape used to create the array is the baseline reference for all the other shapes in the array. For all parameters, the user can either leave them constant or vary them along the array. If a parameter is kept constant, it will use the value found in the baseline reference shape. By disabling the “Constant” checkbox for any of the parameters, the user will be able to vary a particular parameter for each shape along the i and j (Column and Row) elements in the array.

All variations in the patterning parameters section are relative to the original baseline shape used to create the array and are not absolute values. For example, the dose of the element in i,j can be calculated as:

$$Dose_{ij} = Dose_0 + i \times \Delta Dose_i + j \times \Delta Dose_j,$$

where

$Dose_0$	Dose of the shape used to create the array
$\Delta Dose_i$	Dose increment along the columns
$\Delta Dose_j$	Angle increment along the rows
i	Column index
j	Row index

The horizontal spacing and vertical spacing can be set as Absolute Spacing in μm , Overlap % or Spacing % using the Spacing Mode Switchers.

Certain beam parameters such as Focus and Stigmation may also be varied. If they are varied, the beam parameters will be changed by the requested values only during NanoPatterning and will be set back to their initial values when NanoPatterning is complete.



Figure 103: Array Builder used to create a 6 X 6 array. (A) The original shape. (B) The 6 X 6 array.

Figure 104 demonstrates what this tool can do with a single Square as the baseline reference shape.

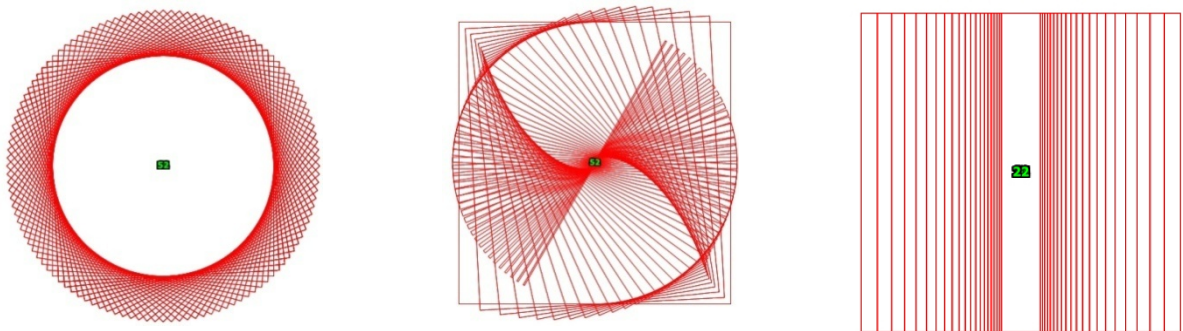


Figure 104: Interesting shape patterns created using the Array Builder acting on a single Square shape.

Save and Load Arrays

It is possible to Save and Load the Array Builder parameters in order to be able to quickly repeat the same experiment. The NPVE reads and writes .AB (Array Builder) files.

The information from all three different types of array will be included in the saved files.

Radial Array Builder

Arrays having a circular distribution can be created using the Radial Array Builder. Like the Rectangular Array Builder, the Radial Array Builder is composed of 3 main sections: Geometry Parameters, Patterning Parameters and Beam Parameters.

Figure 105: The Geometric Parameters of the Radial Array Builder.

Geometric Parameters

Initial Radius

The Initial Radius is the radius of the innermost ring of the array. By default, it is set to 3 times the width of the shape used to generate the array.

Initial Count

The Initial Count is the number of shapes on the innermost ring of the array.

Number of Rings

The Number of Rings is the total number of rings that will be created. The rings are all centred.

Initial Offset

The Initial Offset is an angle that is applied to all of the rings of shapes.

Orientation

The Orientation offers two choices: Normal or Constant. As Figure 106 shows, when Normal orientation is selected, the angle of the shapes is adjusted so the shape is perpendicular to the normal of the arc of the ring. However, when the Constant orientation is selected, the angle of the shapes remains constant all along the ring.

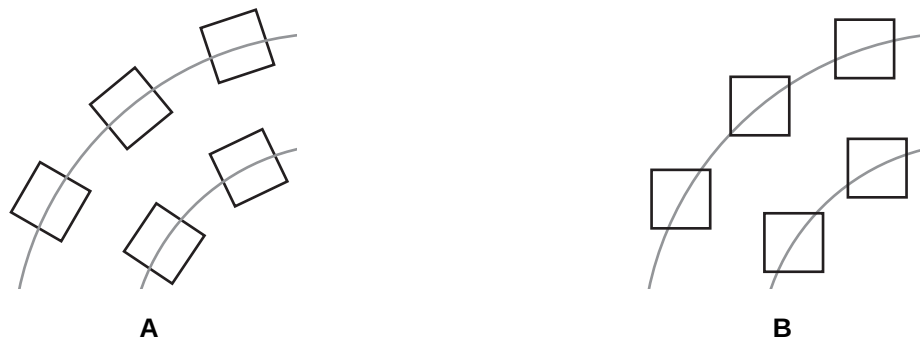
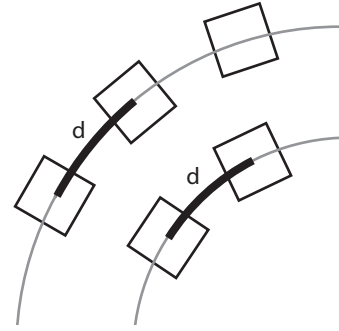


Figure 106: Orientation difference of 2 rings of a radial array. (A) Constant Orientation. (B) Normal Orientation.

Constant Spacing

The Constant Spacing option is helpful to create arrays where the arc segment between each of the shape centres is equal. NPVE will then calculate the appropriate number of shapes on each ring to maintain a constant distance between elements of a ring. For this reason, when Constant Spacing is enabled, the Count/Ring variation is disabled.



Note:

Since NPVE can only fit an integer number of shapes on a ring, under the Constant Spacing condition the distance d between the shapes of different rings may slightly vary.

Count/Ring Variation

The Count/Ring Variation is used to change the number of shapes per ring. This field is disabled if the Constant Spacing is enabled.

Radius/Ring Variation

The radius of the rings can be linearly incremented by using the Radius/Ring Variation. The radius of the i^{th} ring is given by:

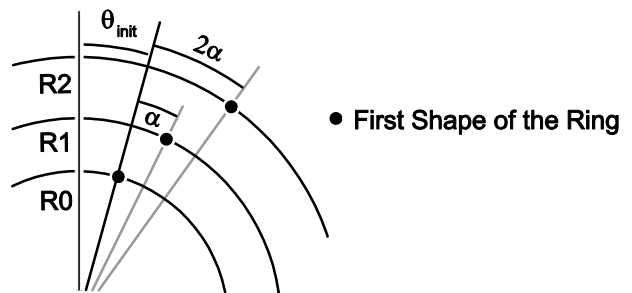
$$R_i = R_0 + i \times R_{\text{ring}}$$

where R_0 is the radius of the initial ring, i is the index of the ring, and R_{ring} is the Radius/Ring Variation.

Angle/Ring Variation

The Angle/Ring Variation allows variation in the position of the first element of each ring. The offset of the first element is given by:

$$\theta_i = i \times \alpha + \theta_{\text{init}}$$



Where θ_i is the offset of the i^{th} ring, i is the index of the ring, α is the Angle/Ring Variation and θ_{init} is the initial offset angle.

Scaling/Ring Variation

It is possible to scale up or down the size of the original shape used to create the array. The scaling on the i^{th} ring is given by:

$$S_i = (100 + S_{\text{ring}}) \% \times i$$

where S_i is the scaling of the i^{th} ring, S_{ring} is the Scaling/Ring Variation and i is the index of the ring.

Scale Radius

When the Scale Radius option is disabled, the radius of the i^{th} ring can be calculated using this equation:

$$R_i = R_0 + i \times R_{ring}$$

where R_0 is the radius of the initial ring, i is the index of the ring and R_{ring} is the Radius/Ring Variation.

However, when the Scale Radius is enabled, the radius of the i^{th} ring is given by:

$$R_i = R_{i-1} + i \times R_0 \times S_{ring}$$

where S_{ring} is the Scaling/Ring Variation.

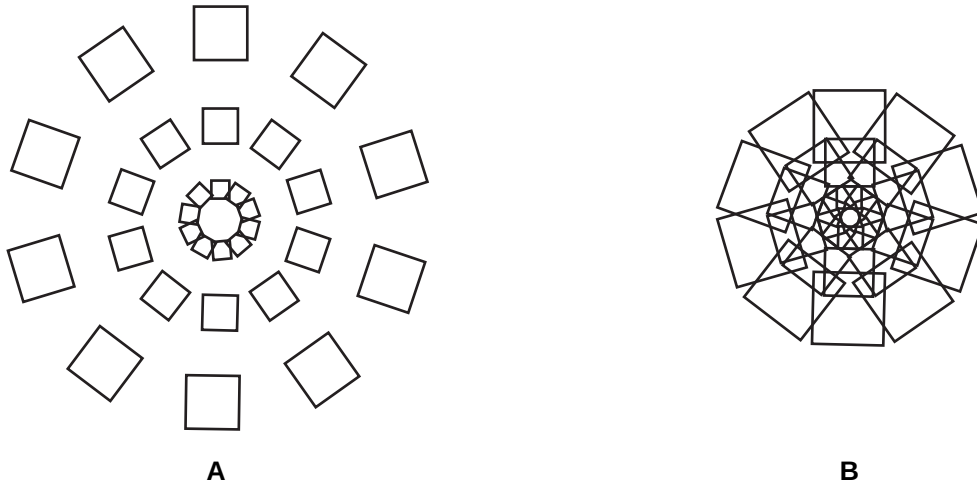


Figure 107: Same array build with (A) Scale Radius enabled and (B) the Scale Radius disabled.

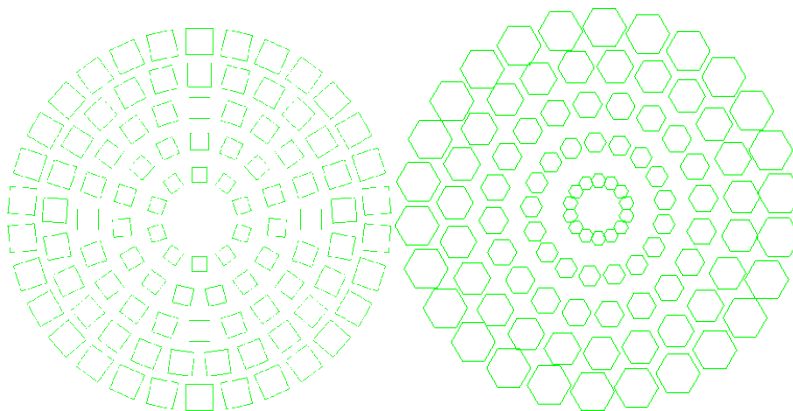


Figure 108: Example of patterns created using the Radial Array Builder.

Custom Array Builder

Unlike the Regular Array or the Radial Array, the Custom Array Builder lets the user edit any parameters of the shapes in the array manually (see Figure 109). Geometric and Patterning parameters can all be changed independently by selecting them from the list of parameters at the bottom of the array.

To ease the creation process, the Custom Array Builder can be populated using three different sources:

- Regular Array Builder
- Radial Array Builder
- A Microsoft Excel™ File

The Custom Array Builder is only available when a single shape is selected.

When a row of the Custom Array Builder is being edited, the corresponding shape is highlighted in the preview Viewport.

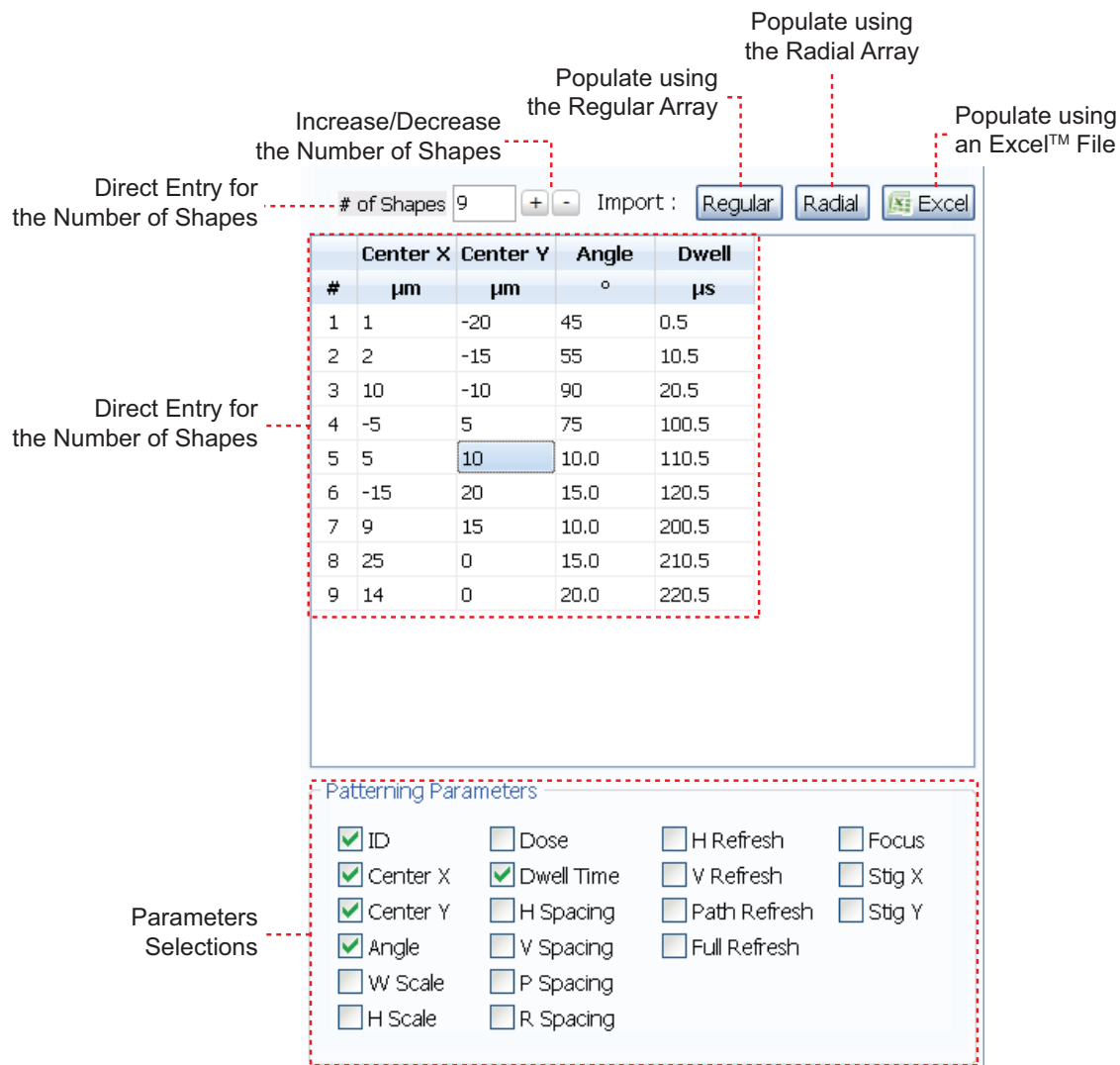


Figure 109: The Custom Array Builder Panel.

Save and Load

It is possible to pattern the same array in subsequent sessions by saving and reloading the parameters of an array. The NPVE saves the data from all 3 types of array in one file. The original shape(s) used to build the array is(are) also embedded into the Array Builder file (.ab).

Loading an array can also be done using the **Load** button located in the toolbar.

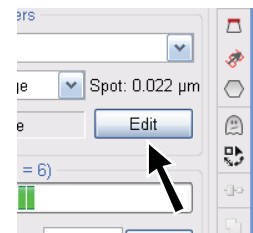
9.8. Operation Editor ☒ADV

The NPVE introduces the concept of “Operations” in order to improve ease of use and repeatability. Operations can be considered as pre-defined tables of values that set all nanopatterning parameters required to execute a shape. In general, different parameter settings are required to achieve different patterning behaviours, but once an expert user has set up and optimized a particular Operation for a given behaviour, such as material deposition, any user can easily select and employ this Operation without concern for the exact advanced parameter settings. The Operation Selector in the NanoPatterning Parameters Panel (see Figure 44) is used to automatically set the patterning parameters for a particular behaviour (i.e. milling, etching, depositing, etc.) in a straightforward manner that does not require advanced knowledge of the microscope or the particular advanced settings required in order to achieve the desired behaviour. Assigning an Operation to a single shape or to a set of shapes will automatically fill the correct advanced patterning parameters for the intended purpose of the shape as a function of the selected beam. However, before users can apply Operations using the Selector, the Operations themselves must be defined by expert users, and this process is greatly simplified through the use of the Operation Editor.

Operations aid in the reproducibility of results of a similar action over time (milling, deposition, gas assisted etching, etc...) at any FIB current setting. Operations modify patterning parameters as a function of the beam settings (beam current, spot size, accelerating voltage) and gases used in a particular patterning application. For example, the optimal dwell spacings and dwell time required for a successful and efficient carbon deposition using a 10 pA beam will be different than those for a 27 nA beam. Operations manage these differences by varying the patterning parameters based on beam conditions, etc. according to formula input by expert users.

The Operation Editor ☒ADV is accessible through the **Edit** button in the NanoPatterning Parameters Panel.

Defining an Operation consists of entering values for all the editable parameters of the Operation for a given beam condition. An Operation need not be defined for all beam conditions, but it must be defined, in terms of entering optimal values, for the conditions the user plans to invoke for a given Operation. Undefined parameters or non-optimal parameters will give unexpected results. As mentioned above, defining Operations should be left to expert users. Considerable systematic experimentation and development is often required to determine optimal settings.



Note:

By default, the NPVE system ships with a number of standard Operations. It is not necessary to apply an Operation to a shape – the values of the Advanced Parameters may always be entered by the user. However, succeeding in this manner can be a daunting process for the novice.

The Operation parameters are grouped into three sections: Global Parameters, Editable Parameters and Static Parameters.

Static Parameters concern only the beam and are not editable. They are retrieved from the SmartSEM[®] software and are displayed to give information to the user about the beam conditions already registered in SmartSEM[®] – see the SmartSEM[®] user manual for further details. They include:

- Beam Condition Name
- Beam Current
- Spot Size
- EHT

The Global Parameters are parameters that will affect all beam conditions for a given Operation and they can be seen as core parameters for that Operation. They do not depend on the beam condition. They are:

- Operation Name
- Gas
- Scan Style
- Gas Wait Time
- Description
- Minimum Start Pressure
- Verification Colour

Editable Parameters are distinct to each beam condition. There are a total of six different Editable Parameters in the Operation Editor:

- Dwell Time
- Dose
- H Spacing
- V Spacing
- H Refresh
- V Refresh

Note:

Automated control of gases can only be accomplished through Operations. It is not possible to automatically use gases for a shape that does not have any Operation assigned to it. However, the user may always manually control the gas valves through the SmartSEM® interface.

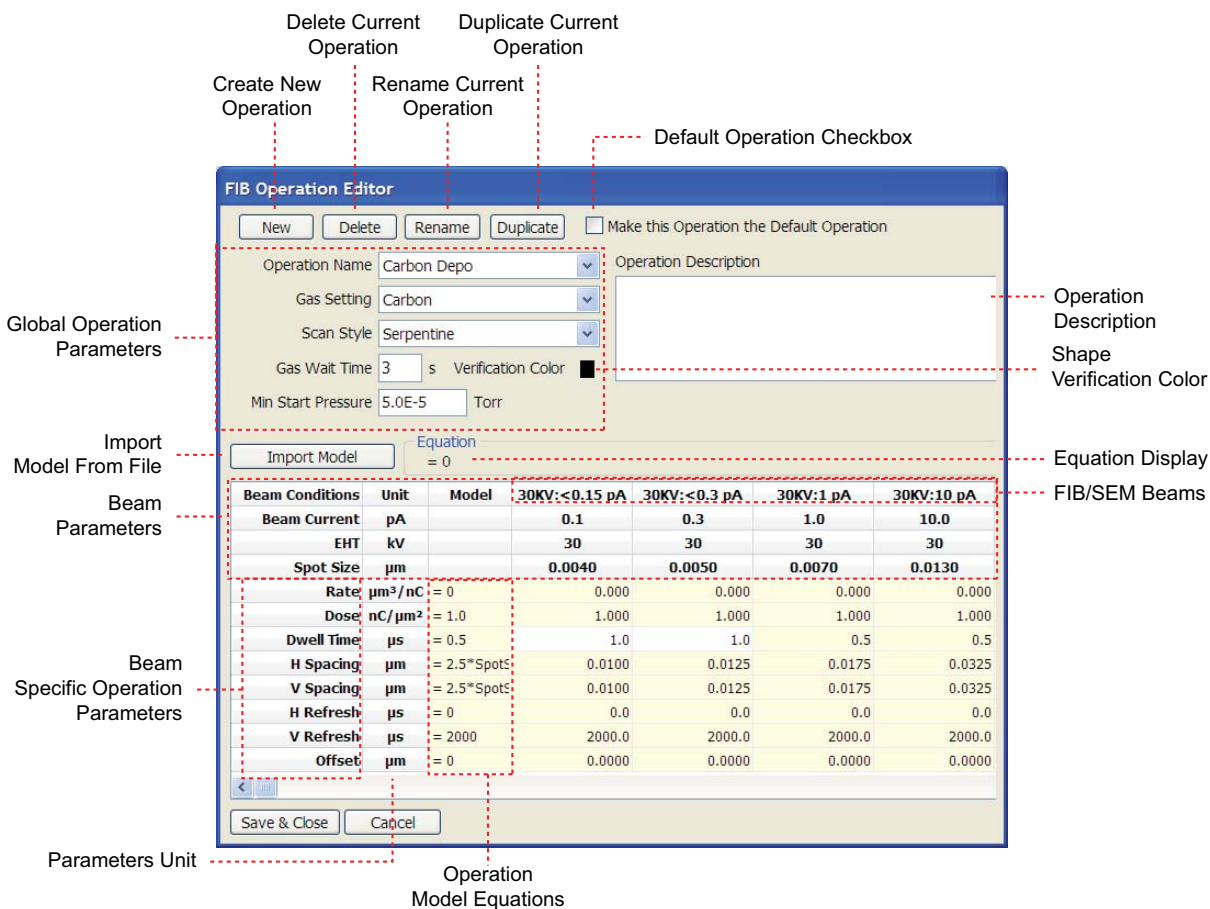


Figure 110: Fields in the Operation Editor.

The minimum start pressure is a setting that will prevent the patterning from starting before the chamber pressure has reached the value specified.

Verification Colours

It is possible to assign a colour to be associated with a given Operation. This Verification Colour is used to fill the Shape when the NPVE is under the Verification Mode. It is useful to see, at a glance, which Operation has been assigned to which shapes. The NPVE can toggle the Verification Mode by pressing F11 or F12.

F11 Draw the verification colour on all the shapes.

F12 Draw the verification colour on the selected shapes only.

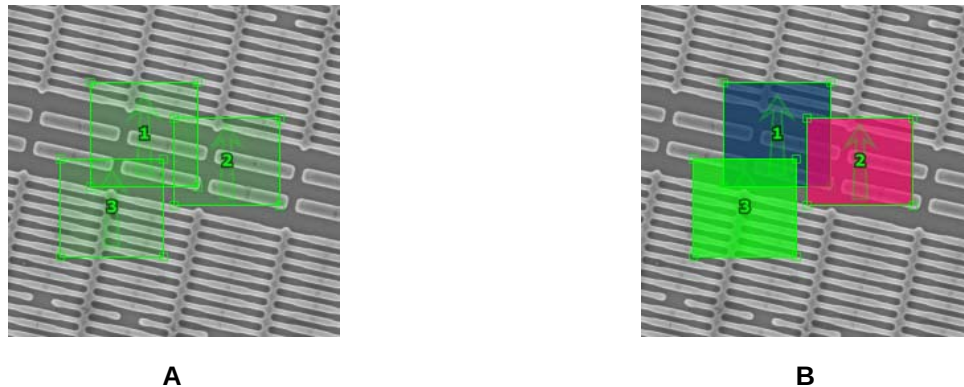


Figure 111: Illustration of the Verification Mode with 3 different verification colours. (A) Standard shape display mode ("Operation Check" turned off). (B) "Operation Check" turned on.

Create New Operation

It is possible to create as many Operations as the user desires. In fact, it is strongly recommended that a unique Operation is created for each application type. For example, a FIB user may desire separate Operations for: "Coarse Milling", "Fine Milling", "Polishing", "Carbon Deposition", "Tungsten Deposition", "XeF2 Low-Dose Etching", et cetera.

To create a new Operation click on the **New** button next to the Operation Selector in the NanoPatterning Parameters Panel. This will display the form shown in Figure 112. Enter the Name and the Description of the Operation and click "Create".

Once the new Operation is created, the Global Operation parameters should next be assigned. The Global Operation parameters are the same for each beam; this is why they are called global. Once this is complete, the Specific Operation parameters for each of the beam conditions can be entered.

The image shows a Windows-style dialog box titled "New Operation". It has a standard title bar with minimize, maximize, and close buttons. Inside the dialog, there are two input fields: "Operation Name" and "Description". The "Description" field is a larger text area. At the bottom of the dialog, there are three buttons: "Create", "Import Model", and "Cancel".

Figure 112: New Operation Creation Form.

Definition of the Specific Operation Parameters

For a given Operation, there are two ways to define the Specific Operation parameters for each beam condition:

- Enter the values directly in the table
- Use the Operation Equation Editor

Double clicking of the name of the row of the Specific Operation parameters will enable the Operation Equation Editor (Figure 113). As its name suggests, by using this tool, the user can quickly and efficiently assign values to a given parameter through equations.

Spot Size	μ
Dwell Time	μ
Dose	nC/
H Spacing	μ
V Spacing	μ
H Refresh	μ
V Refresh	μ

Cells to double-click to show the equation editor helper

Operation Equation Editor

The Operation Equation Editor is shown in Figure 113.

Equation Input Box

The Equation Input Box can accept two types of entry: constant values or equations. Equations can include static Operation Parameters as variables whose values depend on the beam conditions. For example, it is possible to assign the dwell spacing of the Operation to be a function of the spot size.

$$H \text{ Spacing} = 0.5 \times \text{SpotSize}$$

The Variables Picker lists the variables which can be inserted in the equations. Clicking on one of the variables will append its name at the end of the current equation.

The Equation Input Box also accepts % values, i.e. $H \text{ Spacing} = 50\% \times \text{SpotSize}$

Validation Panel

As the user types in an equation, the Validation Panel gives feedback as to whether the expression in the equation field is valid or not.

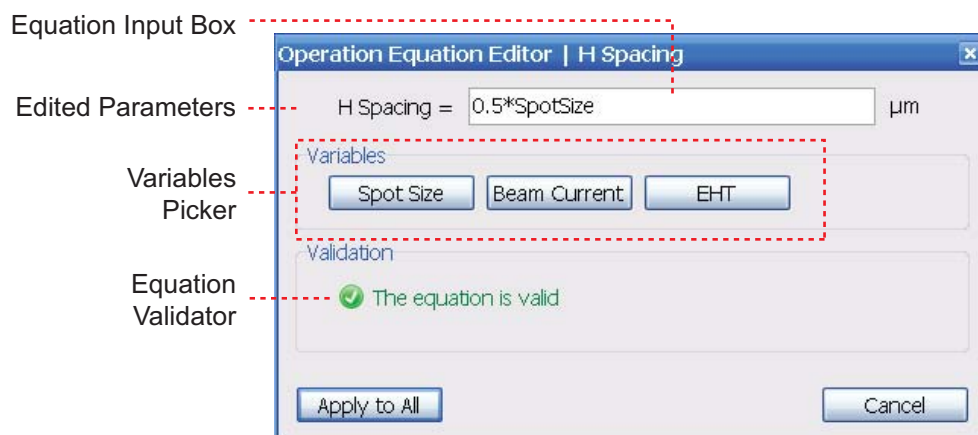


Figure 113: Operation Equation Editor annotated.

Part II – Visualization of Imaging & NanoPatterning

Part II of this manual is focused on the Visualization portion of the system that allows the user to process both imaging and patterning data to maximize the usefulness of the visual display. The following sections describe a number of additional tools and features that aid in the visualization, display and storage of images. These functions are located in the Layout Panel, Tools Panel, and the Parameters Panel, as shown schematically in Figure 114.

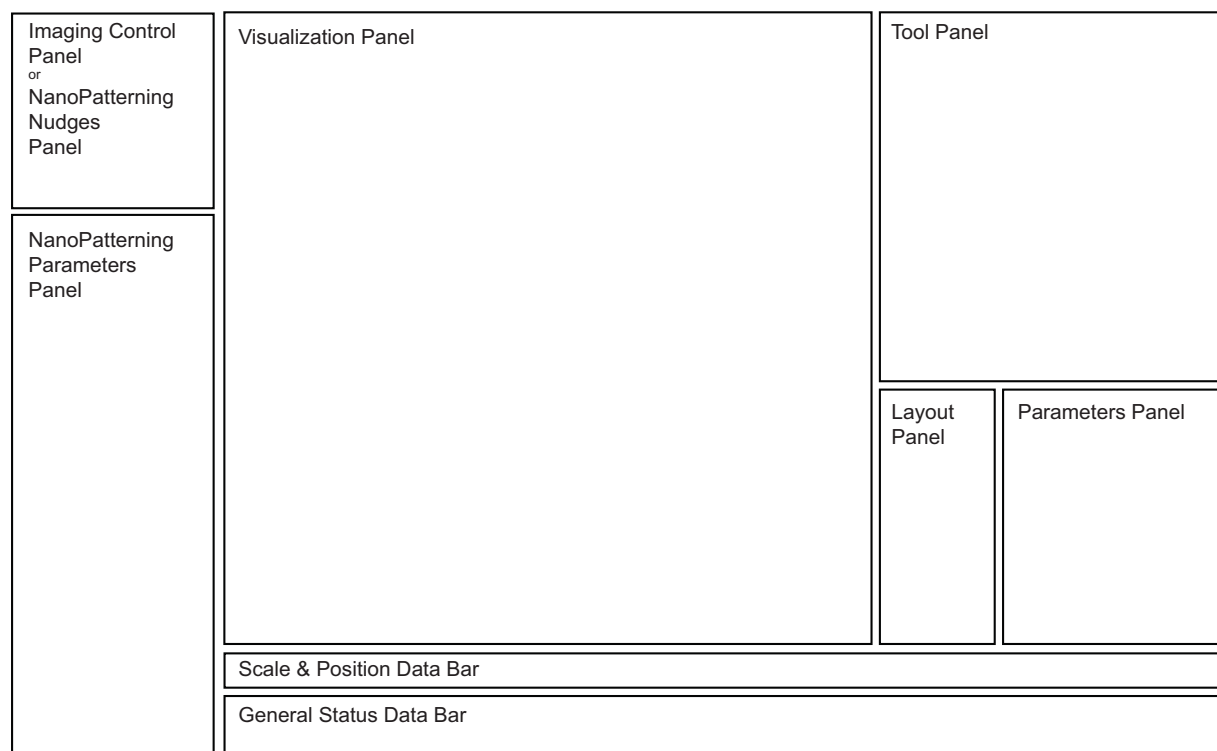


Figure 114: Panels related to the Visualization of Imaging and NanoPatterning portion of the NPVE.

Once again, as in Part I, While the **Standard** NPVE system is quite powerful, a number of additional capabilities are available in the **Advanced** version of the software, which provide specific solutions and enhancements to address a wide range of applications. Throughout this manual, the features that are only available in the **Advanced** version are described where most relevant, and are indicated by the symbol **ADV** so as to indicate to the user that these features are available only in the **Advanced** version. Note that the “Mosaic Tool” capability described in Part III is virtually identical to the “ATLAS” system available for Zeiss Scanning Electron Microscopes, and is only available in the **Advanced** version of the NPVE. The **Advanced** version includes a powerful add-on that adds many capabilities that enhance the visualization and recording of patterning details that are of particular interest to the advanced FIB user, or SEM users investigating gas chemistry effects or electron beam induced transformations. In both the **Standard** and **Advanced** version, the NPVE displays all patterning true to scale in the Visualization Panel, within the bounds of the shape(s) being patterned. With the **Advanced**, additional visualization modes are enabled that allow the user to dynamically view and process each pattern individually and display graphical endpoint signals from all or selected patterned shapes, and from regions within patterned shapes. Details of the parameters for each shape are recorded by the Project Analysis and Recording engine that is standard in the NPVE, but this data is significantly enhanced with

the addition of automatic recording of endpoint graphs and synchronized thumbnail representations of selected patterned shapes throughout the patterning process. For reference, features that are enabled or greatly enhanced in the **Advanced** version will be denoted by the ☒ADV symbol.

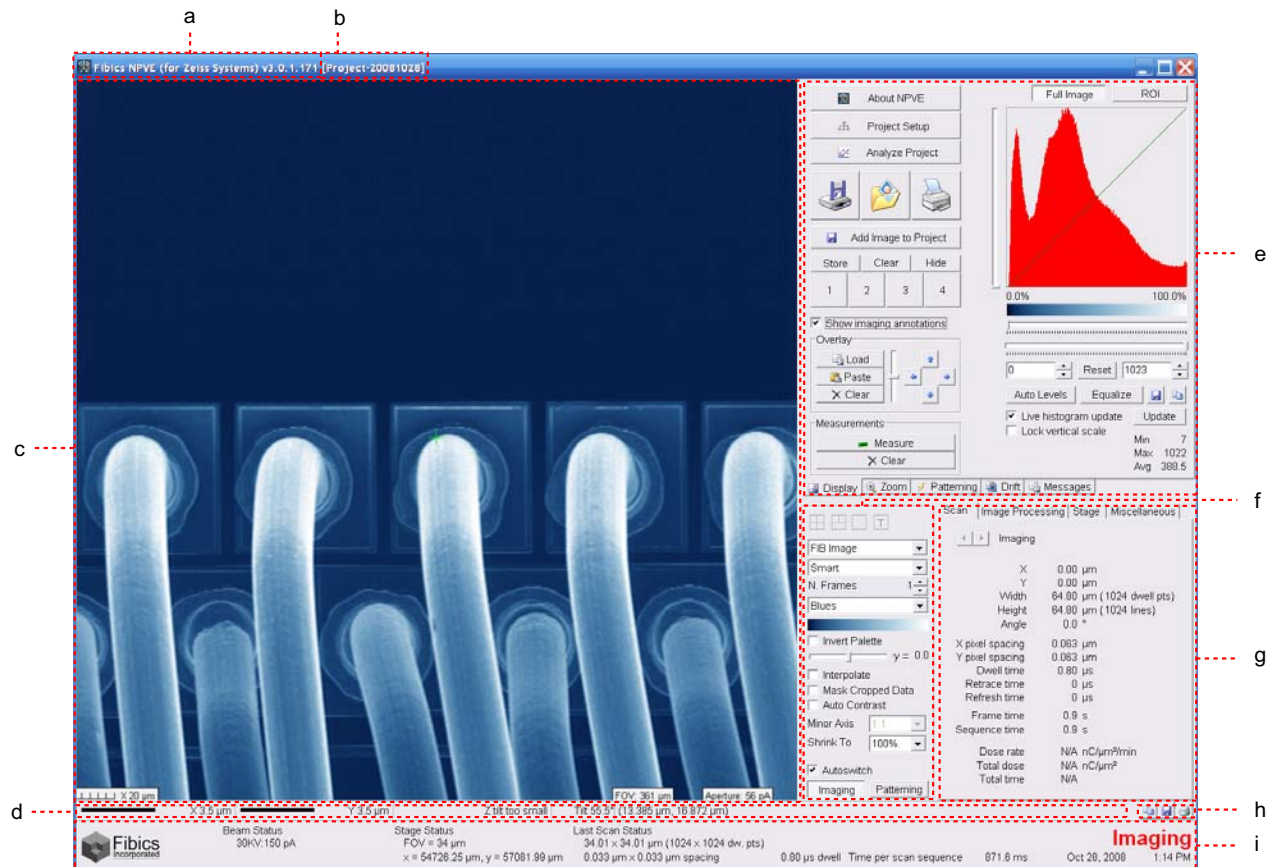


Figure 115: Overview of the panels related to Visualization in the NPVE.

- (a) NPVE version and build information
- (b) Project Name: User entered or auto-generated
- (c) Visualization Panel, which can display image, patterning, and graphical data
- (d) Scale and Position status bar
- (e) Tools Panel with Multiple Tabs
- (f) Layout Panel
- (g) Parameters Panel with Multiple Tabs
- (h) Full-window **Quick Save** buttons
- (i) General status bar

The NPVE version and build information (a), above) is relatively self explanatory, consisting of a multi-decimal number reflecting the exact revision of the software. When contacting Fibics with any inquiries, please be sure to note this number so we can determine exactly which version and build you are running.

The Project Name (b), above) is displayed either as the value entered by the user (described in the section on the Tools Panel: Display Tab: **Project Setup** Button) or as an auto-generated value based on the date in the format "Project-YYYYMMDD".

The Visualization Panel (c) is a flexible display area that shows either imaging or overlaid patterning data or, in the Advanced version of the NPVE, can also show detailed patterning data in several forms depending on the settings selected in the Layout Panel (f), described below. When displaying patterning

data ☒ADV, each nanopattern is displayed in a square area that can contain multiple tabs where data can be displayed using different processing algorithms and settings.

In the sections that follow, the function of each of the panels named above will be described in detail, as well as the interaction between panels.

To best understand the operation of the software, we will start with a description of the Layout Panel ((f), above) and each of its subpanels, and how they affect the display in the Visualization Panel. Following this, a description of the tabs in the Tools Panel ((e), above) will describe the various tools available in the NPVE that assist in both graphical and visual endpointing, Project Analysis and Recording, et cetera, and finally a discussion of the speed buttons and FIB status information available in the NPVE interface.

10. Tools Panel

The Tools Panel interacts closely with the Layout Panel to determine the majority of the functionality of the NPVE Visualization modules. From the Tools Panel, the following tabs are available:

10.1. Display Tab

From the Display Tab the user can extensively control how images or patterns are displayed in the Visualization Panel.

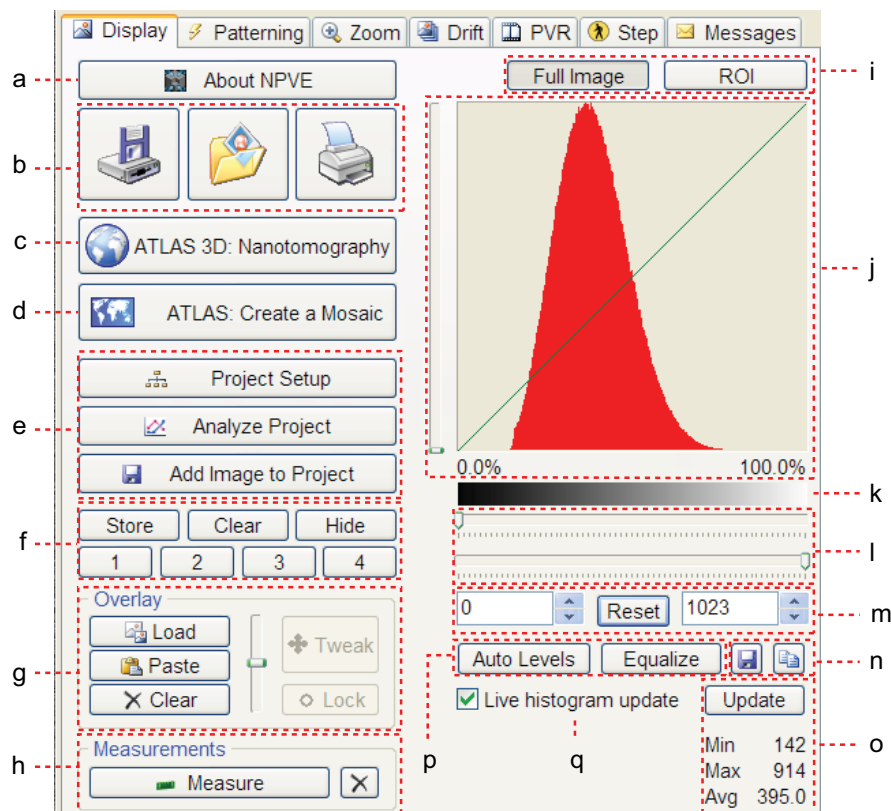


Figure 116: Display Tab annotated.

- a. **About NPVE** Button
- b. Save/Load/Print Image Buttons
- c. **ATLAS 3D: Nanotomography** button
- d. **ATLAS: Create a Mosaic** Button
- e. NPVE Project Buttons
- f. Image Memory Buttons
- g. Image Overlay Controls
- h. Measurement Tools
- i. Histogram Region Controls
- j. Histogram
- k. Applied Palette Representation
- l. Min and Max Sliders
- m. Min and Max Enter Values
- n. Save and Copy Histogram
- o. Min, Max and Average Readings

- p. Contrast Enhancement Buttons
- q. Live Histogram Update Checkbox

The Display Tab provides the user with controls for manipulating the image display and recording or performing analysis of a Project. The NPVE defines a Project as a series of images and shape patterning processes that are either named by the user (through the **Project Setup** button) or named chronologically by date and time, automatically ending one project and starting another at midnight.

The many components of the Display Tab are labeled “a” through “q”, above. The functions of these components are described below.

About NPVE (a)

This button provides information about the current version of the NPVE software.

Save/Load/Print Image Buttons (b)

These three buttons allow the user to save and print the current view (Imaging or Patterning view) as well as load a prior image into the Imaging view in order to view or print it. The NPVE is capable of generating very high resolution images that can be one gigabyte (or more) in size. These large images can be viewed with the VE-Viewer software provided with the NPVE software installation package.

Note:

When images are saved via the **Save Image** button in the Display Tab, NPVE always saves the image at the display resolution (1024 by 1024 pixels) for scan resolutions of 1024 by 1024 dwell points and smaller.

NPVE Project Buttons

The three controls of the **Project** Buttons allow the user to get information on the version of the NPVE software and hardware that is running (About NPVE), to setup details related to project recording / job tracking (Project setup), and to access a separate form to analyze recorded details about a project (Analyze Project).

ATLAS 3D: Nanotomography Button (c)

ATLAS 3D is a tool by which a sequence of cross-section images (stack of images) is acquired using the SEM beam while the FIB beam progresses/mills through the sample. Once acquired, the stack of images can be used for volume reconstruction.

See the ATLAS 3D User Guide for more detailed information regarding ATLAS 3D.

ATLAS Mosaic Tool (d)

The ATLAS Mosaic Tool provides the ability to define and execute a large sequence of images which can subsequently be stitched together (commonly referred to as an image “mosaic” or “montage”) to form a high resolution “extreme field of view” image of the sample. Please refer to the ATLAS User Guide for further discussion on the ATLAS Mosaic Tool.

NPVE Project Buttons (e)

The three **Project** buttons allow the user to set up details related to project recording / job tracking (Project setup) and open a separate form to analyze recorded details about a project (Analyze Project). When **Project Analysis and Recording** is enabled, the one-click **Add Image to Project** button saves the current image to the project for later analysis.

Image Memory (f)

Four image memories are accessible via these buttons. The user can store up to 4 images (Imaging view) in memory and recall each of them with a single click at a later time, much like the memory of a hand calculator.

To store the current image, first select the **Store** button, and then click on the **Memory** button (1 to 4) where you wish to store the image. Once an image is stored, pressing the **Memory** button recalls that image.

Clear removes all images from stored memory and **Hide** simply restores the most recent Imaging view.

Hint:

When **Project Analysis and Recording** is enabled, all single scan images acquired during a given project are saved to disk. These stored images can be viewed by using the **Page Up** (to move backwards) and **Page Down** (to move forwards) keys to step through the acquired images in sequence. The **Home** and **End** keys can also be used to go to the first or last image acquired during the current project. When viewing images other than the most recent single scan, the relative “Index” of the image being viewed, i.e. “(last image –1)” appears in the NPVE title bar at the top of the application window next to the version and project information.

Imaging Annotations Checkbox

The **Show Imaging Annotations** checkbox allows the user to display or hide information related to the imaging (e.g. scale bars, FOV, imaging mode, etc.) When checked, the image annotations are displayed directly on the image and will be visible when the image is saved and recalled.

Overlay Image Buttons and Overlay Image Transparency Slider (g)

The **Overlay Image** buttons allow the user to load an image from disk and overlay that image onto the Visualization Panel. Once an image file is selected for loading, the software will automatically bring up a tool to register the Overlay Image onto the imaged sample. This is done by allowing the user to select identical points on the “live” and “loaded” images. Once registered, the overlaid image is displayed over the live image, with the degree of transparency controlled by the transparency slider (f). The **Clear** button is used to completely remove the overlaid image.

Note that, once registered, the overlaid image is locked to the relevant stage position (if it is available) and will stay locked to that position as the stage is moved or the field of view (FOV) of the microscope is changed. Changes to the image overlay alignment can be made using the **Tweak** and **Lock** buttons.

Measurement Tool (h)

This tool (h) allows the user to make simple measurements on the image displayed on screen. The measurements are overlaid on the image when the user clicks, drags and releases the left mouse button. Pressing the **Clear** button removes any overlaid measurements.

Note

These measurements do not change (i.e. they are not dynamically updated) as the FOV is changed.

Histogram and Related Controls (i through q)

The right side of the Display Tab shows the intensity histogram of the current view and other controls related to the display of the image intensity.

The histogram is calculated from either the Full Image or a Region of Interest (ROI). Pressing the **ROI** button and dragging the mouse across the image will select a region of interest from which the histogram is calculated (i).

The histogram (j) shows the relative distribution of intensities from low to high in red, with the shape of the user selected intensity palette shown in overlay as a green line. The slider at the left of the histogram changes the vertical scale that is displayed.

The Applied Palette Representation (k) is displayed below the histogram, and is graphically identical to the palette representation which is displayed in the Layout Panel. Here, however, the Applied Palette can be manipulated, which results in a change in the way all images in the Visualization Panel are displayed, without requiring adjustment of the microscope's own Brightness and Contrast values.

The most basic manipulation involves changing the Min or Max sliders (l) or direct input as numeric values (m). Changes here adjust the minimum and maximum range over which the image is displayed, directly affecting the apparent brightness and contrast of the image. Pressing the **Reset** button (m) returns to the full range. Clicking and dragging the mouse on the Histogram will also select a Min to Max range over the extent of the histogram that the user drags the mouse.

Note that the histogram image can also be copied to the clipboard () or saved to a file () , with the speed buttons in section (n).

10.2. Patterning Tab

The Patterning Tab allows the user to view (and review) thumbnail images from the patterning of each shape, and the pattern-induced signal intensity endpoint graphs and reconstructed cross-sections to examine the three dimensional nature of the data resulting from shape patterning operations.

The Patterning Tab consists of four regions (as seen in Figure 117) :

- First Multi-Display pane capable of displaying Endpoint Graphs, patterned shape Thumbnail Images or Cross-Section Reconstructions digitally created from patterned shape data
- Second Multi-Display pane
- Multi-Display Pane Selector** buttons
- Control** Buttons

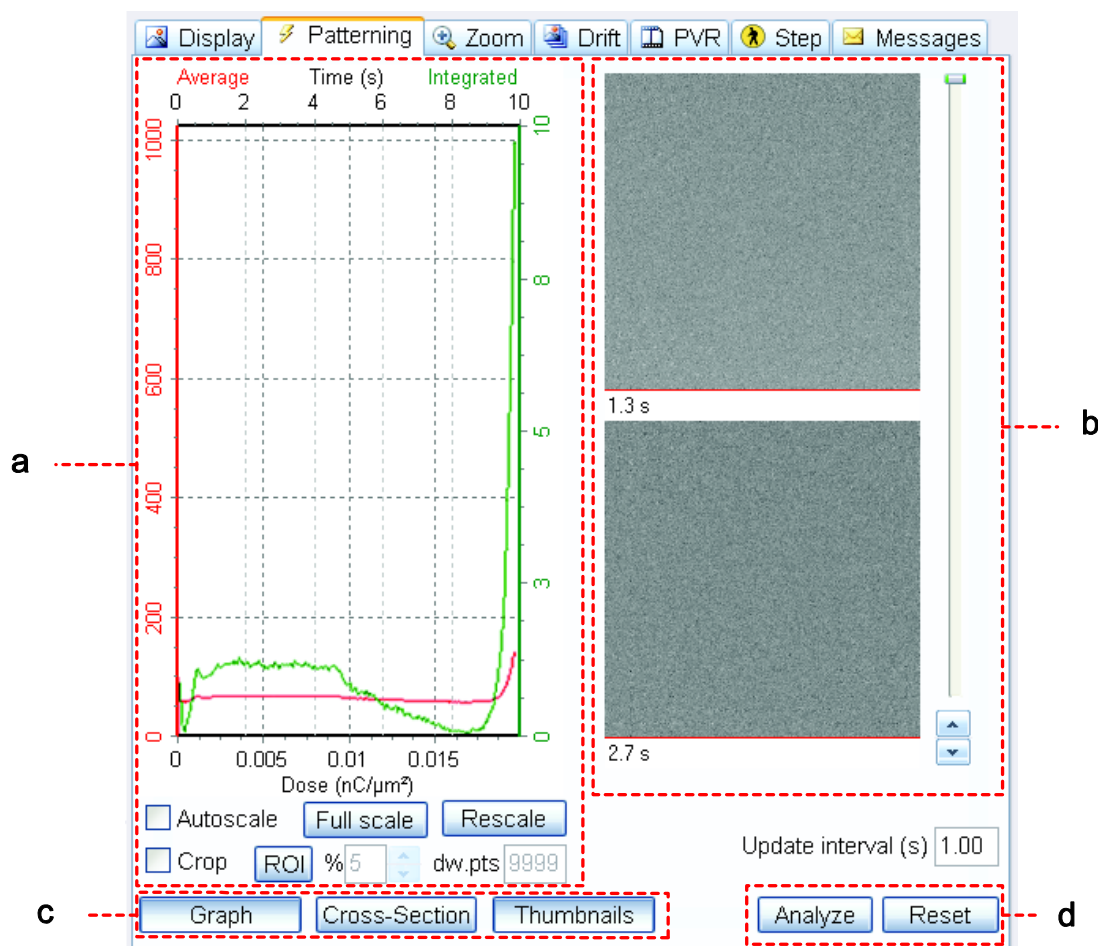


Figure 117: Patterning Tab annotated.

Each Multi-Display pane can show one of three options. Pressing a **Pane Selector** button (c) adds the desired display (Graph, Cross-Section or Thumbnails) to the left hand pane (a), moving the previous contents of the left hand pane to the right hand pane (b).

The two **Control** Buttons (d) are **Analyze** and **Reset**. **Reset** simply resets the data in the Multi-Display panes back to zero, as if patterning a new shape has been started, i.e. the Endpoint Graph will be cleared and restarted at a reading of 0.0 nC/μm². The **Analyze** button will be “greyed out” during patterning or imaging, but becomes active when the execution of patterning or imaging is paused. When active, pressing the **Analyze** button will transfer the user to the Analysis page for more detailed analysis of the current patterning operations than can be performed in the Multi-Display tabs. A description of the operation of the Analysis page can be found in the Project Analysis section.

The Endpoint Graph ☒ADV

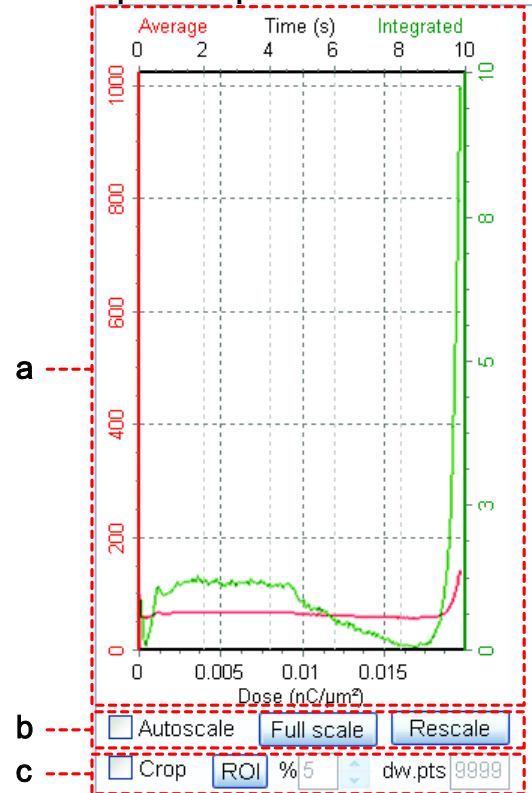


Figure 118: The end point graph under the Patterning Tab.

The Endpoint Graph displays the image intensity as a function of **Dose** and **Time**.

The graph display (a) for Intensity graphs shows two lines – red for the Averaged values (read off the left hand axis, always between 0 and 1023) and green for the Integrated values, (read off the right hand axis). The Integrated number corresponds to the sum, over all pixels, of their brightness (0-1023), so it reflects the brightness as well as the size of the mill box.

Checking **Autoscale** in (b) causes the Integrated graph range to dynamically re-scale the “y” axis from the minimum to the maximum values of the data displayed on the “x” axis. The **Full Scale** button re-plots the Integrated data from the minimum to maximum “y” axis values for *all* data points resulting from the execution of this pattern, regardless of whether they are currently displayed, while the **Rescale** button performs the same re-plot, but only for the range of “y” axis values presently displayed. In all cases, if the Integrated intensity graph were to exceed the upper or lower “y” axis limits, the graph will re-scale.

The **Crop** controls, (c), above) cause the Intensity graph to be generated from a subset of the patterned shape data, discarding the outer perimeter dwell points. The number of perimeter dwell points to be discarded are specified as a fixed percentage and an absolute number. Only the percentage can be edited, but the number of actual dwell points is automatically updated. Also, it is possible to select an arbitrary rectangular region of interest (ROI). To do so, select the **ROI** button and then click and drag the left mouse button over the region of interest on the display of the active pattern in order to select a rectangle. Once a **ROI** is selected, only dwell points within it will be used to calculate the graph. Note that checking the **Mask Cropped Data** checkbox in the Layout Panel will cause the selected cropped region to be removed from the display of the image derived from the patterned shapes.

The Thumbnail View ☒ADV

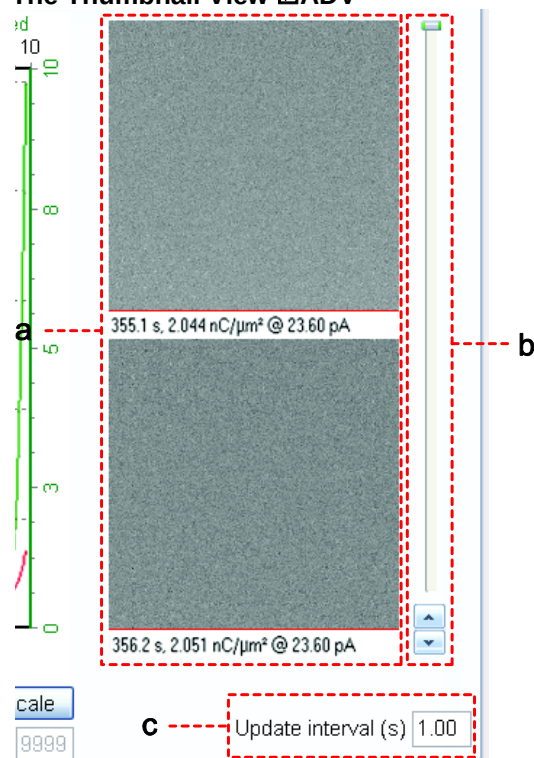


Figure 119: The thumbnail view available under the Patterning Tab.

The **Thumbnail** view allows the user to review a series of thumbnail images that are acquired in sequence during shape patterning operations. The display portion (a) of the **Thumbnail** view shows two thumbnail images, annotated with the time, in seconds, since the start of patterning execution corresponding to the point when the thumbnail was stored, the cumulative dose at that time and the most recently measured/registered beam current. The **Position Slider** (b) can be used to move through the thumbnails. When the slider is moved fully to the bottom, the most recent thumbnail is displayed. The list automatically increments to keep the most recently acquired thumbnail in view.

Reviewing thumbnails by scrolling up or down during patterning can be extremely helpful in terms of obtaining a perspective on the patterning process.

The collection of Thumbnails is cleared upon execution of each new pattern; however, the thumbnails are automatically saved when **Project Analysis and Recording** is enabled.

The thumbnails are generated using the display settings defined for the Thumbnail imaging view.

The frequency with which new thumbnails are added to the list is controlled by setting the **Update Interval**, (c). The user may select any value, in seconds, from 0.25 seconds or greater. The default is 1.0 seconds. If the user performs a very long patterning sequence with a short update interval (i.e. over 20 minutes of continuous patterning at a 0.25 second update time), the list will fill the internal buffer such that the NPVE engine will automatically double the update interval and purge every other thumbnail at this point, so as to constrain the memory requirements. Note that this still represents many thousands of thumbnail images and has not proven to be a limitation in normal operations.

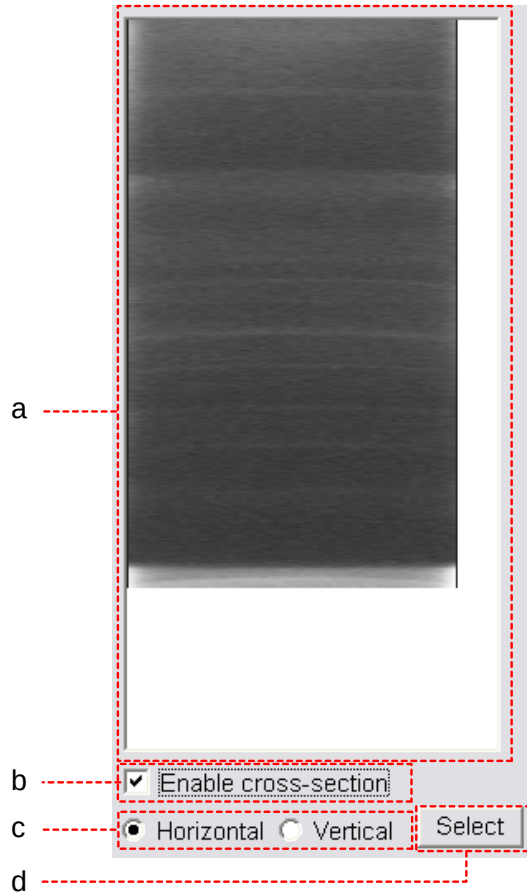
The Cross-Section View ☒ADV

Figure 120: A screenshot of the cross-section view.

The **Cross-Section** View reconstructs a cross-sectional image (a) using the data recorded from the image thumbnails throughout patterning. The reconstructed cross-section can be very useful when combined with the endpoint graph and thumbnail view data in order to understand the three dimensional nature of where the information (endpoint signal) is coming from. In the example shown to the left, a metal line has appeared at the bottom right of the reconstructed cross section, and individual capping layers present between each dielectric layer can be seen. A line in the cross-section is added every time a thumbnail is produced, and corresponds to a vertical or horizontal slice of the thumbnail. As such, the horizontal axis of the cross-section corresponds to space and the vertical axis corresponds to time (or equivalent dose and is thus somewhat related to depth). The cross-section is only produced during a milling operation if the **Enable Cross-section** box (b) is checked. **Thumbnails** must also be enabled. The user can also select whether the cross-section is created from horizontal or vertical (c) slices of the thumbnail images. Pressing the **Select** button (d) will bring up the Thumbnail view on which the location of the desired cross-section can be defined with the mouse. With the left mouse button down, select the centre location, drag the mouse cursor, and then release the mouse to select a width for line averaging and thus a better signal to noise ratio).

10.3. Zoom Tab

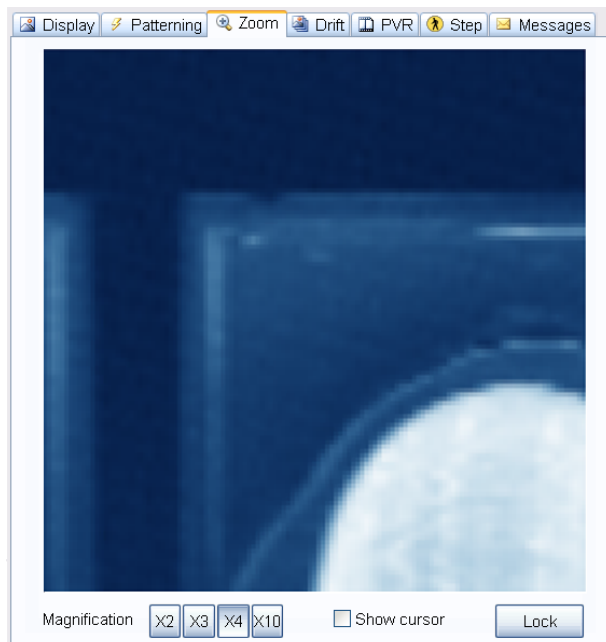


Figure 121: The Zoom Tab.

The Zoom Tab shows a magnified image of either the Imaging or the Patterning view. A magnification factor of 2, 3, 4 or 10 can be selected.

When the Zoom Tab is selected, the magnified image follows the mouse cursor from view to view. To Lock the image to a specific location, click on the Lock button with the left mouse button, followed by the target location on the Visualization Panel. To Unlock, press the **Unlock** button on the tab or click the view with the right mouse button. The zoom image is updated approximately 5 times per second so it does update during patterning, but not as fast as in the Visualization Panel, which can be updated at least 20 times per second. In practice this is not an issue, but can lead to momentary flicker in the Zoom image.

10.4. Drift Tab

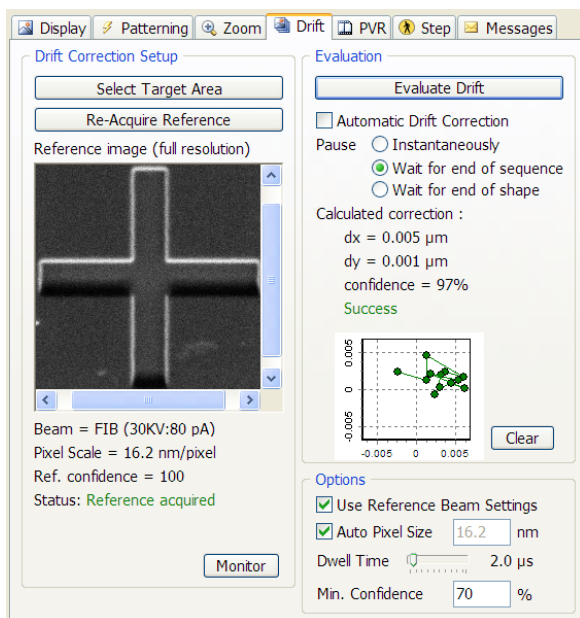


Figure 122: The Drift Correction Setup Tab.

Drift correction can be set up through the Drift Tab. Setup consists of defining the pixel resolution and dwell time used to acquire the region of interest, as well as selecting the actual area used to measure the amount of drift.

By default, when the Auto Pixel Size checkbox is checked, the pixel size is 1/4096 of the current FOV. When unchecked, the user can specify a desired pixel size.

Select the target region of interest by clicking the **Select Target Area** button and clicking on the Image view. The size of the target area can be modified by scrolling the mouse wheel. When selected, a reference image will be acquired and used if it is adequate.

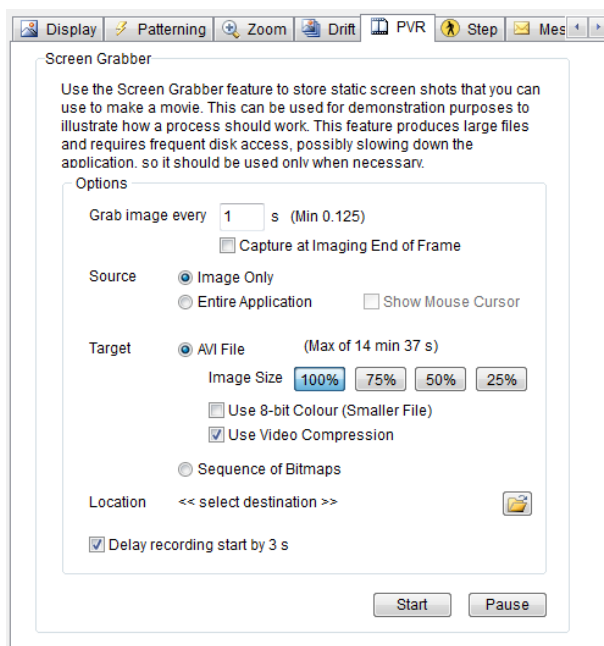
You may re-acquire a new reference image of the same region of interest by clicking the **Re-Acquire Reference** button. This new image will now be used as the reference.

After a reference has been acquired, the user may manually check and correct for drift by clicking the **Evaluate Drift** button. Radio buttons provide the option to pause patterning and evaluate the drift immediately, pause the patterning at the end of the current patterning frame, or pause the patterning at the end of the shape. After the patterning is paused, the evaluation sequence will acquire the new image of the region of interest, evaluate the drift and apply the necessary correction if the confidence of the algorithm is greater than the Min Confidence parameter set in the Options section. Once the correction is done, patterning is resumed. If the correction fails, the patterning will remain in a paused state, waiting for the user to resume.

By checking the Automatic Drift Correction checkbox, the user will be asked to enter a time interval between corrections. Once setup, the system will automatically pause, correct for drift and resume patterning at the specified interval (pausing always occurs at the end of a frame).

When a drift value is measured, the results are displayed as the Calculated correction, with a message indicating the success or failure of the calculation. All corrections are plotted on the associated graph to allow the user to visually track the history.

10.5. PVR Tab



The PVR (personal video recorder) Tab allows the user to capture and store static screen shots taken according to a specified timeframe.

Images can be stored individually as bitmap files, or AVI file format movies can be automatically generated.

Figure 123: The PVR Tab.

10.6. Step and Repeat Patterning Tab

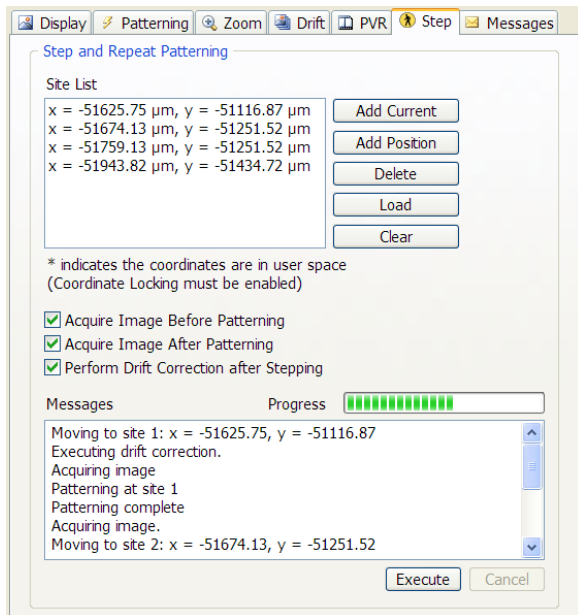


Figure 124: The Step and Repeat Patterning Tab.

The Step and Repeat Patterning tool allows the user to execute a series of patterns at various stage locations.

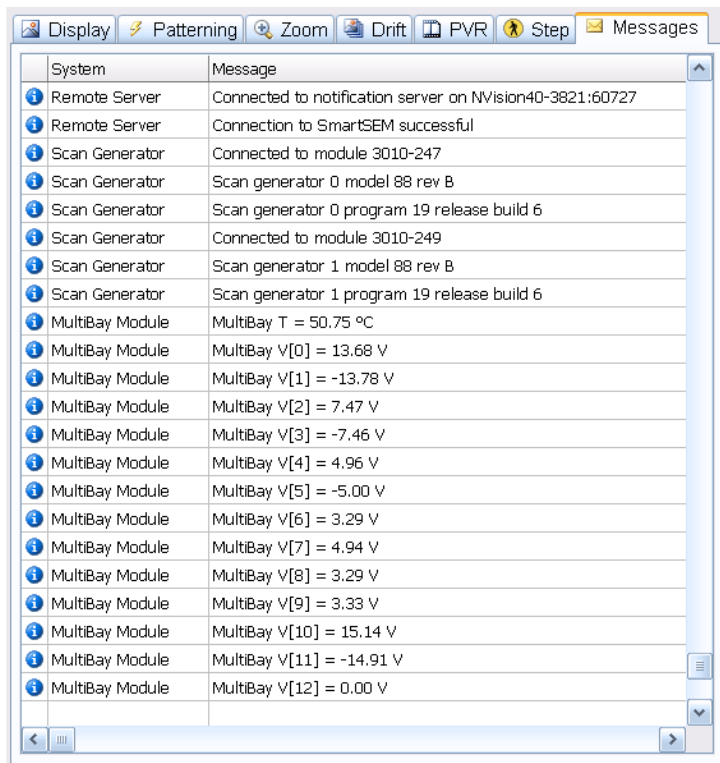
The Site List shows the list of stage positions at which the patterning will be repeated. To add a site, move to the desired site and click the **Add Current** button. Alternatively, the user can click the **Add Position** button to manually input the position, or load a list of sites from a comma delimited text file.

To execute, select all the shapes that you wish to pattern and click the **Execute** button. Currently it is only possible to do all shapes in series or in parallel. It is not possible to define a sequence where some shapes are milled in parallel and then other shapes in series.

If the Acquire Image Before and After Patterning boxes are checked, an image will be taken after moving to the new stage position and after the patterning is complete to provide a record in the current project. The option to perform drift correction after stepping is also available.

10.7. Messages Tab

The Messages Tab allows a user to view various configuration information, communication steps and commands sent between the microscope and the NPVE system. It will display the start and end of processes, flag communication errors and provide a method through which one can check the tool status.



System	Message
Remote Server	Connected to notification server on NVision40-3821:60727
Remote Server	Connection to SmartSEM successful
Scan Generator	Connected to module 3010-247
Scan Generator	Scan generator 0 model 88 rev B
Scan Generator	Scan generator 0 program 19 release build 6
Scan Generator	Connected to module 3010-249
Scan Generator	Scan generator 1 model 88 rev B
Scan Generator	Scan generator 1 program 19 release build 6
MultiBay Module	MultiBay T = 50.75 °C
MultiBay Module	MultiBay V[0] = 13.68 V
MultiBay Module	MultiBay V[1] = -13.78 V
MultiBay Module	MultiBay V[2] = 7.47 V
MultiBay Module	MultiBay V[3] = -7.46 V
MultiBay Module	MultiBay V[4] = 4.96 V
MultiBay Module	MultiBay V[5] = -5.00 V
MultiBay Module	MultiBay V[6] = 3.29 V
MultiBay Module	MultiBay V[7] = 4.94 V
MultiBay Module	MultiBay V[8] = 3.29 V
MultiBay Module	MultiBay V[9] = 3.33 V
MultiBay Module	MultiBay V[10] = 15.14 V
MultiBay Module	MultiBay V[11] = -14.91 V
MultiBay Module	MultiBay V[12] = 0.00 V

Each line in the Messages Tab contains an icon identifying the purpose of the message, the System the message relates to, and the message itself.

The icons and their corresponding meanings are listed below:

-  Information
-  Notice
-  Warning
-  System Critical

It is also useful to scroll this list to the right in order to view a timestamp for each message.

Figure 125: The Messages Tab.

11. Layout Panel

Using the Layout Panel, the user can select a desired Visualization mode as well as control a selection of the display parameters to optimize the visual quality of the imaging and patterning. The different regions of the Layout Panel are identified below, and the operation of each region is described in detail in the section that follows.

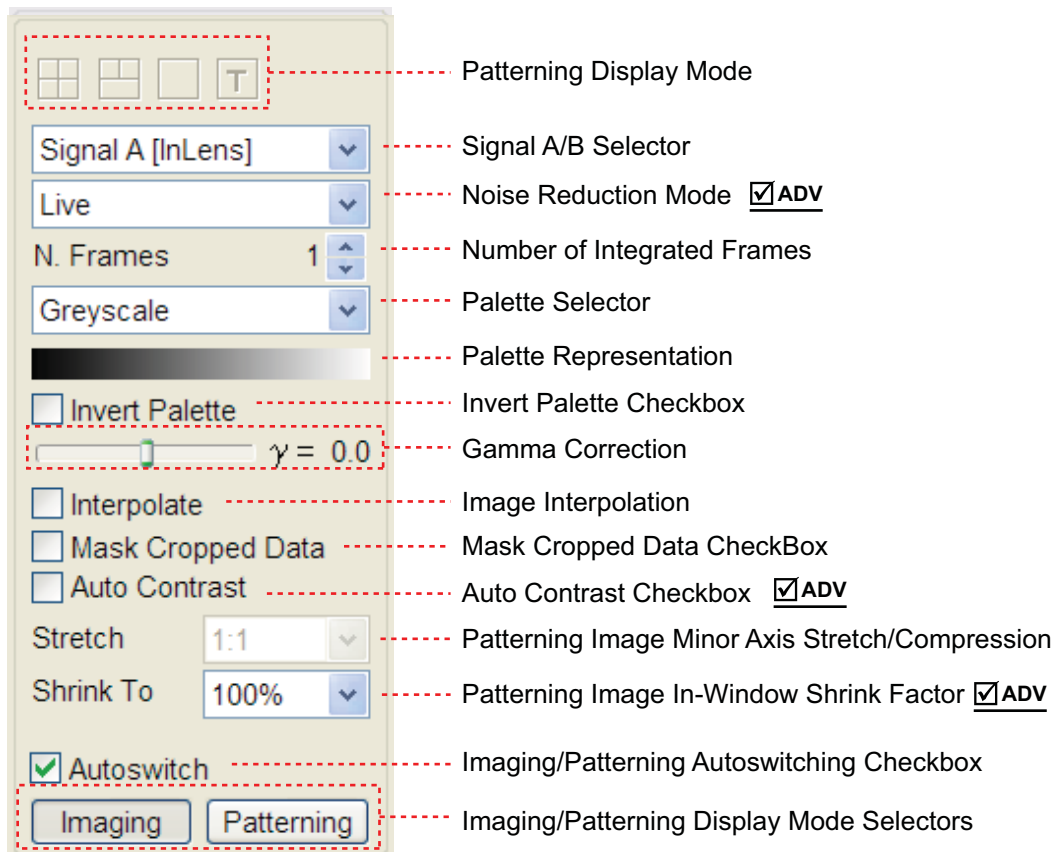
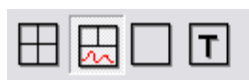


Figure 126: Layout Panel Annotated.

Patterning Display Mode **ADV**



Selects the visualization mode used when patterning is displayed in detail.

The options are, from left to right, quadrant view where each quadrant view can be customized, a mixed mode where two quadrant views and a large version of the endpoint graph are displayed, a single view where a large image of a selected pattern is displayed in the large view, and a thumbnail view that defines the display properties of the generated thumbnails and a live digitally generated cross-section.

Many factors come into play in selecting the optimal **Patterning Display** mode. For some patterning operations, it is preferable to enlarge the display of a given patterned shape to as large as possible, making the single view ideal. For the case of shapes with patterning occurring in parallel, it may be preferable to see as much detail on as many shapes at one time as possible, making quadrant view the best choice. Most of the time, however, it would appear that users prefer the “mixed mode”, with two quadrant boxes allowing detailed display of two shapes (or one shape displayed with two different

parameters sets, such as different palette and image processing options, discussed in detail below) plus an extended graph of the selected intensity signal.

Desired Imaging Mode



A drop down listbox selects the imaging mode used to display the imaging or optional patterning data ☒ADV. There are several modes that can be selected:

Smart

Displays individual “frames” of data. When imaging, a live frame is a single rastered scan. When milling with **Smart** display mode selected, a displayed “frame” corresponds to either a single patterning scan through a shape or an average of the signal over $1/30^{\text{th}}$ of a second during multiple scans, whichever takes the longest time. For example, when patterning shapes, the time to complete a single pass of a given shape can be very short, so a ~ 33 milliseconds ($1/30^{\text{th}}$ of a second) average is taken as the shortest time slice. When the actual patterning time is longer, then a full frame is the shortest time slice.

Integrated

Displays a rolling average of frames of data. In imaging mode, at any given time, the displayed image corresponds to the average of the previous N frames, where N is defined by the user with the **Num Frames** (number of frames) selector. In patterning mode, the **Integration** time is specified instead of the number of frames. The actual number of frames used in the average will then depend on the duration of the actual milling frame.

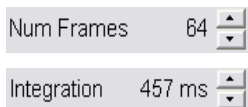
Snapshot

Displays integrated frames of data in quarter second intervals. The data displayed in the intensity graph is obtained from these images. They are only updated every quarter second (this is not a rolling average). *(Not available in imaging mode).*

Differential

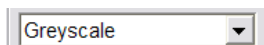
Displays the difference between the current Snapshot data and the 4 previous *Snapshot* data, weighted by 50%, 25%, 15% and 10% respectively. This results in persistence of changes for around 1 second. A region of a patterned shape that becomes brighter appears bright in the Differential image and a region of a patterned shape that becomes darker appears dark in the Differential image. In the absence of change, the data is centred around channel 511 on the 0-1023 range. This view is particularly good for determining regions of a patterned shape that have recently appeared or disappeared. *(Not available in imaging mode).*

Number of Frames



Specifies the number of frames used in the rolling average in the Integrated display mode while imaging, or the integration time while patterning shapes. Depending on the version of the NPVE software installed, the metric may be a number of frames or a time (in milliseconds or seconds).

Palette Selector



The base palette specifies which colours are assigned to the measured intensities. The default Greyscale palette provides a linear scale of greys between black (for no intensity) and white (for full intensity). Several other palettes are supplied for false colour rendering of the patterned shapes which can make small variations in the intensity significantly more noticeable. Palettes of particular interest include a Thermal palette ranging from black to white through the range of reds and yellows; and a Red-Green-Cyan palette where low signal is a deep red, medium signal is a solid green and high signal is a bright cyan.

Palette Representation



A graphical representation of the colours used to display the signal intensity from the lowest intensity (left hand side of the bar) to the highest intensity (right hand side of the bar). This Palette Representation is dynamically updated as settings such as the **Gamma** and the **Minimum** or **Maximum** of the Tools Palette Display Tab are changed, regardless of whether the NPVE is actually scanning the beam.

Invert Palette



Selecting this checkbox inverts the order of the palette. In the case of the greyscale palette, this results in white being used for no intensity and black for full intensity.

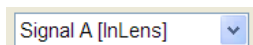
Gamma Correction



Provides gamma (γ) correction for the palette. Gamma correction is used to induce a non-linear relationship between the actual signal intensity and the way it is displayed. This can make the image more sensitive to small changes at the low or high intensity end of the signal range. A gamma of 0.0 results in a linear relationship. A positive γ results in low intensities appearing brighter more rapidly with the response at high intensities remaining relatively unchanged. A negative γ results in high intensities appearing brighter more rapidly with low intensities being relatively unchanged.

The user can select any gamma value from -2.0 to +2.0. The default is 0.0 (linear relationship). The shape of the gamma “curve” is displayed as a green line on the Histogram of the Tools Panel Display Tab.

Signal Selector



The signal selector allows the user to specify whether Signal A or Signal B is displayed in the Visualization Panel. The Signal Selector also displays which detector is assigned to each signal.

Hint:

To output Signal B from the microscope plinth, Dual-Channel must be activated in SmartSEM.

To control Signal A, the anchor in SmartSEM must be set to the main form. Signal B can be controlled when the anchor is placed on the zone B display (secondary display). When switching between both detector inputs, NPVE takes care of sending the anchor to the appropriate zone.

Image Interpolation☒ Interpolate

Selects whether the image is scaled to fit the window with or without “smart” interpolation. Without interpolation turned on, each dwell point results in a distinct rectangle of uniform intensity on the image. If “smart” interpolation is turned on, a smoother image is obtained by interpolating values for the pixels on the display that fall between the exact locations of the dwell points in the displayed image, to a maximum number of interpolation steps, after which the image is uniformly scaled. This “smart” interpolation results in a smooth image which still gives the user a perception as to where the actual dwell points fall in the representation.

Mask Cropped Data ☒ADV☐ Mask Cropped Data

Selects whether the pixels that are cropped out for the endpoint graph should also be masked out from the display of a patterned shape. For example, this allows the user to hide the bright edges of a high aspect ratio mill to allow for better visual endpointing.

Auto Contrast ☒ADV☒ Auto Contrast

Selects whether the contrast should be automatically adjusted dynamically during patterning to maximize the visibility of the visual data from the patterning process without affecting the physical detector gain. When Auto Contrast is enabled, the user cannot change the histogram parameters as they automatically get recalculated every $\frac{1}{4}$ second. If Mask Cropped Data is checked, then the contrast is based only on the visible data. Note that it is often very effective to display two versions of the same patterned shape – one with and one without Auto Contrast enabled so as to maximize the information available.

Patterning Image Minor Axis Stretch / Compression ☒ADV

Minor Axis: 1:1

Displays the image of a patterned shape as stretched in the direction of the minor patterning axis. This is sometimes useful to expand the pixels perpendicularly to the scan direction, in particular when a large aspect ratio box (for example, one that is quite wide but not very high) is being milled, as often occurs during cross sectioning or the production of TEM samples with the FIB.

Shrink To ☒ADV

Shrinks the image of a patterned shape to a fraction of its full size. This can be useful to reduce pixilation when interpolation is turned off and when shapes with few dwell points are being patterned, as often occurs with high aspect ratio milling using gas assisted etching.

Autoswitch ☒ADV

When checked, the Patterning view is automatically selected when patterning starts and the imaging mode is restored when an imaging pass is performed. If the Autoswitch box is checked and the user selects the imaging view during patterning, the NPVE will automatically restore the milling view after 10 seconds.

Imaging / Patterning ☒ADV

Determines what view is currently displayed. The user can, at any time, select the Patterning view to view the last (or current) patterning images or the Imaging view to view the most recent image pass.

All these settings can be modified temporarily during a session or adjusted permanently as a layout scheme in the Preferences dialog box. Once defined as a scheme, the user can select a specific scheme at any time to customize the interface. Defining and selecting schemes are discussed in the section entitled Setting NPVE User Preferences.

12. Parameters Panel

12.1. Image Processing Tab ☒ADV

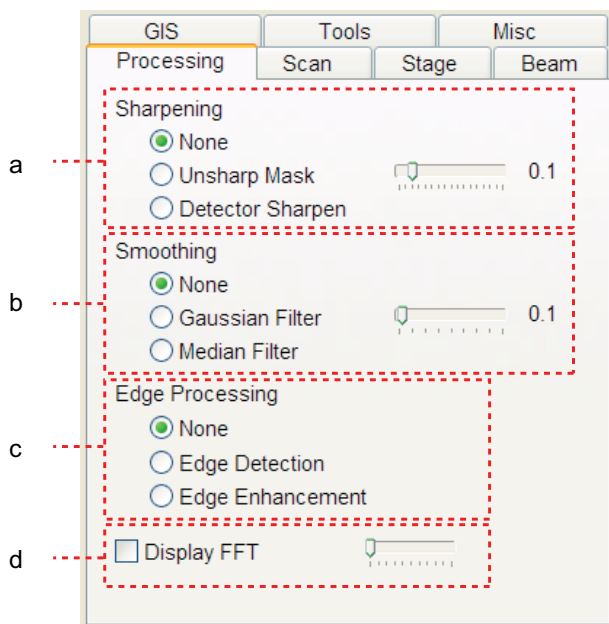


Figure 127: The Image Processing Tab under the Parameters Panel.

The Image Processing Tab provides the user with additional parameters for processing the display of data on the screen.

Selecting **Detector Sharpen** (a) enhances the perceived spatial resolution of the image but also tends to increase the amount of noise in the image.

Gaussian Filter (b) provides smoothing using a convolution with a Gaussian function for varying width. This is useful for reducing noise, but it also tends to blur the image. **Median Filtering** (b) allows the removal of speckled noise without degrading the spatial resolution.

Edge Processing (c) allows the user to detect or enhance changes in brightness in the image. Under certain conditions, edge detection can allow the user to see subtle changes in an image that are otherwise difficult to see. When imaging a highly textured sample, **Edge Enhancement** can be used to artificially promote edges for better visual recognition of features.

Real-Time Fast Fourier transform (FFT) is available (d). FFT can be used to grasp the frequency composition of an image and consequently assess the focus and stigmatism quality. The slider associated with the Display FFT control adjusts the brightness / contrast threshold setting of the displayed FFT.

Selecting **Gaussian Filter** combined with **Detector Sharpen** often reduces the amount of noise without compromising the spatial resolution.

Note:

FFT processing is only displayed with images whose dimensions are a power of 2, which is the standard imaging mode in the NPVE system.

12.2. Scan Environment Tab

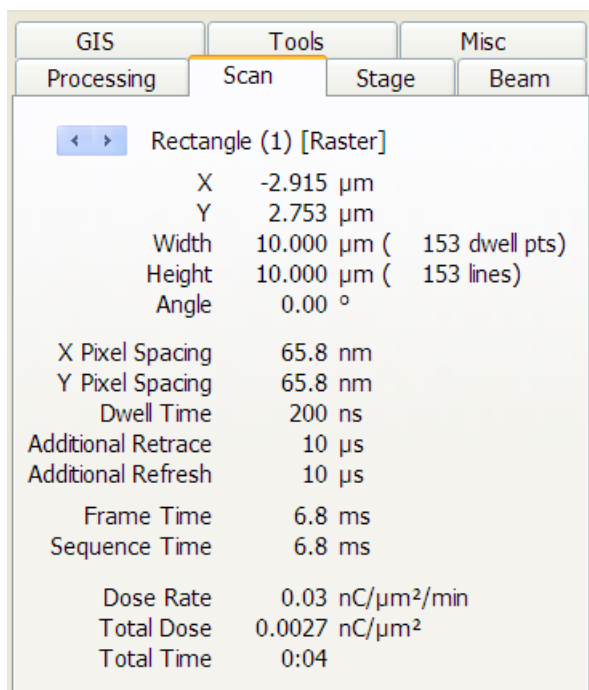


Figure 128: The Scan Environment Tab displaying information regarding the current scan.

The Scan Environment Tab provides information about the imaging and patterning parameters that are being used by the FIB to manoeuvre the ion beam over the sample.

This example at left shows raster conditions for imaging with a box of 10 μm by 10 μm . The arrow keys allow the user to step through either the Imaging Scan Environment or the currently active patterns, in sequence, with each pattern identified as "Pattern 1 - Rectangle", "Pattern 2 - Trapezoid", etc. in the form of "Pattern [ID] – [Name]".

X and Y Pixel Spacing and Dwell Time are the step sizes and dwell times of the raster. Retrace Time is the delay between successively patterned lines. Refresh time is the additional delay between successive frames.

Dose Rate gives the calculated dose per unit area delivered in one minute.

Total Dose shows the dose per unit area accumulated in the selected shape during execution of the current patterning request. Note that the dose is not calculated for imaging.

Total Time shows the total duration of the current patterning sequence.

12.3. Stage Tab

Parameter	Value
X	-51157.77 μm
Y	-51115.94 μm
Z	49106.37 μm
Rotation	347.29 $^{\circ}$
Tilt	-0.99 $^{\circ}$
M	5.000 mm

Pre-Tilt = 0.0 $^{\circ}$

Buttons: Move Stage, Eucentric, Centre Feature Vertically, Do Backlash, Initialize Stage, Define Sample Holder

The Stage Tab gives real-time information about the actual position of the stage and tilt compensation applied. This can be used to adjust the scan based on sample orientation.

If a sample is loaded on an angled stub, the stub tilt angle can be entered and accounted for using the **Pre-Tilt** button.

Centre Feature Vertically

The function is only available in FIB mode. The stage will move along the z and m axes to centre a given feature on the sample at the centre of the screen.

This tool can be used to bring a feature to the microscope coincidence point.

Figure 129: The Stage Tab in the Parameters Panel, displaying information regarding the current stage position.

Note:

The user is responsible for ensuring any stage motion will not harm the microscope or the sample.

Move Stage Panel

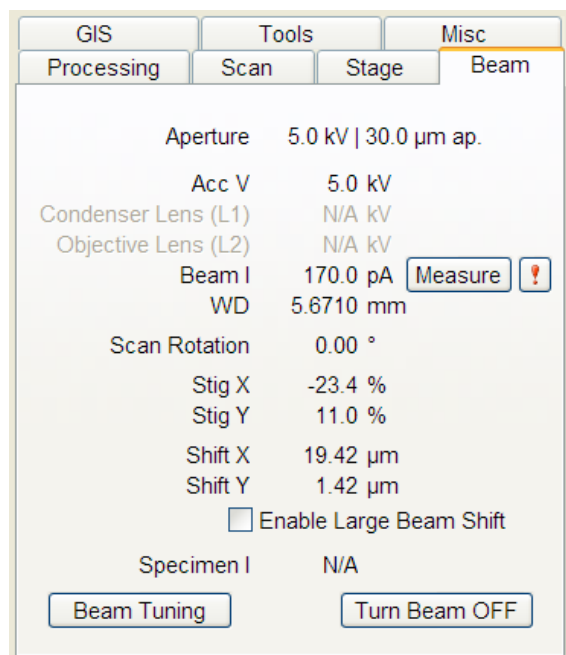
	Current	Target	Delta
X	-51160.9 μm	-51160.9	0.0
Y	-51115.8 μm	-51115.8	0.0
Z	49106.4 μm	49106.4	0.0
Rotation	347.3 $^{\circ}$	347.3	0.0
Tilt	-1.0 $^{\circ}$	-1.0	0.0
M	5.000 mm	5.000	0.000

Buttons: Go, Cancel

The NPVE allows the user to control the microscope stage position using the Move Stage Panel.

The stage can move to a target position or it can be moved by a given amount by entering values into the "Delta" column.

Beam Tab



Relevant information about the beam and microscope conditions are displayed under the Beam Tab.

Beam current can be measured by clicking the **Measure** button, or a value can be input by clicking on the red exclamation mark followed by the **Assign** button.

The **Enable Large Beam Shift** checkbox toggles the SmartSEM large beam shift functionality.

The **Turn Beam OFF** button automatically turns off the active beam.

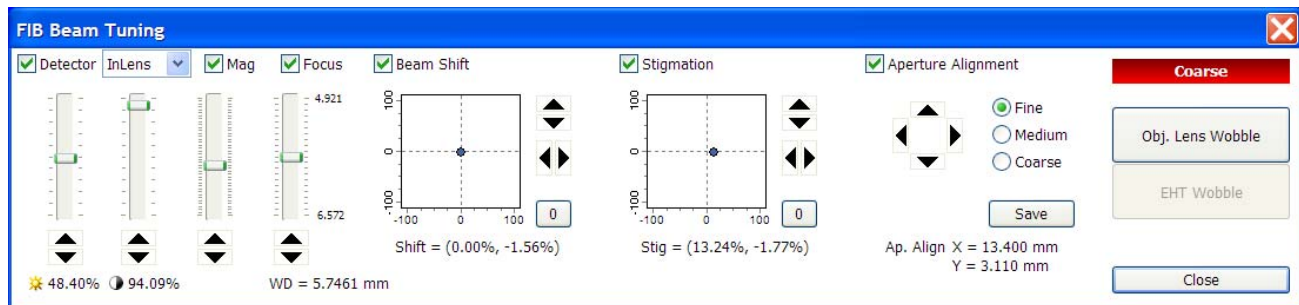
Figure 130: The Beam Tab in the Parameters Panel.

Beam Tuning Panel

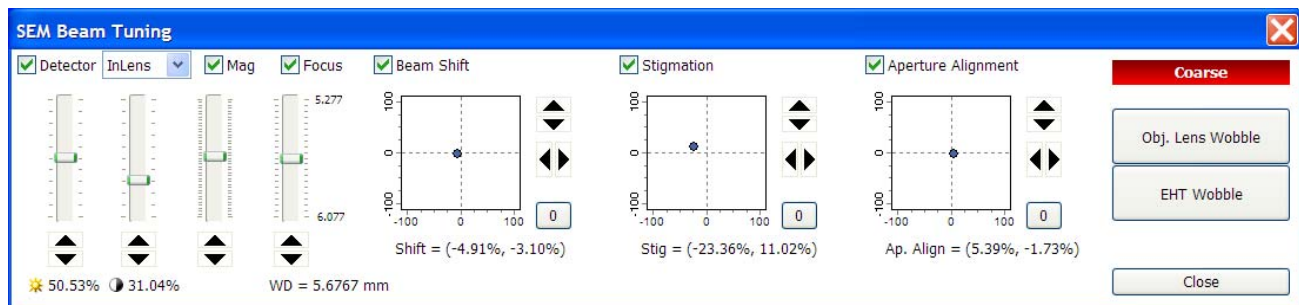
The Beam Tuning Panel is a convenient way to digitally adjust the microscope beam settings. The different panels or trackbars can be clicked and dragged to change the beam parameters. A more intuitive way of changing the parameter values is to position the mouse cursor over the black arrow heads and to scroll the mouse wheel.

The parameters that can be changed are:

- Brightness & Contrast
- Magnification
- Focus
- Beam Shift
- Stigmation
- Aperture Alignment



A



B

Figure 131: The Beam Tuning panels. A) For the FIB Beam. B) For the SEM Beam.

The mouse wheel step size and scrollbar sensitivity can be changed by toggling between “Coarse” and “Fine” mode using the **Coarse/Fine** toggle button in the top-right of each panel.

Gas Injection System (GIS) Tab

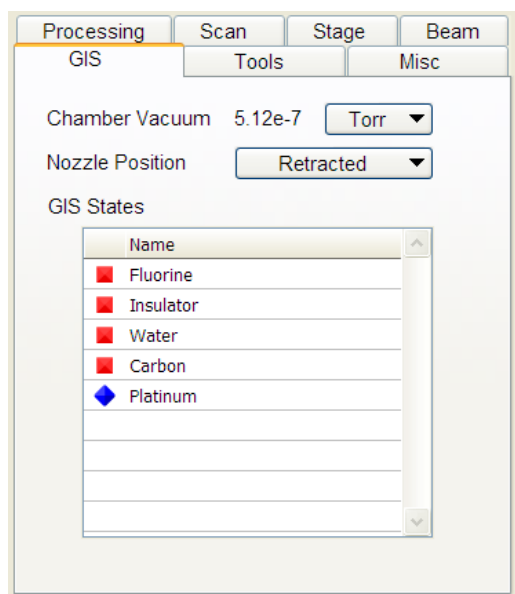


Figure 132: The GIS Tab

The GIS Tab displays chamber pressure information and provides the user with the ability to control the GIS system.

The Chamber Vacuum pressure units can be toggled between Pa, atm, mbar, and Torr using the drop down pressure unit list.

Nozzle Position can be changed between Retracted and Inserted using a drop down list. The GIS gas states are also listed.

Each gas state can be toggled between “not ready”, “valve close”, and “valve open” by left-clicking the target gas type in the list. When toggling from the not ready state, a dialogue box will appear asking the user whether the gas in question has been outgassed. An option to automatically outgas the line is available.

Note:

It is the user's responsibility to ensure inserting or retracting the nozzle will not damage the microscope or the sample.

Miscellaneous Tab

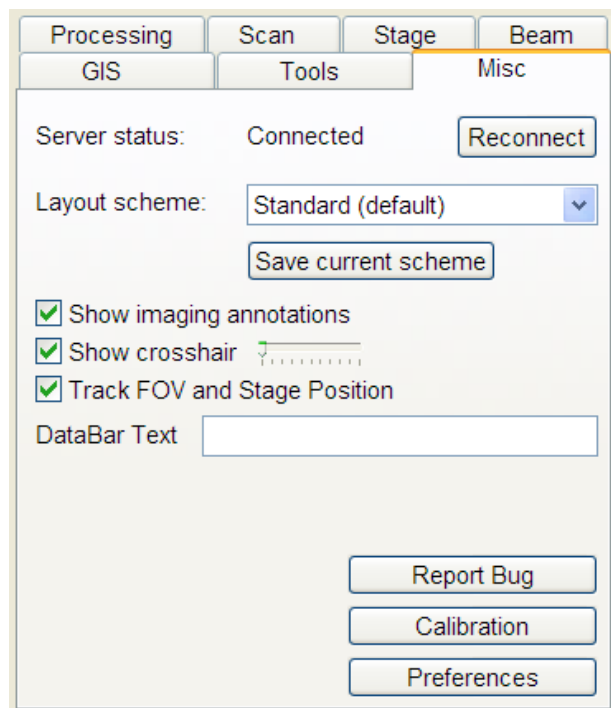


Figure 133: The Miscellaneous Tab.

The Miscellaneous Tab displays information and provides the user with access to interface settings that are less frequently used by the average user.

In general, most of the options on this tab will be configured at installation; however the user should be aware of the existence of these options.

Layout schemes provide the user with a list of predefined user interface configurations which may be selected at any time. The current layout can also be saved by pressing the **Save current scheme** button. New schemes can be defined in the Preferences dialog box.

The checkboxes for **Show imaging annotations**, **Show crosshair**, and **Track FOV and Stage Position** can all be toggled to disable each individual Visualization Panel attribute.

Any DataBar Text entered in the Miscellaneous Tab will appear in the DataBar of the Visualization Panel.

The **Calibration** button launches the Calibration Wizard which allows the NPVE system to be calibrated to the SmartSEM[®] image.

The **Preferences** button brings up the Preferences dialog box. The NPVE user interface can be customized to provide each user with his or her preferred viewing layout or even to allow a single user to set different parameters for various imaging conditions. In addition to defining various layout schemes, the basic default display parameters can be set on a per machine basis.

The **Report Bug** button copies the system information to the PC's clipboard which can then be pasted into an email, along with a detailed description of the bug, and sent to support@fibics.com.

12.4. The Tools Tab

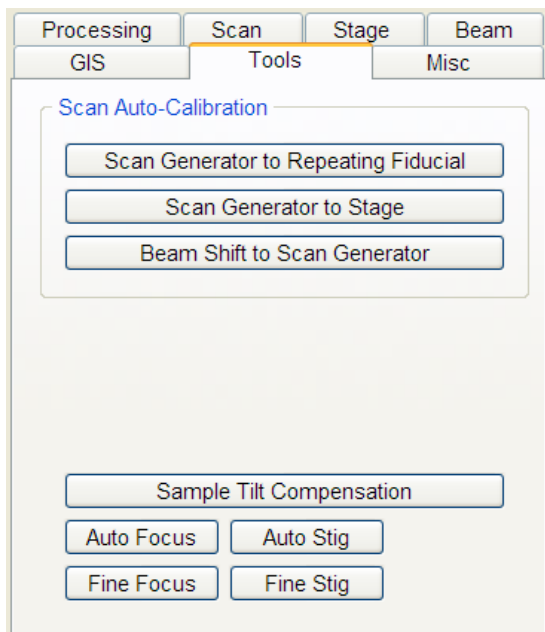


Figure 134: The Tools Tab

The Tools Tab contains several features to help the user accomplish tasks such as focusing the beam or calibrating the beam shift. They are:

- Scan Generator to Repeating Fiducial
- Scan Generator to Stage
- Beam Shift to Scan Generator
- Auto Focus
- Auto Stig
- Sample Tilt Compensation

Auto Focus and Auto Stig

The Auto Functions, “Auto Focus” and “Auto Stig”, are useful to assure that the SEM probe is properly focused and stigmated. Note that before starting the execution of the autotune functions, it is advisable to manually focus and stigmatize the beam so it is close to the ideal values. If the Auto Functions are run and the initial conditions are too far from the ideal values, the Auto Function will not find the optimal values.

The Auto Focus and Auto Stig functions are executed in two steps: a coarse step with a wide range of values swept to find the approximate values and a fine step to fully optimize the focus and stigmatization. If they fail during the coarse steps, the Auto Functions will not start the fine steps.

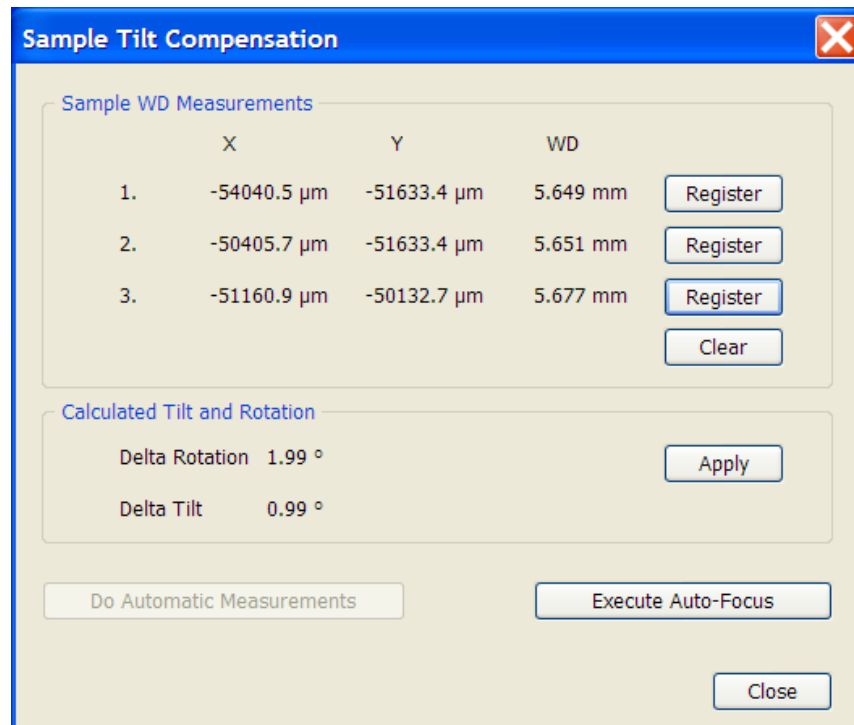
Sample Tilt Compensation

The Sample Tilt Compensation can be used to bring a tilted sample normal to the SEM column. This can be useful for imaging large areas since the focal plane of a tilted sample will vary spatially. The user must register three points on the sample’s surface. To register the three points, move the stage to a given location, focus the beam and click the **Register** button in the Sample Tilt Compensation Form (see Figure 135). Repeat this step two more times to populate the full list. Make sure that the registered points are several hundreds of microns away. This will ensure that the calculation is more reliable.

Note:

The user is responsible for ensuring any stage motion will not harm the microscope or the sample.

Once all three points are registered, the user can click the **Apply** button and it will calculate how much tilt and how much rotation it needs to make the sample completely flat vis-à-vis the SEM beam. Please note that the tilt plane of the stage is fixed. As such, the sample must be properly oriented (sample mounting and stage rotation) to successfully execute the Sample Tilt Compensation. Once corrected, use the scan rotate of the SEM beam to modify the imaging orientation.



The image shows a software window titled "Sample Tilt Compensation" with a blue header bar and a red close button in the top right corner. The window is divided into two main sections. The first section, "Sample WD Measurements", contains a table with three columns: "X", "Y", and "WD". It lists three measurement points, each with a "Register" button to its right. Below the table is a "Clear" button. The second section, "Calculated Tilt and Rotation", displays "Delta Rotation 1.99 °" and "Delta Tilt 0.99 °", with an "Apply" button to the right. At the bottom of the window, there are three buttons: "Do Automatic Measurements", "Execute Auto-Focus", and "Close".

	X	Y	WD	
1.	-54040.5 μm	-51633.4 μm	5.649 mm	Register
2.	-50405.7 μm	-51633.4 μm	5.651 mm	Register
3.	-51160.9 μm	-50132.7 μm	5.677 mm	Register

Clear

Delta Rotation 1.99 °

Delta Tilt 0.99 °

Apply

Do Automatic Measurements

Execute Auto-Focus

Close

Figure 135: The Sample Tilt Compensation Panel.

Beam Shift Calibration

The Beam Shift Calibration is used to ensure that the actual beam shift of the SEM beam is close to the target beam shift. The microscope SEM beam shift calibration is typically done at an EHT of 10 kV. Unfortunately this calibration may not be sufficiently accurate at a different EHT. For this reason, the Beam Shift Calibration should be repeated for each instance where a different EHT is used, particularly for low EHT (3 kV or less).

13. Setting NPVE User Preferences

The NPVE user interface can be customized to provide each user with his or her preferred viewing layout or even to allow a single user to set different parameters for various patterning conditions ☒ADV. In addition to defining various layout schemes, the more basic default display parameters can be set on a per machine basis. These settings are defined in the Preferences dialog box that is described in this section and are accessible through the Miscellaneous Tab of the NPVE Parameters Panel.

13.1. General

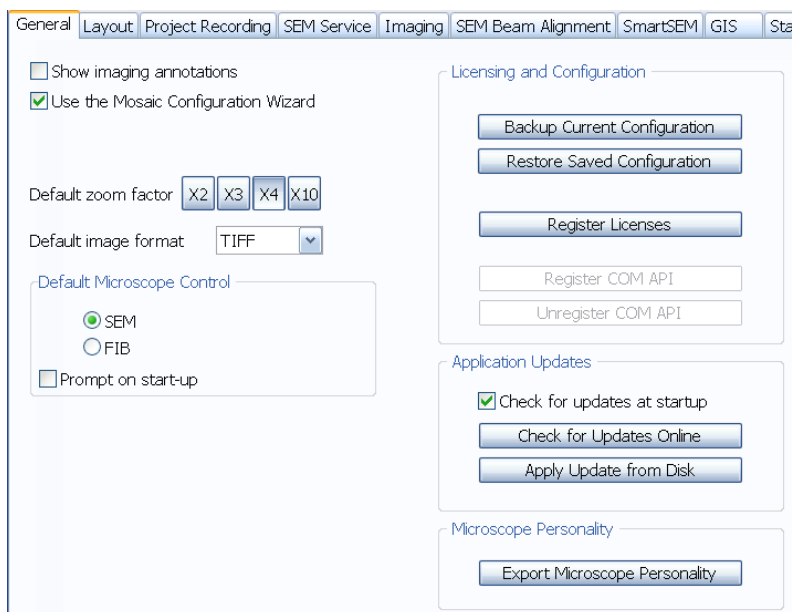


Figure 136: The General Tab of the User Preferences Panel.

The General Tab allows the user to further customize the interface. The user can choose to display or hide the imaging annotations (center target, FOV and size marker, etc.). The default zoom factor for the Zoom tool page can also be specified, as well as the default image format when data is saved. TIFF is the default image format as it provides lossless compression, but BMP or JPEG can also be selected. The **Export Microscope Personality** button allows the user to save the microscope personality in XML format.

13.2. Layout

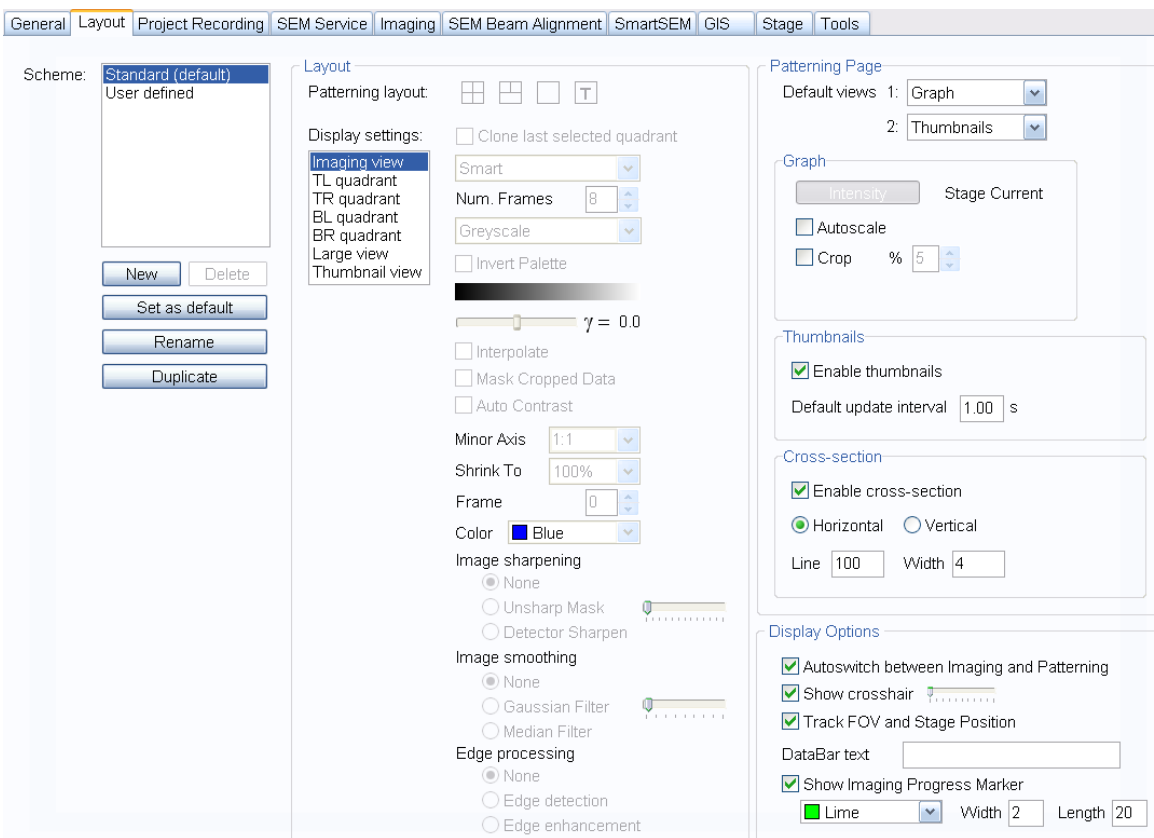


Figure 137: Screen shot of the Preferences dialog box.

A “Scheme” defines what the operating parameters of the imaging tabs are. In other words, defining a Scheme means setting up default parameters to use in the Layout Panel.

The Standard scheme is shipped with all machines and cannot be edited. However, any new scheme created by the user can be modified and used as the default when the NPVE software is started.

To create a new scheme, simply click on the **New** or **Duplicate** buttons. The **New** button will create a scheme based on the Standard scheme, whereas the **Duplicate** button will create a scheme by copying the one selected in the list. Schemes can be deleted or renamed. When set as default, the scheme will be loaded when the NPVE software starts up.

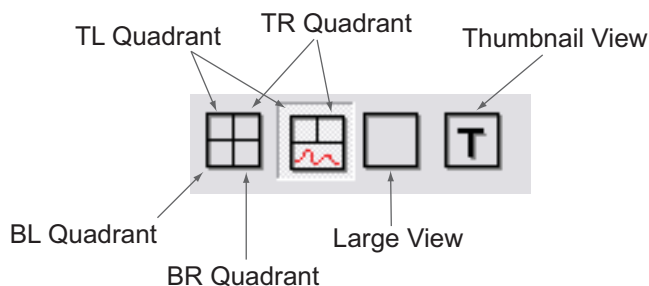


Figure 138: Parts of the different Layouts.

The Imaging view parameters can be adjusted in the Layout Panel, which is found in the Project Analysis main window. Additionally, the user can select the width, in pixels of the “Frame” displayed around the edge of each pattern box in the patterning views, as well as the colour of the background.

13.3. Project Analysis and Recording

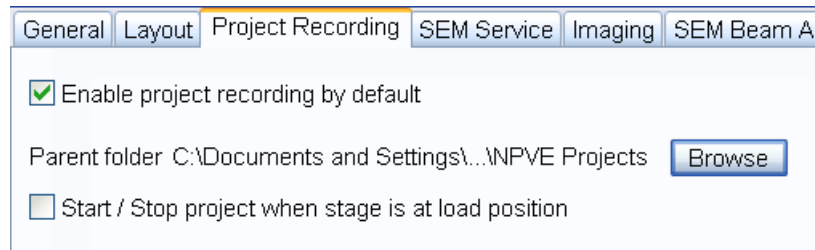


Figure 139: The Project Recording Tab of the User Preferences Panel.

The Project Analysis and Recording Tab controls the default behaviour of the NPVE's Project Analysis and Recording function, which stores all images (and optional patterning information along with their associated thumbnail and intensity data ☒ADV).

Project Analysis and Recording can be turned off if desired from this tab by selecting Enable Project Recording by Default. Although each job can be saved in a specific folder, it is possible to assign a default parent folder that would contain all standard job folders. Because of the high data throughput required for Project Analysis and Recording, this folder must reside on a local drive (it cannot be a networked or removable drive). The Parent Folder is set by default to be C:\NPVE\Project Recording\.

Additionally, project recording can be set to start and stop when the stage is at the load position.

Part III Project Analysis

14. Introduction

Project Analysis requires that Project Recording be enabled using the **Project Setup** button of the Display Tab on the Tools Panel, and it can be accessed with the **Analyze project** button of the same tab.

At each preset beam location, the beam is left to dwell for a given period of time before moving to the next point in the raster. At its simplest, a raster pass consists of deflecting the beam at fixed increments along one axis from a start point to an end point, dwelling for a fixed dwell time at each point. At the end of a line, the beam awaits a fixed retrace time before moving an increment in a second axis. The beam then returns to the start point in the first axis and begins again. This process continues until all increments in both axes have occurred, and the beam has dwelled at all points in the scan. The NPVE engine considers the completion of this process in both axes to be the completion of one frame or raster. The system then awaits a predetermined refresh time before starting over, repeating the process as many times as necessary until the desired dose per unit area has been delivered.

During each patterning operation (milling, gas assisted etching, deposition), the NPVE Advanced system records a dynamically re-scaled graph of the intensity signal, both average and integrated. As the intensity data used for this graph is taken directly from the dwell data generated from the FIB, it is quite sensitive, and dynamic re-scaling of the data improves its usability.

Along with generating graphs of intensity, the Advanced NPVE engine also creates and dynamically displays “thumbnails” of the patterning operation, recording a sub image of the mill box every quarter second, along with the cumulative dose and beam current. These thumbnail images can be stepped through or smoothly replayed forwards or backwards, either during a mill (whether paused or milling) or afterward as part of the review process.

The “Digital Lab Book” Project Analysis and Recording enabled by the NPVE Advanced system is a user selectable feature that, when enabled, records all aspects of the job - each image pass is recorded, in sequence, as well as each patterning operation (milling, gas assisted etching, deposition), and with the full set of NPVE generated thumbnails for the mill and the endpoint graph signals (both intensity and stage current, if available). The beam conditions, beam currents and cumulative dose throughout the mill, x and y spacing as well as dwell times, box size, position and raster style (raster, bitmap, etc.), stage position, etc. are recorded to disk and can be played back at any time - during the job, or at a later date for training or a “post mortem” of the job results.

The job review software within the Advanced NPVE system makes reviewing the job straightforward and quick. The job can be stepped through by following the sequence of images that were acquired before and after a patterning operation, drilling down to the mill(s) themselves where the user is shown the box positions, endpoint graph and mill thumbnails. Clicking at any point on the graph brings up the associated thumbnail, and the mill can be reviewed (“played”) forward or backward from any point.

There are several other features of NPVE Advanced, including the ability to selectively crop a region so that only data from that part of the mill is graphed and/or displayed, the ability to generate reconstructed cross-sections from the plan-view thumbnail mill images, and a host of image processing options and usability enhancements. These will be covered in the following sections, starting with an overview of the layout of the main software application itself.

The NPVE Project Analysis application allows the user to analyze all the mills and images that were acquired during a project using NPVE. In this application, one can review the parameters of the mills, and the endpoint graph of the mill, as well as reconstruct digital cross-sections through the acquired mill thumbnail data to, for example, explore the uniformity of the mill or the structure of the milled layers.

14.1. Loading a Project

All the project information is stored in a folder structure on the file system. Upon startup, the user is presented with a dialog box to select the folder that contains the project. Once selected, the images will start populating the interface as the project is being loaded in the background. It is not necessary for the user to wait to the end to start looking at mills and images, but the PowerPoint presentation can only be created when the loading is complete. For long projects loading all the images may take some time. Also, selecting a mill results in all of the thumbnails being loaded, which may take quite a bit of time for long mills. To load another project, click the **Load Project** button and select a different folder. Alternatively, the user can drag and drop a folder from the Windows' Explorer onto the Project Analysis application.

14.2. Creating a Power Point Presentation

The NPVE Project Analysis application enables the user to create a simple yet effective project report in the form of a PowerPoint presentation. Clicking on the **Start Presentation** button at the bottom of the form will start PowerPoint and create a new presentation using the default NPVE template. Some information regarding the project is inserted in the title slide at this point. Slides corresponding to the project images, mill information and endpoint analysis can be inserted by clicking on the associated buttons marked with the PowerPoint icon, as described in the following sections. Before exiting the project analysis application, the user will be required to save or close the presentation because PowerPoint will also be shut down at this point. Once a presentation is complete, the user is free to change the Slide Master template or colours and add and remove text as they desire. For information on these processes, consult the Microsoft PowerPoint online help.

14.3. Description of the Interface

The Project Analysis features three control regions:

- (a) Mill Chronology List
- (b) Main Tabs (Project, Info, and Analysis)
- (c) Control Buttons

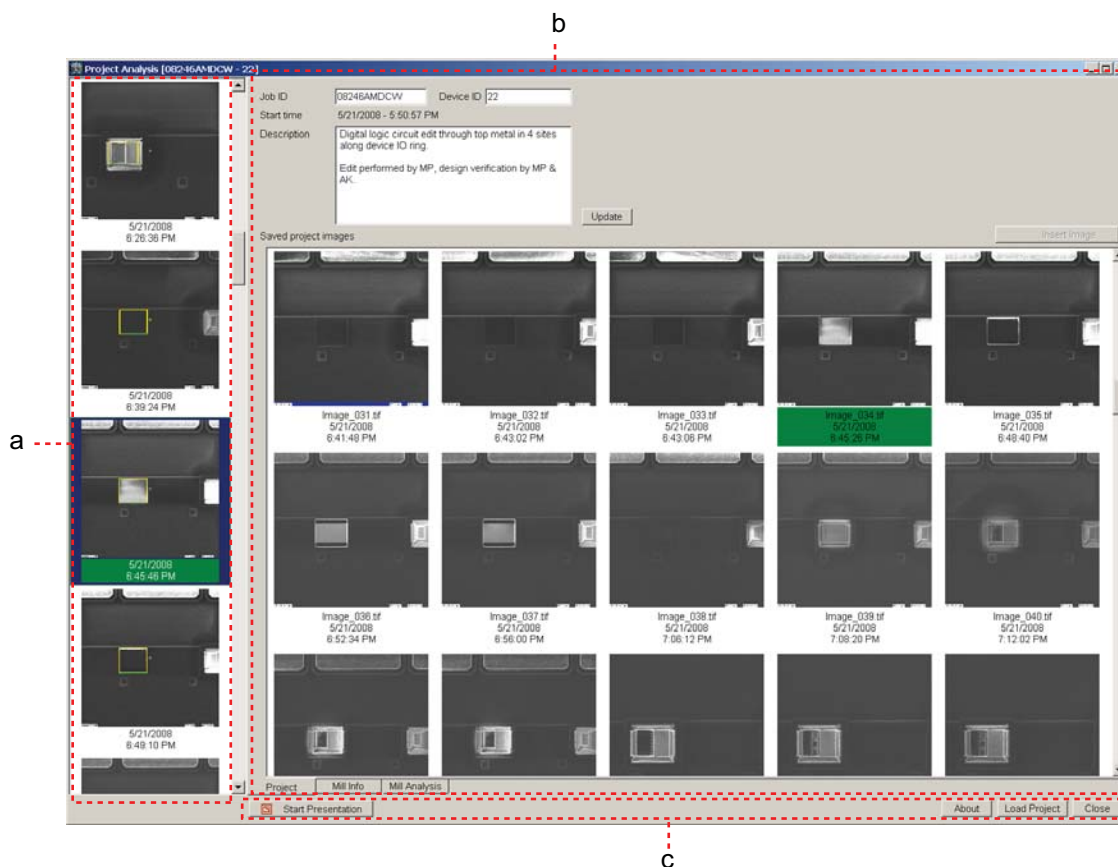


Figure 140 : Screenshot of the complete interface of the Project Analysis Application.

Mill Chronology List

The Mill Chronology List ((a), above) is a vertical list, controlled by a scroll bar, showing a thumbnail view of the images acquired immediately before the mill. The date and time that each patterning event starts is recorded below the image thumbnail. The currently selected mill is displayed with a gray background. In addition, if an image is selected in the main Project display, then the mill that was started after that image is highlighted in green. This allows quick visual cross-referencing between mills and images between the two lists.

Project Analysis Main Tabs

The Project Analysis main tabs ((b), above) contain the three main analysis functions: The Project Tab, containing a list of all recorded images, the Info Tab, containing information of all repairs active in

the selected mill, and the Analysis Tab, which allows the user to review the endpoint graph, associated thumbnail images of repairs, and reconstruct cross-sectional views of the repair.

Project View

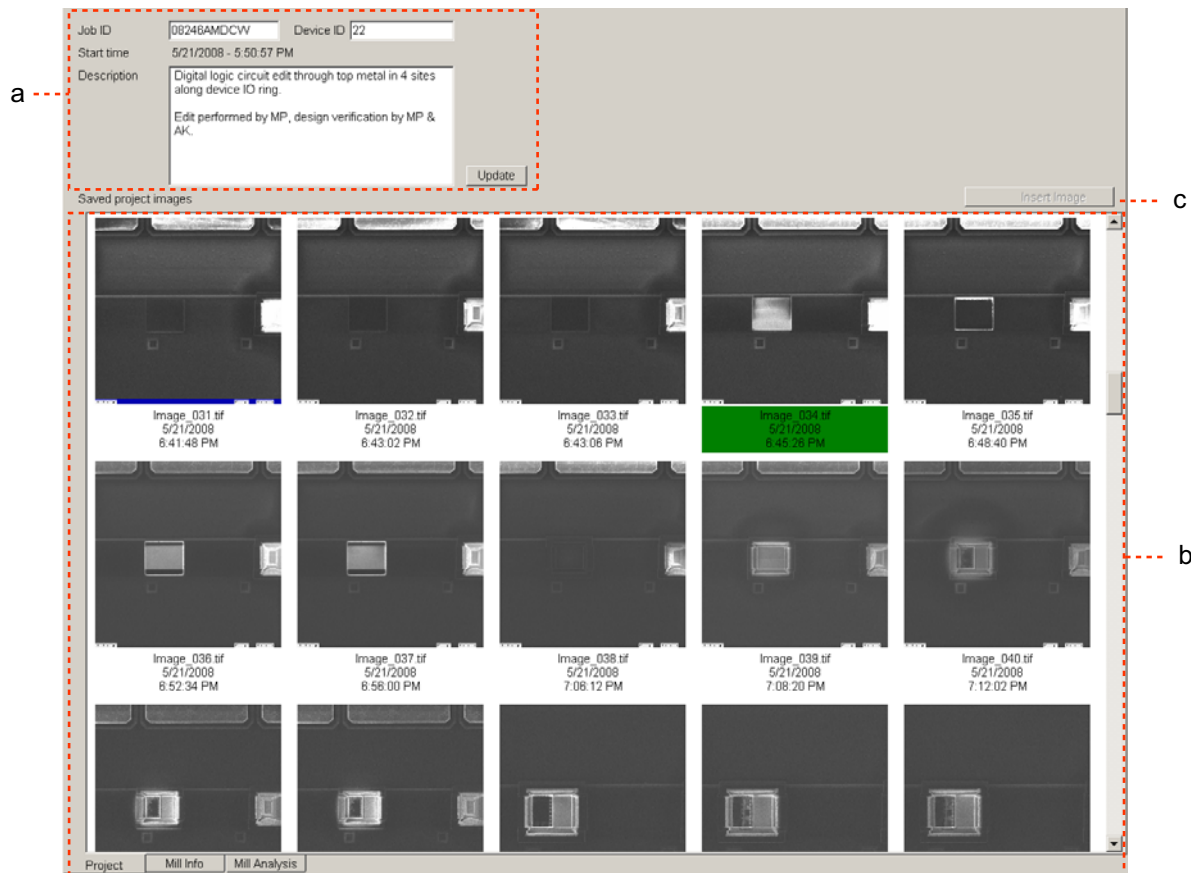


Figure 141: Overview of the Project View Panel of the Project Analysis application.

The Project view provides global information regarding the job, such as the assigned Job and Device IDs, the time at which the project was started and the description entered by the user (a). Although the Job and Device ID are read-only, the description can be edited. Clicking the **Update** button saves the edited description to the job details file. In addition, the **Project** view displays thumbnails of all images saved in the project folder (b). Details of the image, such as its filename and date/time of creation are displayed below the thumbnail. Any of these images can be viewed in full size to save, print or copy to the clipboard by double-clicking on the thumbnail. The image file can also be deleted by pressing the **DELETE** key on the keyboard. If a PowerPoint presentation was started for this project, the image can be inserted into the presentation as a new slide by clicking the **Insert Image** button. Once inserted, a small PowerPoint icon appears below image thumbnail.

Mill Info View

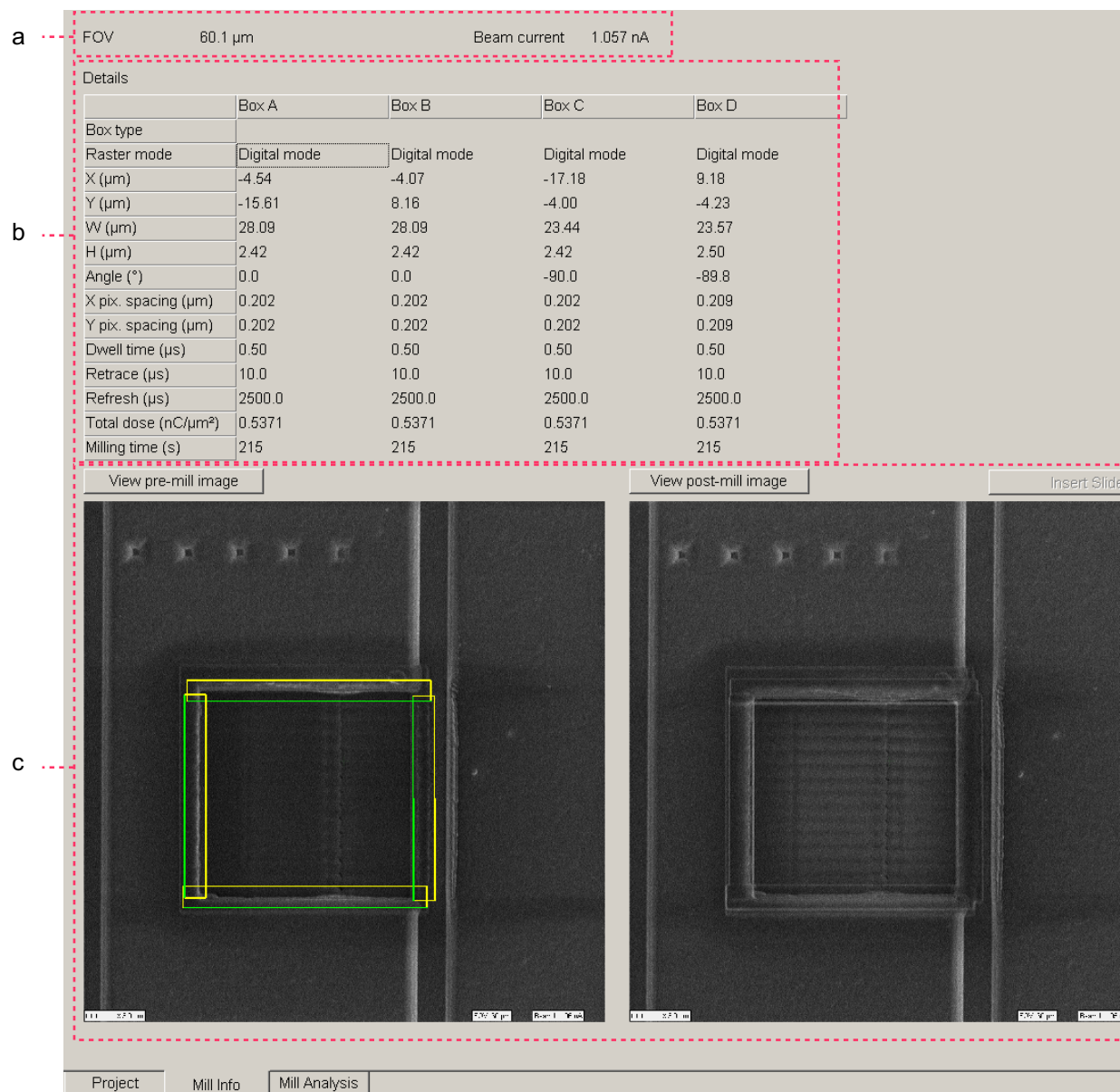


Figure 142: Screenshot of the Patterning Info view of the Project Analysis.

The Mill Info view provides detailed information about the currently selected mill. It includes general information such as the FOV and beam current used during the mill, in region (a) above. Details regarding the rastering of the mill box are provided in (b), as well as the image taken before the mill (pre-mill image) and after the mill (post-mill image). Clicking on the **View pre/post-mill image** buttons or double clicking on an image displays the full size image and allows saving/printing and copying to the clipboard. If a PowerPoint presentation was started, the pre and post mill images, along with some mill info can be inserted as a slide by clicking the **Insert Slide** button.

Mill Analysis View

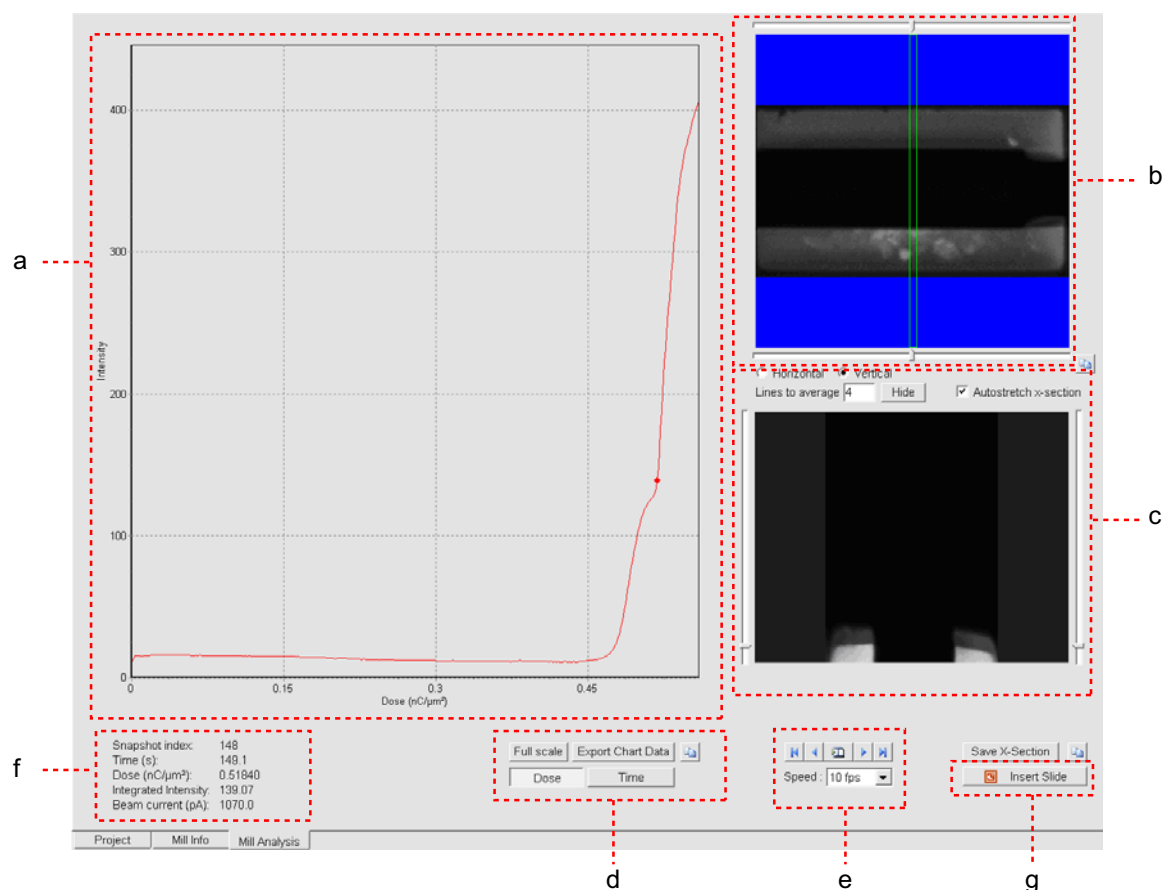


Figure 143: The Mill Analysis Panel

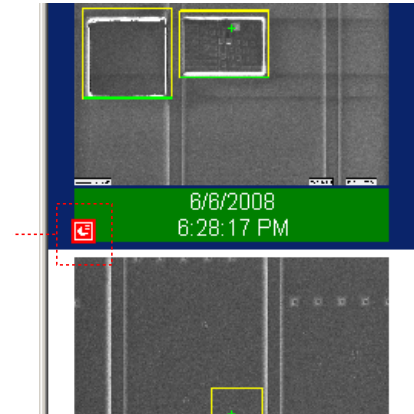
The *Mill Analysis* view is the heart of the Project Analysis Form. It displays endpoint information related to the currently selected mill and allows the user to fully investigate the patterning process. The user can simultaneously view: an endpoint graph (a), a thumbnail image from a selected point in the endpoint graph (b) and a reconstructed cross-section (c).

Viewing these three pieces of information simultaneously provides the user with a multi-dimensional analysis of the mill. The thumbnail provides, at a given time (or dose, and thus loosely related depth), spatial information. The endpoint graph provides temporal information of the entire image brightness (signal intensity) and the cross-section combines both spatial and temporal information.

The x-axis of the endpoint graph can read either time or dose, as selected in (d). A red diamond on the endpoint graph selects the dose (or time) for which the thumbnail will be displayed. The vertical track bars around the cross-section also reflect this position. As such, the user can click on the graph at a specific location and view the thumbnail acquired at that time. In addition, the user can scan through the snapshots manually or play back the mill as a movie with the movie controls (e). As the snapshots scroll, their information (f) is also updated. The user can also select the speed at which the snapshots scroll, whether related to the real patterning time or simply at 10 updates per second.

The cross-section can be taken horizontally or vertically, and the user selects which lines of the thumbnail are used for the reconstruction. A green outline on the thumbnail indicates the selected slice. Its position can be adjusted via the track bars around the thumbnail. Any of the images in the page can be copied to the clipboard or saved to a file, and if a PowerPoint presentation was started, the entire view can be inserted as a slide by clicking on the **Insert Slide** button (g). If this form gets inserted as a slide, then a PowerPoint icon gets appended to the mill thumbnail in the Mill Chronology List, as illustrated below.

The PowerPoint icon displayed on the relevant image from the Mill Chronology List, indicating that details from this mill have been inserted as a slide in the active PowerPoint presentation.



APPENDICES

A. Validating Images in the NPVE Mosaic Tool

After acquiring an image, the NPVE Mosaic Tool can pass the image filename to an external program that analyzes the image to identify if it is adequate or if it should be re-acquired because of the presence of imaging artefacts.

1. Program Requirements

The external program requirements are as follows:

- It is a console (command-line) application that outputs to the standard output.
- It accepts the full path of the image file as the last argument that is passed to it. The path should be enclosed in double quotation marks if it contains a space.
- The last line of its output should be either PASS or FAIL. Prior to this, it can output debugging or processing information which will be displayed in the mosaic log but is otherwise ignored. Only the final line is relevant to identify success or failure.

The ImageVerify.exe application supplied with the NPVE package adheres to these requirements and can be used for this purpose.

2. Configuration of the Mosaic Tool

The Validation Tab of the Mosaic Batch Processing window contains an Image Verification section.

In this section, if the checkbox labelled **Verify image quality using script** is not checked, images will not be verified. Otherwise, the mosaic tool will use the additional information provided to determine if it should repeat the tile image or proceed to the next tile. The options are the following:

Script	Provide the full path to the script that will be executed to verify the image quality. You can browse for an executable file by clicking on the Browse button ... next to the edit box.
Script options	Specify the optional command line parameters in the edit box. These will be passed to the script before the image file path.
Number of repeats allowed	Specify the number of times the image acquisition should be attempted before proceeding to the next tile even if the verification fails.
Script Timeout (s)	Specify the amount of time in seconds given to the script to complete the analysis, before automatically proceeding to the next tile. If an instance of the script fails to complete within this time, an error will be posted to the log and the tool will proceed to the next tile, and the script process will be unconditionally terminated.

3. Using ImageVerify.exe

ImageVerify.exe processes an image using the same algorithm as the Matlab script written by Daniel Berger of MIT for the MIT and Harvard groups. It can be used by the NPVE Mosaic Tool or can be run as a stand-alone application from the command line prompt.

USAGE: ImageVerify [-tX] [-x0] ImagePath, where

-t (Optional)	Sets the threshold to X (floating point). The default value is 10.0.
-x0 (Optional)	When applied, ignores the first row of the image. By default the first row is included.
ImagePath (Required)	This is the full path name of the image file. The path name should be enclosed in double quotation marks (" ") if it contains a space, otherwise no quotation marks are required. Please see the Example below.

The output of this application includes:

- The calculated threshold (product of the user defined threshold and the standard deviation of the standard deviations of each line pair difference)
- The average standard deviation of each line pair difference
- The standard deviation of the standard deviations of each line pair difference
- The largest calculated standard deviation for a given line pair difference. This also includes how much larger than the average it is and for which row this value was observed.
- The final line reads FAIL or PASS, whether the largest calculated standard deviation exceeds the threshold above the average or not.

Example #1

```
C:\>IMAGEVERIFY "I:\2009 WORK\09065 ATLAS NPVE FOR SEM\HARVARD
DATA\VARIOUS DEFECTS\W6_S02_CROP.TIF"
THRESHOLD = 20.800 [10.0]
AVG STDEV = 28.536
STDEV OF STDEV = 2.080
MAX STDEV = 42.550 [+14.014] AT ROW = 84
PASS
```

B. NPVE Module Power Up/Down and Reset Procedures

1 Power ON Procedure

1. Ensure all cables are properly connected
2. Ensure power supply connection is secure
3. Turn green power switch on unit to ON position
4. Turn computer ON (if it was not ON)
5. Start NPVE

2 Power OFF Procedure

1. Exit NPVE
2. Turn green power switch on unit to OFF position
3. Turn computer OFF if necessary
4. Disconnect cables if necessary

3 Partial System Reset after blackout or systematic NPVE data acquisition error

1. Exit NPVE
2. Turn green power switch on unit to OFF position
3. Verify USB cable connections
4. Wait 10 seconds
5. Turn green power switch on unit to ON position
6. Start NPVE

4 Full System Reset for conditions not resolved by Partial System Reset

1. Turn computer OFF
2. Turn green power switch on unit to OFF position
3. Wait 10 seconds
4. Turn green power switch on unit to ON position
5. Turn computer ON (if it was not ON)
6. Start NPVE

C. General Troubleshooting

Symptom / Message	Problem	Solution
Message Box <i>'Verify Power and Connections to Acquisition Module'</i> or System message <i>'Could not find or configure data acquisition module'</i>	Module is not powered on	Perform system power ON procedure
	USB cables are not connected	Verify that all 3 (NPVE) or 2 (ATLAS) USB cables are properly connected to the module and the computer. Perform Partial System Reset
	USB device is not recognized	Verify / correct USB device registration Perform Partial System Reset
	Unknown problem	Level 1: Perform Partial System Reset Level 2: Perform Full System Reset Level 3: Contact system vendor
System message <i>'Unable to locate data acquisition configuration file'</i>	Application was not installed properly	Reinstall NPVE application
	File is missing	Contact vendor for updated file
System message <i>'Unable to initialize acquisition module clocks'</i>	Unknown problem	Level 1: Perform Partial System Reset Level 2: Perform Full System Reset Level 3: Contact system vendor
System message <i>'Unable to configure acquisition module'</i>	Unknown problem	Level 1: Perform Partial System Reset Level 2: Perform Full System Reset Level 3: Contact system vendor

D. Troubleshooting Communications with Fibics Remote Communication Server

1. Network Configuration

Tip:

How to access the TCP/IP settings

To access the Internet Protocol (TCP/IP) Properties dialog box:

- Click **Start** button
- Right mouse click on **My Network Places**
- Select **Properties**
- Right mouse click on **Local Area Connection**
- Select **Properties**
- In **Local Area Connection Properties** dialog box:
 - In **General** Tab:
 - Select **Internet Protocol (TCP/IP)**
 - Click **Properties** button

Below the user will find the settings to use depending on the configuration and topography of the network the two computers reside on.

The NPVE and SmartSEM computers are on an isolated local area network (LAN) (using a switch or a cable for direct connection between computers).

SmartSEM PC Internet Protocol (TCP/IP) Properties:

Use the following IP address:

IP address: 192.168.1.2
Subnet mask: 255.255.255.0
Default gateway: 192.168.1.1

Use the following DNS server address:

Preferred DNS server: <blank>
Alternate DNS server: <blank>

NPVE PC Internet Protocol (TCP/IP) Properties:

Use the following IP address:

IP address: 192.168.1.1
Subnet mask: 255.255.255.0
Default gateway: <blank>

Use the following DNS server address:

Preferred DNS server: <blank>
Alternate DNS server: <blank>

The NPVE and SmartSEM computers are part of a company-wide LAN with dynamic IP addresses

SmartSEM PC Internet Protocol (TCP/IP) Properties:

Obtain an IP address automatically

Obtain DNS server address automatically

NPVE PC Internet Protocol (TCP/IP) Properties:

Obtain an IP address automatically

Obtain DNS server address automatically

The NPVE and SmartSEM computers are part of a company-wide LAN with fixed IP addresses

SmartSEM PC Internet Protocol (TCP/IP) Properties:

Use the following IP address:

IP address: <ask IT personnel>

Subnet mask: <ask IT personnel>

Default gateway: <ask IT personnel>

Use the following DNS server address:

Preferred DNS server: <ask IT personnel>

Alternate DNS server: <ask IT personnel>

NPVE PC Internet Protocol (TCP/IP) Properties:

Use the following IP address:

IP address: <ask IT personnel>

Subnet mask: <ask IT personnel>

Default gateway: <ask IT personnel>

Use the following DNS server address:

Preferred DNS server: <ask IT personnel>

Alternate DNS server: <ask IT personnel>

The NPVE and SmartSEM computers are on an isolated LAN/directly connected, but one of the two computers is also connected to a separate LAN.

In this case, configure the SmartSEM/NPVE LAN according to scenario A, but leave the **Default gateway** blank for both computers. Configure the Local Area Connection to the separate LAN as required.

2. Firewall Configuration

Both the SmartSEM and NPVE computers may be equipped with firewalls to prevent unintended communication to and from each computer. The firewalls must be configured to allow communication between the NPVE application and the Fibics Remote Communication Server running on the SmartSEM computer.

The following deals with the case where the native Windows Firewall is used on both PCs. If another firewall application is used (Symantec, McAfee, etc.), refer to the person responsible for IT to configure it based on the information below.

Tip: How to access the Windows Firewall Settings

To access the Firewall Settings:

Click the **Start** button

Select **Control Panel**

Double-click on the **Windows Firewall** icon

In the **General** Tab,

To enable the firewall, select **On (recommended)**

Do NOT check the **Don't allow exceptions** box

To disable the firewall, select **Off (recommended)**

It is recommended to keep the firewall enabled to minimize network security risks.

On the SmartSEM PC

The Fibics Remote Communication Server must be allowed access through the firewall on TCP port 4915. In order to allow communication through the firewall, an exception must be created.

When the server application is started for the first time, Windows Firewall will prompt you to **Block** or **Unblock** the program. Select **Unblock**. This will create the necessary exception and will allow communication.

If you want to verify or change the firewall settings for a program, access the **Firewall Settings** and go to the **Exceptions** Tab. From the **Programs and Services** list, locate the *Fibics Remote Communication Server for Zeiss Microscopes* entry. If it is found, check its box to create an exception. If it is not found, click the **Add Program...** button, select *Fibics Remote Communication Server for Zeiss Microscopes* from the list and press **OK**.

On the NPVE PC

The NPVE application must be allowed access to the SmartSEM PC through the firewall. If available, it also requires access to the Internet to check for updates.

After installing the application, create an exception for it as follows: access the **Firewall Settings** and go to the **Exceptions** Tab. Click the **Add Program...** button, select *NanoPatterning and Visualization Engine* from the list and press **OK**.

3. Diagnostics

Troubleshooting Network Configuration

To verify the connectivity between the two computers:

1. Get Names and IP addresses of both computers (see the Tip *How to retrieve the System Name and IP Address*, below).
2. On the SmartSEM computer, open a command prompt (Start ⇒ All Programs ⇒ Accessories ⇒ Command Prompt)
3. Type: *ping NPVE_Computer_Name* and press Enter
4. Verify you are getting replies
5. On the NPVE computer, open a command prompt (Start ⇒ All Programs ⇒ Accessories ⇒ Command Prompt)
6. Type: *ping SmartSEM_Computer_Name* and press Enter
7. Verify you are getting replies

Troubleshooting Ping Request:

Ping Response	Problem	Solution
Ping request could not find host <i>Computer_Name</i> . Please check the name and try again.	Wrong name was entered	Check computer name and try again
	Name is not recognized	Try again by using the IP address
	Cable is disconnected	Verify the connection. Use ipconfig to verify the configuration (see <i>How to retrieve the System Name and IP Address</i>).

Troubleshooting NPVE/SmartSEM Communication

Symptom / Message	Problem	Solution
NPVE cannot connect to Fibics Remote Communication Server (<i>Unable to communicate with request server or Unable to communicate with notification server</i>)	Network is not configured properly	Verify network configuration (see <i>Network Configuration</i>) and use ping to test settings and troubleshoot network configuration problems (see <i>Troubleshooting Network Configuration</i>).
	Fibics Remote Communication Server is not started or is unresponsive	Start the Fibics Remote Communication Server on the SmartSEM computer or restart it from the companion Fibics Watchdog application (see <i>How to start the Fibics Remote Communication Server</i>)
	Wrong computer name was entered in NPVE	In the NPVE software, go to the Preferences Panel and set the SmartSEM Computer Name in the SmartSEM Tab (see <i>How to retrieve the System Name and IP Address</i>)
	Firewall is preventing communication	Configure or disable firewall on the SmartSEM and NPVE computers (see <i>Firewall Configuration</i>)

	Computer name is not recognized	In the NPVE software, go to the Preferences Panel and set the SmartSEM IP address instead of the computer name (see <i>How to retrieve the System Name and IP Address</i>)
	Wrong IP address was entered in NPVE	In the NPVE software, go to the Preferences Panel and set the SmartSEM computer IP address (see <i>How to retrieve the System Name and IP Address</i>)
Unable to initialize session with SmartSEM	NPVE was started before user logged in to SmartSEM and connection timed out	Restart NPVE system after SmartSEM login and initialization are complete
	Unknown problem	Level 1: Check Fibics Remote Communication Server output for description of errors. Level 2: Check SmartSEM server log for errors. Level 3: Contact system vendor

Tip: How to retrieve the System Name and IP Address

To get the System Name from a PC:

- Open **System Properties** (**Windows Logo** + **BREAK** on the keyboard)
- In **Computer Name** Tab,
 - read **Full computer name** entry (not including the final period)

Alternatively, it is possible to use the computer's IP address instead of its name. This is not recommended when DHCP (dynamic IP addressing) is enabled since the IP address may change over time.

To get the IP address of a PC:

- Open the **Run** dialog box (**Windows Logo** + **R** on the keyboard)
- Type: *cmd* and click **OK**
- Type: *ipconfig* and press Enter
- Read the **IP Address** entry from the **Ethernet adapter Local Area Connection** properties.

Tip: How to start the Fibics Remote Communication Server

1. On the SmartSEM computer, verify that the Fibics Remote Communication Server is installed.
2. To start the server, select it from the **Start** menu (Start ⇨ All Programs ⇨ Fibics ⇨ Fibics Remote Communication Server). This will also start the Watchdog program. Both programs will minimize themselves to the system tray.
3. If the server is running, its icon () is visible in the system tray on the right of the screen.
4. To kill/restart the server, click on the Fibics Watchdog icon in the system tray () and press the **Restart** button. This will terminate the Fibics Remote Communication Server. Wait 10 seconds then restart it.

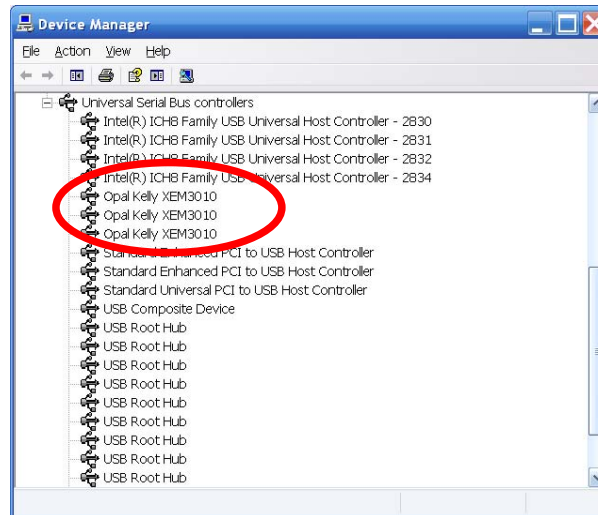
E. USB Device Registration

Windows Logo key + **BREAK** key to open the System Properties

Go to the Hardware Tab

Click on the **Device Manager** button

Expand the list for Universal Serial Bus controllers



Verify there are 2 (ATLAS system) or 3 (NPVE system) entries for Opal Kelly XEM 3010

Option 1:

1. Right click on the entry with a question mark (?)
2. Select Update Driver...
3. Proceed with the reinstallation of the driver
4. Repeat for all entries with a question mark

Option 2:

1. Right click on first entry
2. Select Uninstall
3. Repeat Uninstall for each entry
4. Close Device Manager
5. Close System Properties
6. Perform Partial System Reset
7. When the system is powered ON, complete the registration of all 2 (ATLAS) or 3 USB (NPVE) controllers

F. Using a single Keyboard and Mouse with Synergy

1. Introduction

Synergy is free open source software that allows two computers to seamlessly share a single keyboard and mouse. When the mouse is moved from one screen to the other, the keyboard gets assigned to the computer controlling the active screen. This document explains how to install Synergy on the SmartSEM and NPVE PCs and to configure them to use the Zeiss keyboard/console and mouse. In this configuration, the original NPVE keyboard and mouse are not needed and can be removed from the work area. For complete help and documentation regarding Synergy, please refer to the project's website at <http://synergy-foss.org>.

2. Installation

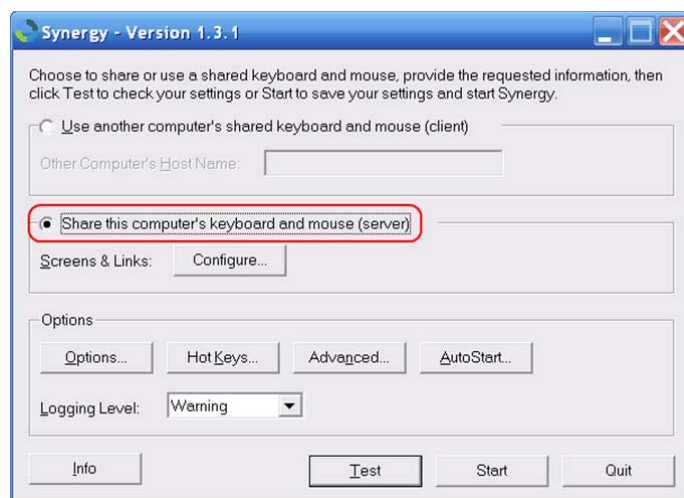
Synergy should be installed from the NPVE Installation CD on both the SmartSEM and the NPVE computers. Prior to configuration, please verify that both computers are properly connected to a network or to each other, as Synergy communicates over the network.

3. Configuration

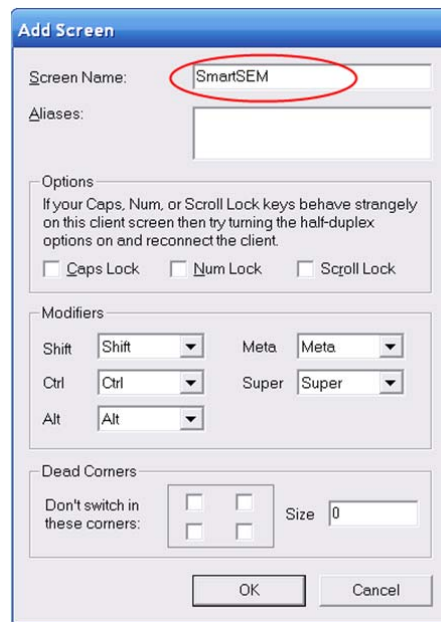
SmartSEM PC

On the SmartSEM PC, Synergy is configured as a Server, since it's this PC's original mouse and keyboard that will be used. After installing Synergy, start the application and follow these steps:

In the main window, select Share this computer's keyboard and mouse (server).

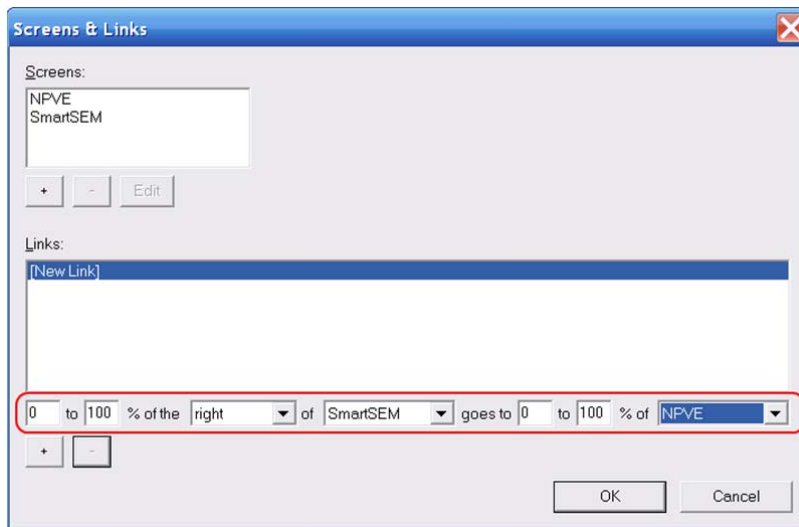


Click the **Configure...** button to open the **Screens & Links** window. Then click the + button to add a screen named "SmartSEM".



The "Add Screen" dialog box is shown. The "Screen Name" field is highlighted with a red circle and contains the text "SmartSEM". Below it is an empty "Aliases" field. The "Options" section contains a text box with instructions: "If your Caps, Num, or Scroll Lock keys behave strangely on this client screen then try turning the half-duplex options on and reconnect the client." Below this are three checkboxes: "Caps Lock", "Num Lock", and "Scroll Lock", all of which are unchecked. The "Modifiers" section contains four dropdown menus: "Shift" (set to "Shift"), "Meta" (set to "Meta"), "Ctrl" (set to "Ctrl"), and "Super" (set to "Super"). The "Alt" dropdown is also set to "Alt". The "Dead Corners" section contains a "Don't switch in these corners:" label, a 2x2 grid of checkboxes, and a "Size" field set to "0". At the bottom are "OK" and "Cancel" buttons.

Click the + button to add a screen named "NPVE". Enter the information below and click the + button to add a new link.



The "Screens & Links" dialog box is shown. The "Screens:" section contains a list box with "NPVE" and "SmartSEM". Below the list box are three buttons: "+", "-", and "Edit". The "Links:" section contains a list box with "[New Link]". Below the list box is a red-bordered box containing a link configuration: "0 to 100 % of the right of SmartSEM goes to 0 to 100 % of NPVE". Below this box are two buttons: "+" and "-". At the bottom are "OK" and "Cancel" buttons.

Enter the information below and click the + button to add a new link.

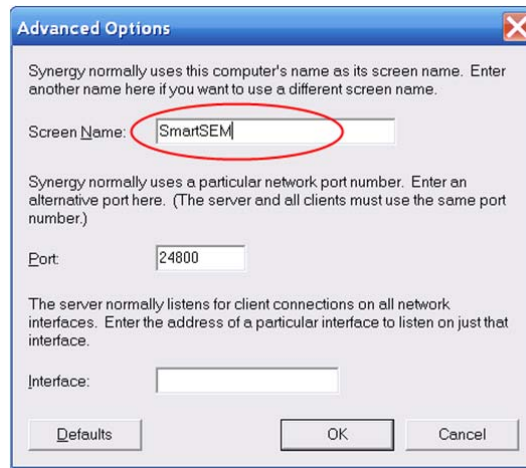
The 'Screens & Links' dialog box has a title bar with a close button. It contains two main sections: 'Screens' and 'Links'. The 'Screens' section has a list box with 'NPVE' and 'SmartSEM', and buttons for '+', '-', and 'Edit'. The 'Links' section has a list box with 'NPVE is right of SmartSEM' and '[New Link]'. Below the list boxes is a configuration area with a red border. It contains two input fields: '0' and '100', followed by '% of the', a dropdown menu with 'left' selected, 'of', a dropdown menu with 'NPVE' selected, 'goes to', another '0' and '100' input fields, '% of', and a dropdown menu with 'SmartSEM' selected. At the bottom right are 'OK' and 'Cancel' buttons.

After configuration is complete, it should look like this:

The 'Screens & Links' dialog box is shown after configuration. The 'Screens' section now has 'NPVE' selected in the list box. The 'Links' section now has two entries: 'SmartSEM is left of NPVE' and 'NPVE is right of SmartSEM', with '[New Link]' at the bottom. The configuration area at the bottom has three red boxes around the dropdown menus: the first contains a hyphen, the second contains a hyphen, and the third contains a hyphen. The 'OK' and 'Cancel' buttons are at the bottom right.

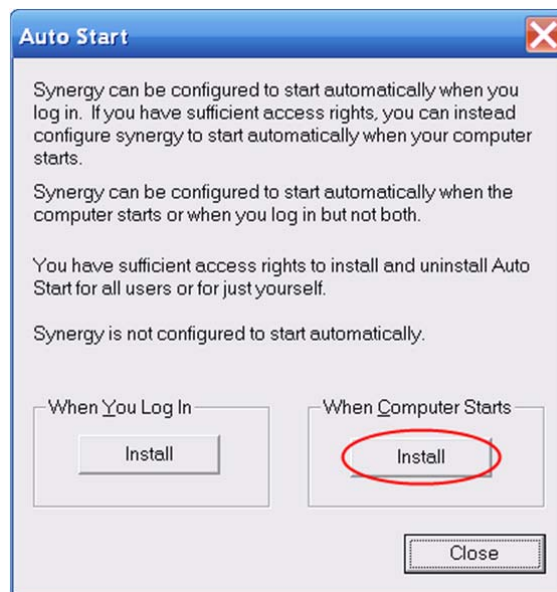
Click the **OK** button.

Click the **Advanced...** button in the Synergy window to open the **Advanced Options** window. Enter "SmartSEM" in the Screen Name field.



Click **OK** to close the window.

Click the **AutoStart...** button in the Synergy window to open the **Auto Start** window. Click the **Install** button in the **When Computer Starts** box.

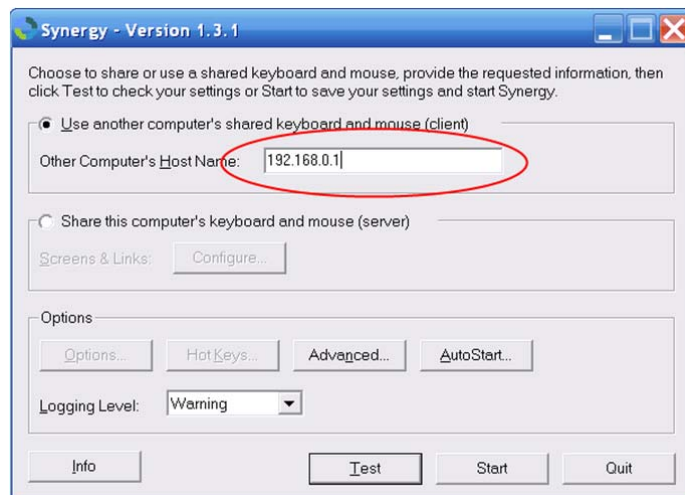


Click **Close** to close the window.

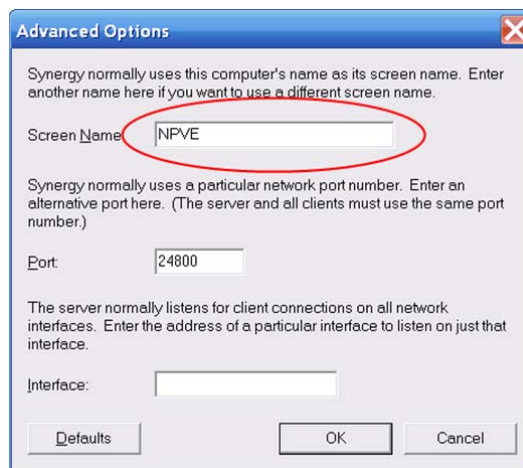
NPVE PC

On the NPVE PC, Synergy is configured as a Client. After installing Synergy, start the application and follow these steps:

In the main window, select **Use another computer's shared keyboard and mouse (client)**. Then, in Other Computer's Host Name field, enter the SmartSEM computer's name or IP address (see Network Configuration and Troubleshooting to identify this).

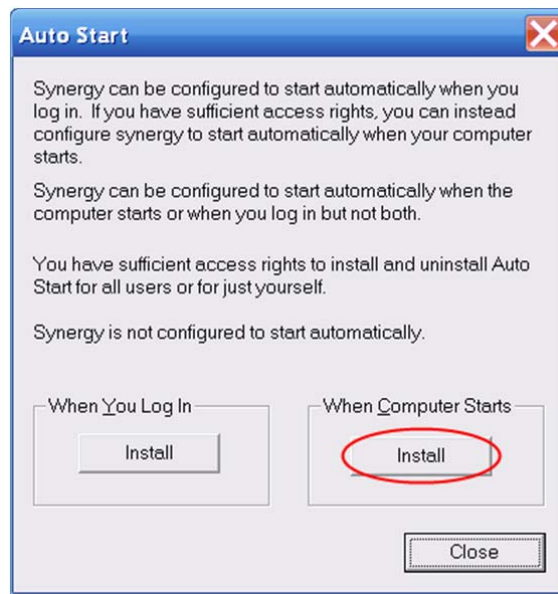


Click the **Advanced...** button to open the **Advanced Options** window. In the Screen Name field, enter "NPVE".



Click **OK** to close the window then click the **AutoStart...** button to open the **Auto Start** window.

Click the **Install** button in the **When Computer Starts** box.



Click **Close** to close the window.

Verify that Synergy is working by clicking the **Test** button in the Synergy window. If the test is successful, click the **Start** button. When Synergy is running, you will see its icon () in the system tray.

4. Deactivate/Uninstall

If the user wishes to stop using Synergy, it can be stopped and prevented from starting with the computer. To do this, open the application by double-clicking on the icon in the system tray and click the **AutoStart...** button. Then, in the Auto Start window, click the **Uninstall** button as illustrated below:



G. Shortcut Keys

Key	Action
Del	Delete selected shapes.
CTRL+Del	Delete all shapes
CTRL + A	Select all shapes
CTRL + C	Copy selected shapes
CTRL + D	Duplicate selected shapes
CTRL + V	Paste shapes
CTRL + X	Cut selected shapes
CTRL + Z	Undo last action
ArrowUp	Move shape by 1 pixel in upward direction
ArrowDown	Move shape by 1 pixel in downward direction
ArrowLeft	Move shape by 1 pixel to the left
ArrowRight	Move shape by 1 pixel to the right
ArrowUp + SHIFT	Move shape by 5 pixel in upward direction
ArrowDown + SHIFT	Move shape by 5 pixel in downward direction
ArrowLeft + SHIFT	Move shape by 5 pixel to the left
ArrowRight + SHIFT	Move shape by 5 pixel to the right
CTRL	Create from centre Stretch from centre
SHIFT	Select/Deselect shape on mouse down
ALT	Lock the aspect ratio of the shape at creation Lock the node positioning to a integer number of 45 ° Lock shape movement at 45 ° increment. Constraint the rotation of a shape to multiple of 15°.
ESC	Cancel shape creation
F10	Toggle Hide/Show All
F11	Toggle Verification Mode for all shapes
F12	Toggle Verification Mode for selected shapes

H. Cursors

	Annotation editing mode
	Mouse on inactive region of the Visualization Panel
	Mouse over a moveable shape
	Mouse over a moveable node
	Moving a shape
	Mouse over a locked shape
	Select Patterning Edge mode, mouse over a valid edge
	Select Patterning Edge mode, mouse not over a valid edge
	Select Patterning/Scan Angle Mode
	Mouse over the bottom-left corner of the shape in rotation mode
	Mouse over the bottom-right corner of the shape in rotation mode
	Mouse over the top-left corner of the shape in rotation mode
	Mouse over the top-right corner of the shape in rotation mode
	Mouse over the top-left or bottom-right stretching node
	Mouse over the middle-left or middle-right stretching node
	Mouse over the top-middle or bottom-middle stretching node
	Mouse over the top-right or the bottom-left stretching node
	Mouse over an edge close to an angle of 45°
	Mouse over an edge close to an angle of -45°
	Mouse over an edge close to an angle of 0°
	Mouse over an edge close to an angle of 90°